

Development of Cardiovascular systems (CVS)



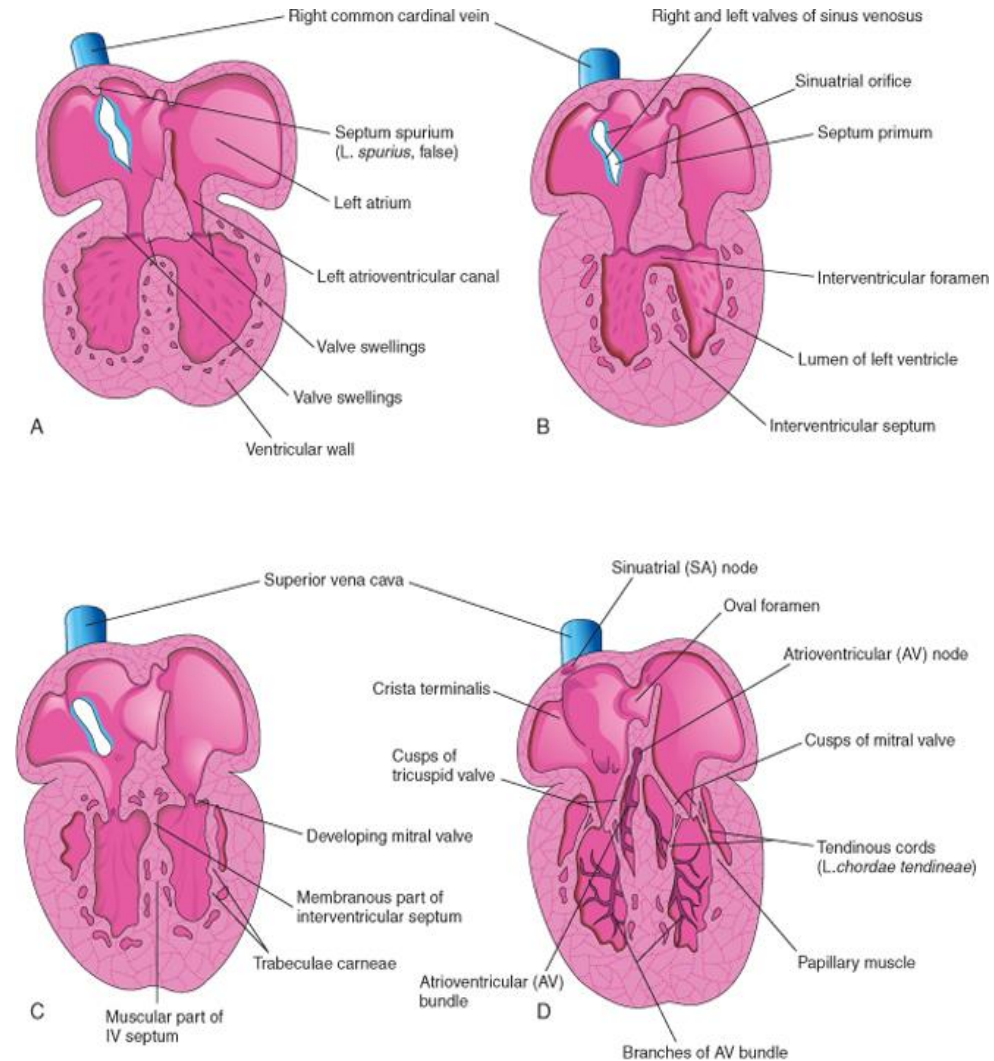
Development of Atrioventricular Valves

- The atrioventricular valves begin to form between the 5th and 8th weeks.
- These valve leaflets or cusps are firmly rooted in the rim of the right and left atrioventricular canals and are thought to arise from proliferation and differentiation of the adjacent endocardial cushion tissues.

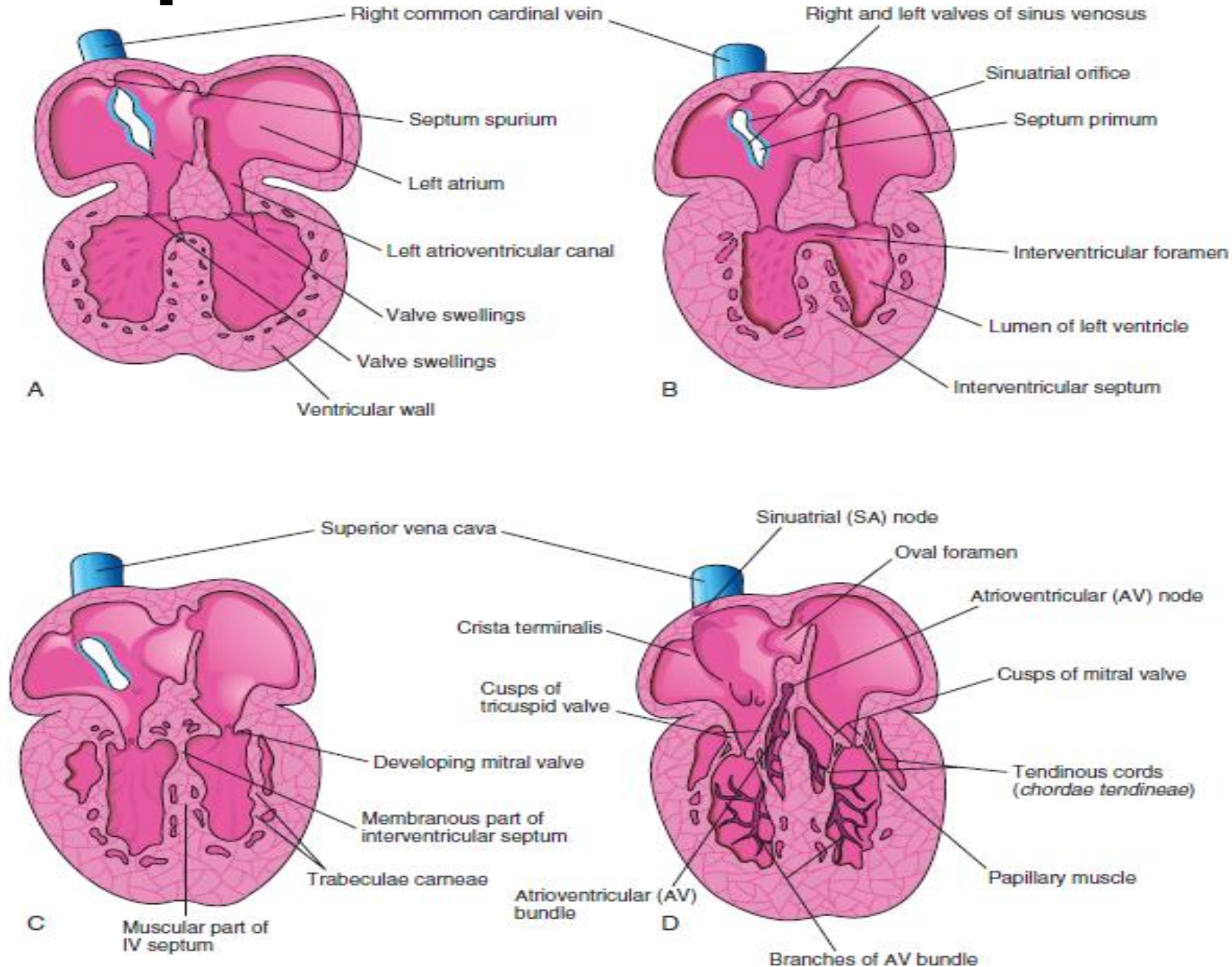
➤ A, 5th wk, showing right & left AV canals and **beginning of valve swellings** due to **proliferations of tissue** (subendocardial tissue) **around AV canals**.

➤ B, 6th week.

➤ C, 7th week complete development of **tricuspid & mitral valves**)



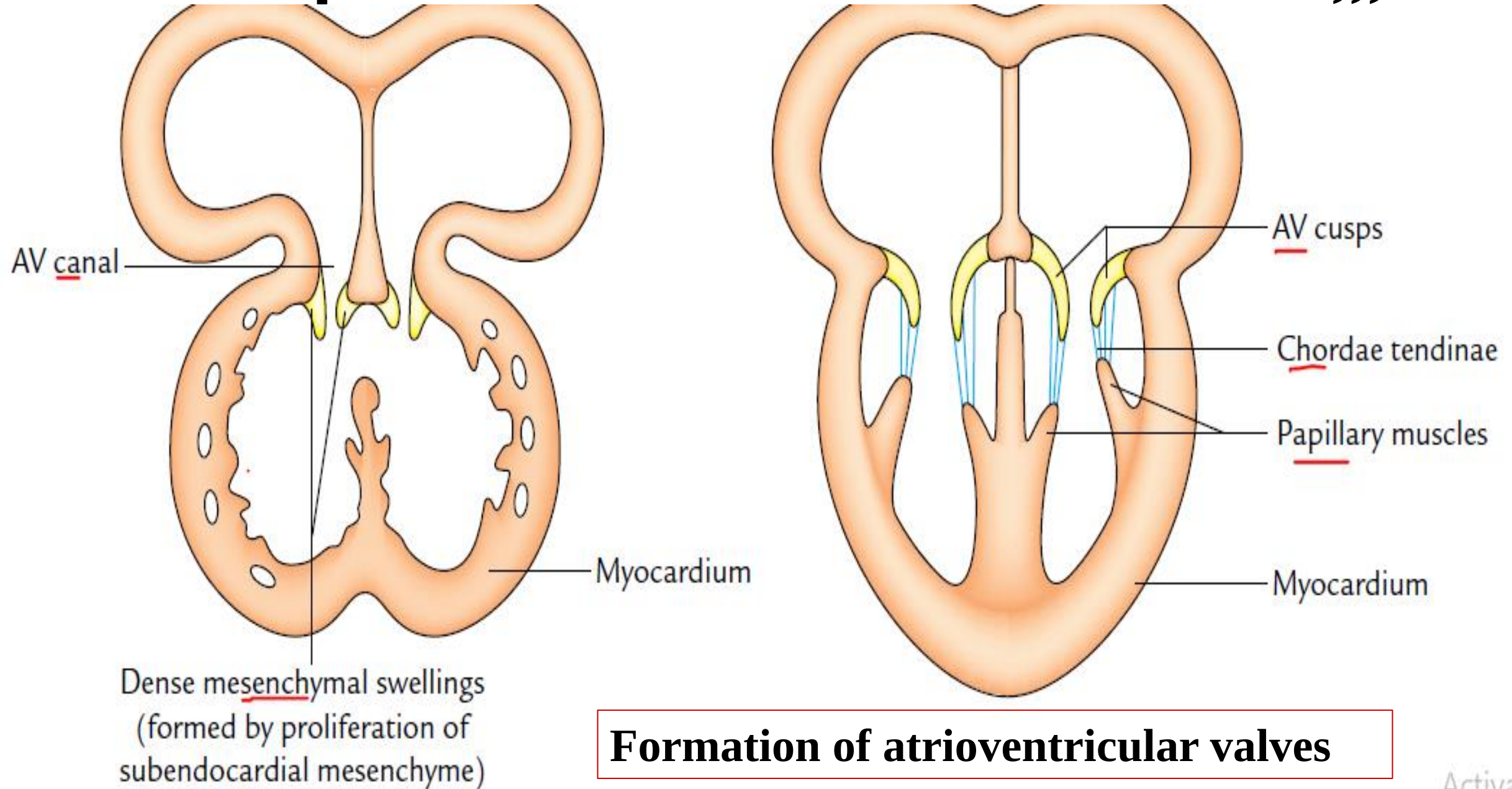
Development of Atrioventricular Valves,,,



Development of Atrioventricular Valves,,,

- **Tricuspid Valve** is present between the **right atrium and right ventricle**, usually has 3 leaflets ,therefore.
- In the AV canal, subendocardial mesenchyme proliferates and **forms three swellings (anterior, posterior, and septal) that project in the AV canal..**
- **The free margin** ventricular surfaces of these valves are connected to thick trabeculae in the wall of the ventricle (the **papillary muscles**) by thin tendinous cords (the **chordae tendinae**) much like the cords attaching to a parachute.
- **Left atrioventricular** valve has only anterior and posterior leaflets and is called the bicuspid valve (mitral valve).

Development of Atrioventricular Valves,,,

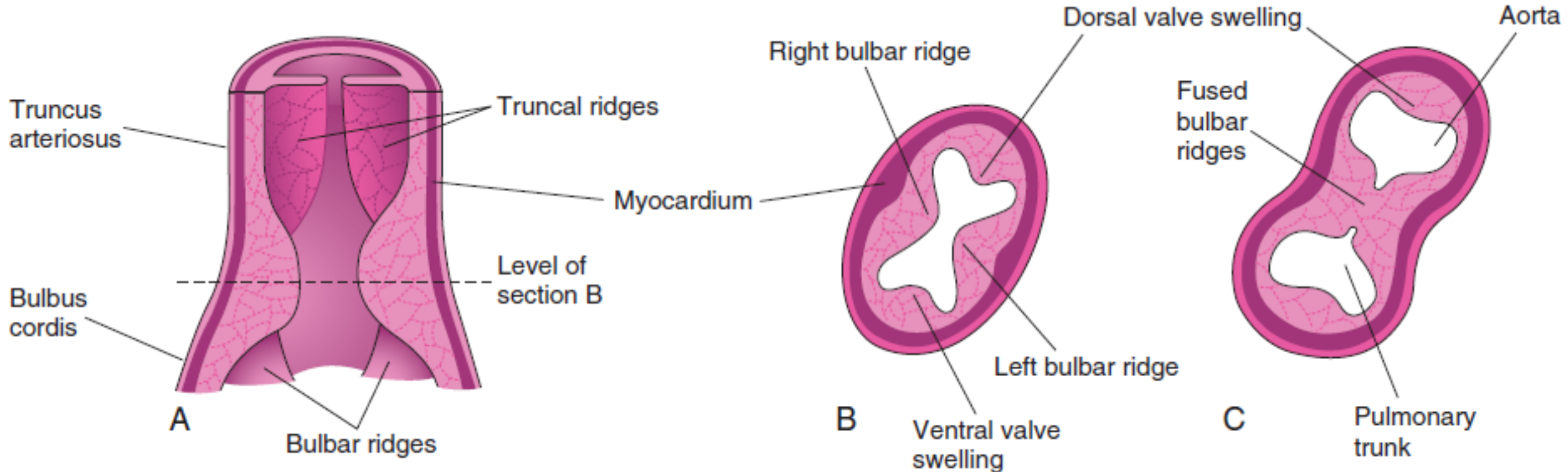


Development of Atrioventricular Valves

- **Mitral Valve** is present between the L/t atrium & L/t ventricle, & guards the left AV canal.
- There are only two cusps in left AV valve (anterior and posterior) that are obliquely placed. It is also called **bicuspid valve**.
- **N.B.** The left AV valve is called **mitral valve** due to valve's resemblance to a bishop's miter (headdress).
- The **atrioventricular (AV) valves** (tricuspid and mitral valves) develop similarly from localized proliferations of tissue around the AV canals.

Development of aortic and pulmonary valves

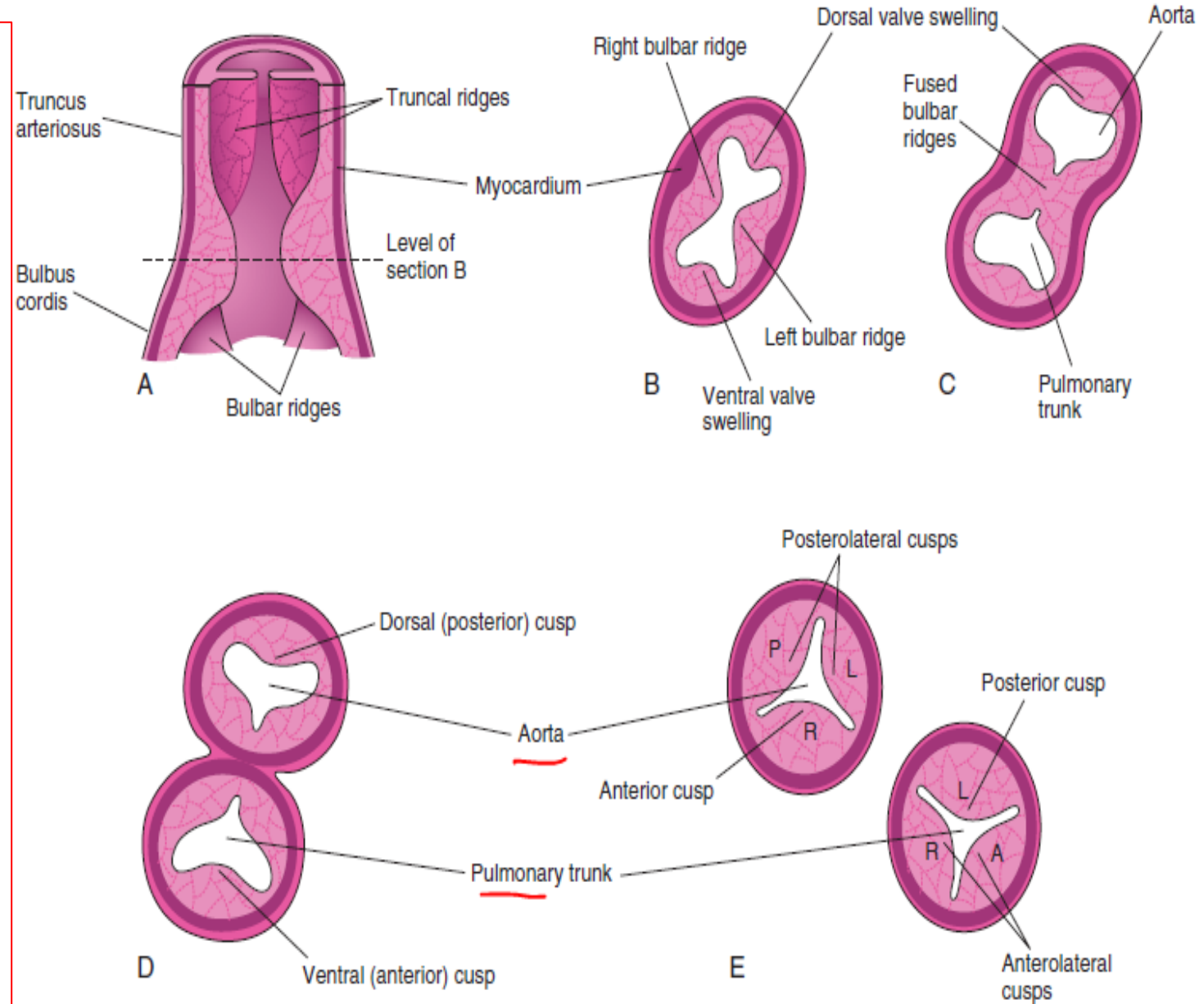
- When partitioning of the truncus arteriosus (TA) is nearly completed, the **semilunar valves** begin to develop from three swellings of subendocardial tissue around the orifices of **the aorta and pulmonary trunk**.



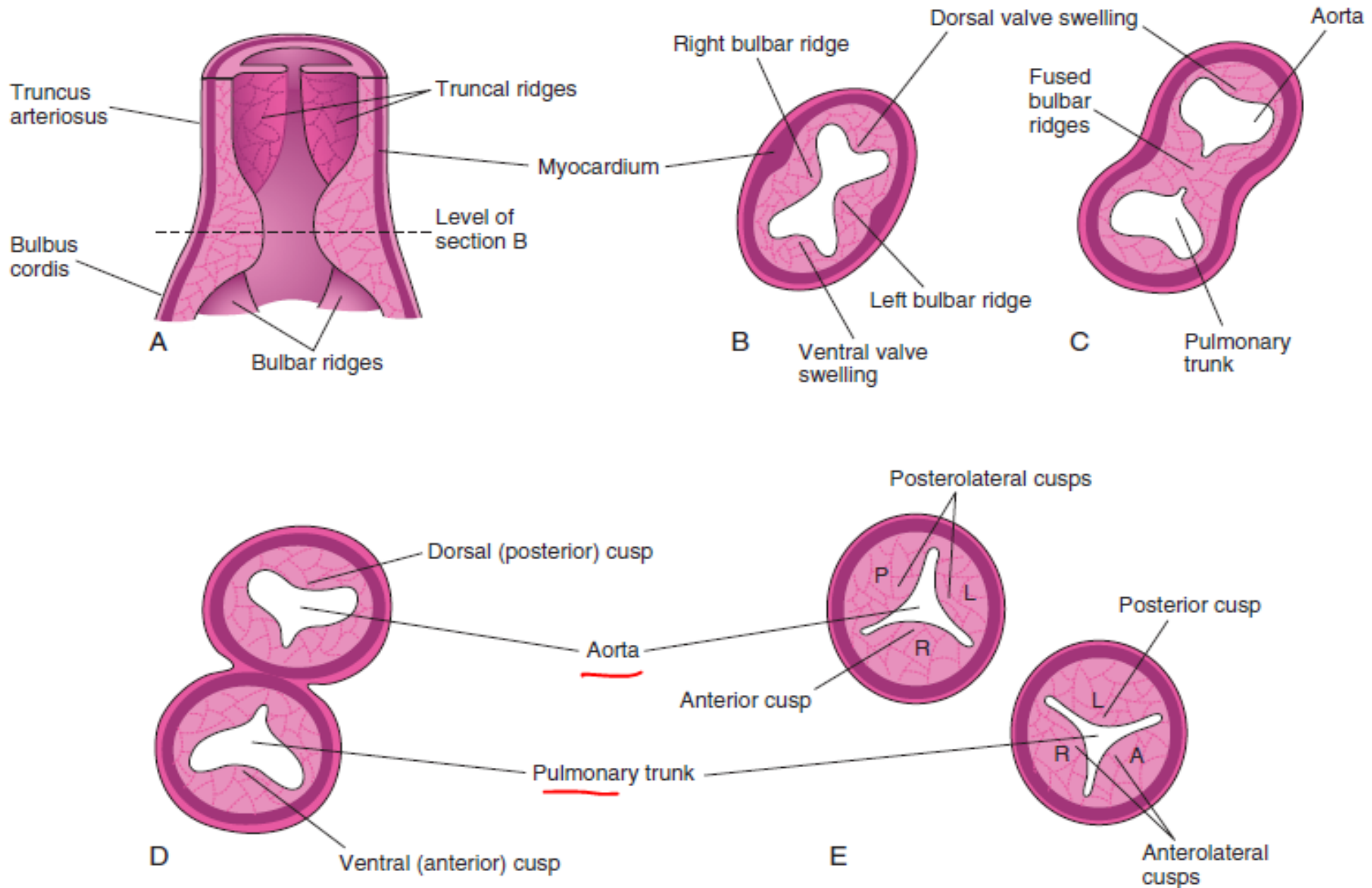
Development of aortic and pulmonary valves,,,

❖ Development of the semilunar valves of the aorta and pulmonary trunk.

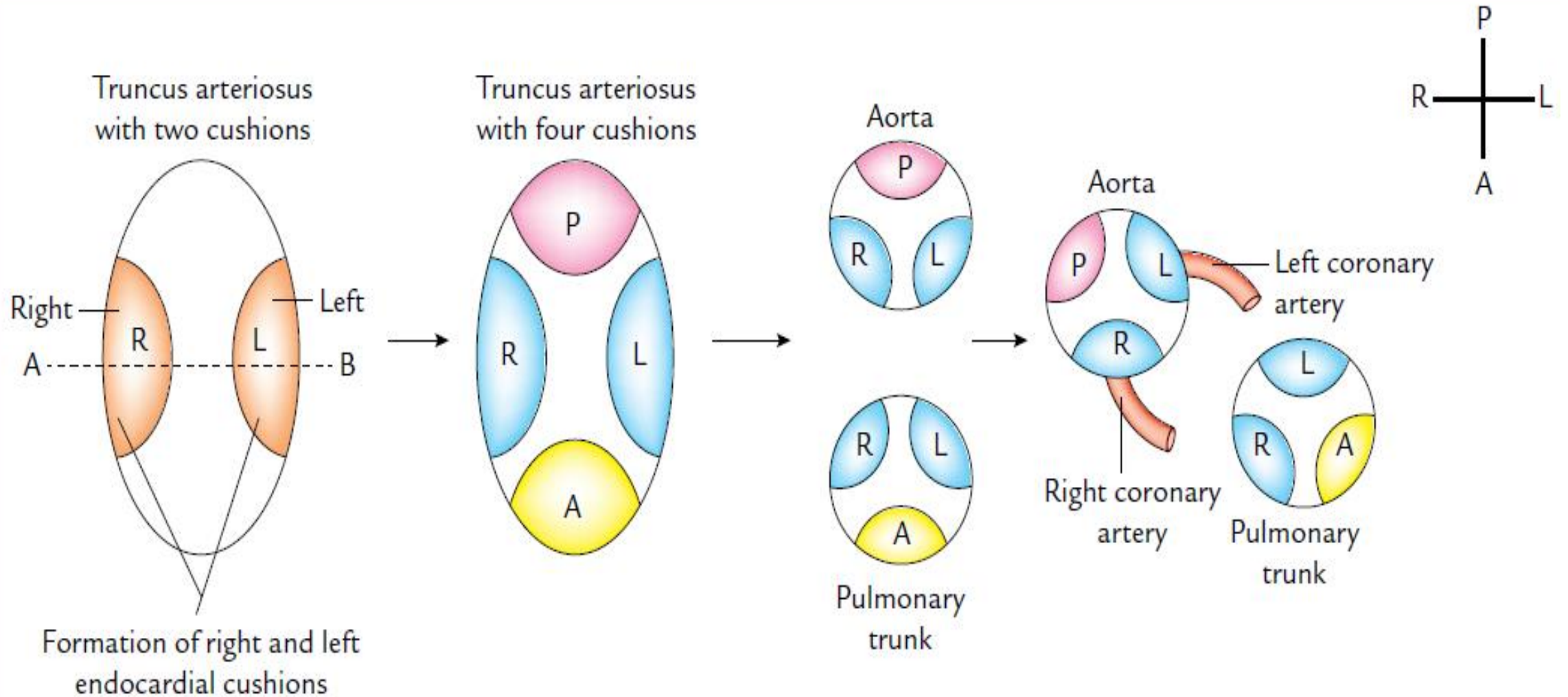
- A, Sketch of a section of the truncus arteriosus and bulbus cordis showing the valve swellings.
- B, Transverse section of the bulbus cordis.
- C, Similar section after fusion of the bulbar
- D, Formation of the walls and valves of the aorta and pulmonary trunk.
- E, Rotation of the vessels has established the adult relations of the valves.



Development of aortic and pulmonary valves,,,



Development of aortic and pulmonary valves,,,

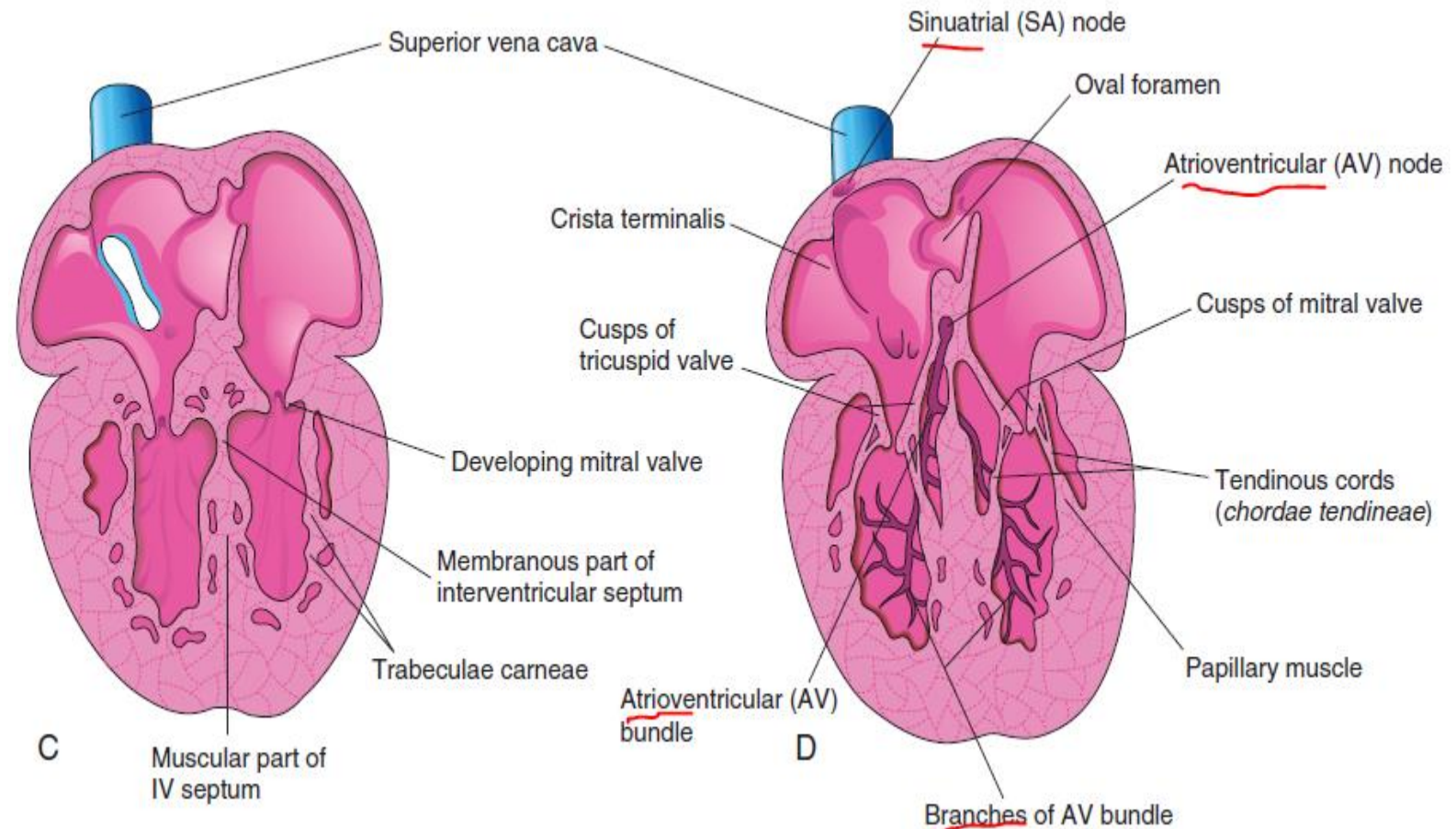


Formation of pulmonary and aortic valves.

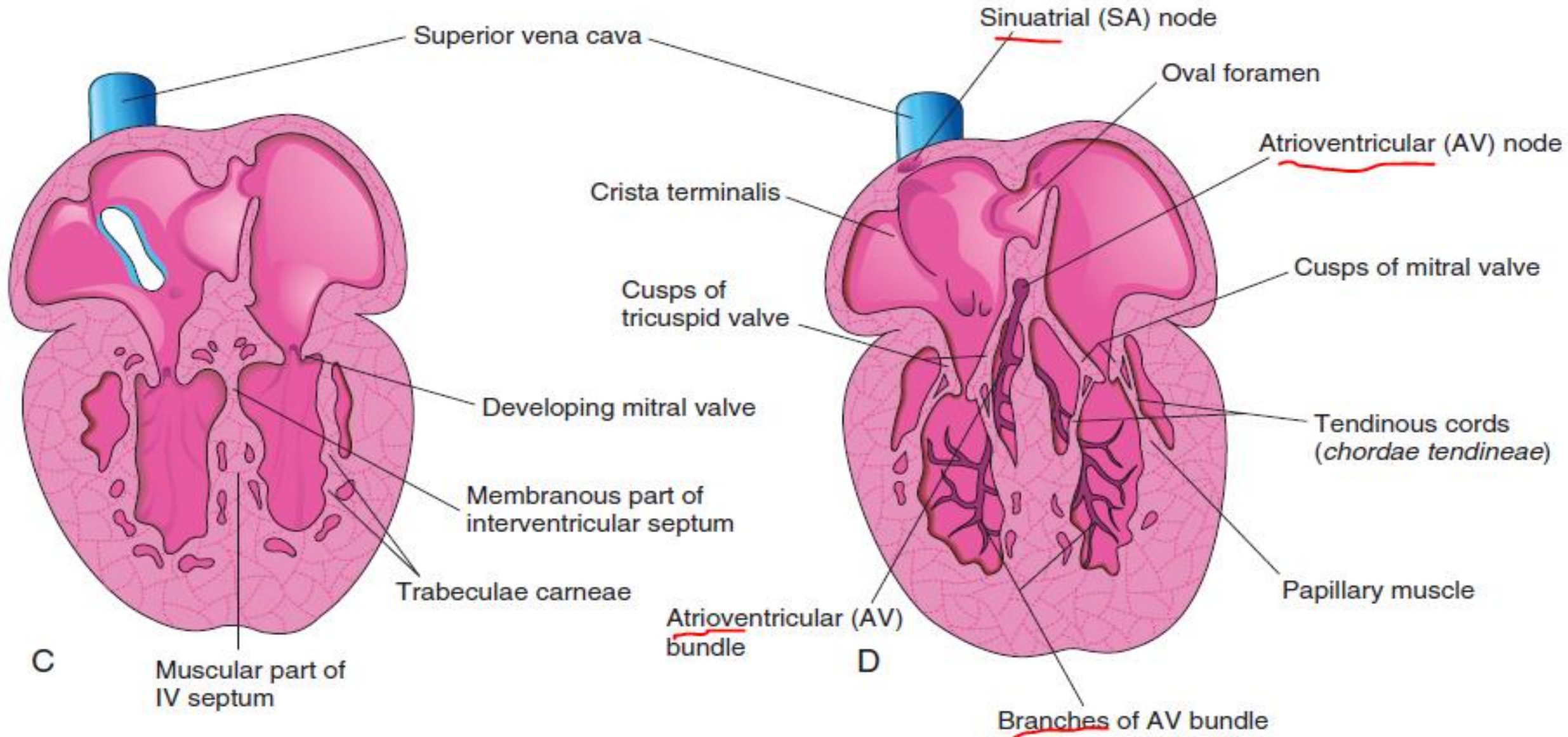
Development of conducting system

- The **sinuatrial (SA) node** develops during the fifth week.
- It is originally in the right wall of the sinus venosus, but it becomes incorporated into the wall of the right atrium with the sinus venosus.
- The SA node is located high in the right atrium, near the entrance of the superior vena cava (SVC).

- The **conducting system of the heart** consists of 4 components:
 - I. SA node (pacemaker of the heart)
 - II. AV node
 - III. Bundle of His
 - IV. Purkinje fibers.



Development of conducting system,,



Development of conducting system,,,

- **SA node:** Sinoatrial node develops during the **fifth week** of IUL. **Initially**, it is located in the right wall of the sinus venosus, but when the sinus venosus is incorporated (absorbed) into the right atrium then it comes to lie in the wall of the right atrium near the opening of the superior vena cava.
- **AV node and AV bundle of His:** They are derived from cells in the left wall of the sinus venosus and AV canal.
- After incorporation of sinus venosus into the right atrium (vide supra), these cells come to lie on the base of interatrial septum just anterior to the opening of coronary sinus. Here these cells form **AV node and AV bundle of His**.
- **Purkinje fibers:** The fibers arising from AV bundle pass from atrium into the ventricle and split into **right and left bundle branches**. The branches from these bundles are distributed throughout the ventricular myocardium and are termed *Purkinje fibers*.

Development of conducting system,,,

❑ **N.B.** The SA node, AV node, and AV bundle are richly innervated by autonomic nerve fibers; however, conducting system is well developed before these nerve fibers enter the heart.

- The conducting system is made up of specialized cardiomyocytes and normally it is the only pathway that initiates impulses & conducts them rapidly throughout the heart.
- Thus, contraction of cardiac muscle is **myogenic** and not **neurogenic**.
- As the 4 chambers of the heart develop, a band of connective tissue grows in from the epicardium, subsequently separating the muscle of the atria from that of the ventricles.
- This connective tissue forms part of the **cardiac skeleton**.

Development of Blood Vessels

- Starting on day 17, vessels begin to arise in **the splanchnic mesoderm** of the yolk sac wall from aggregations of cells called **hemangioblastic aggregates**.
- The blood vessels develop from mesenchymal cells derived from **mesoderm**.

□ **They are formed by 2 processes:** *vasculogenesis* and *angiogenesis*.

I. Vasculogenesis: It is the process of formation of new vessels by **coalescence** (assembly) of angioblasts (**specialized mesenchymal cells**). On day 18, **vasculogenesis** commences in the **splanchnic mesoderm** of the embryonic disc and later in the **paraxial mesoderm**.

II. Angiogenesis: It is the process of formation of new vessels by budding and branching from preexisting vessels.

Development of Blood Vessels,,

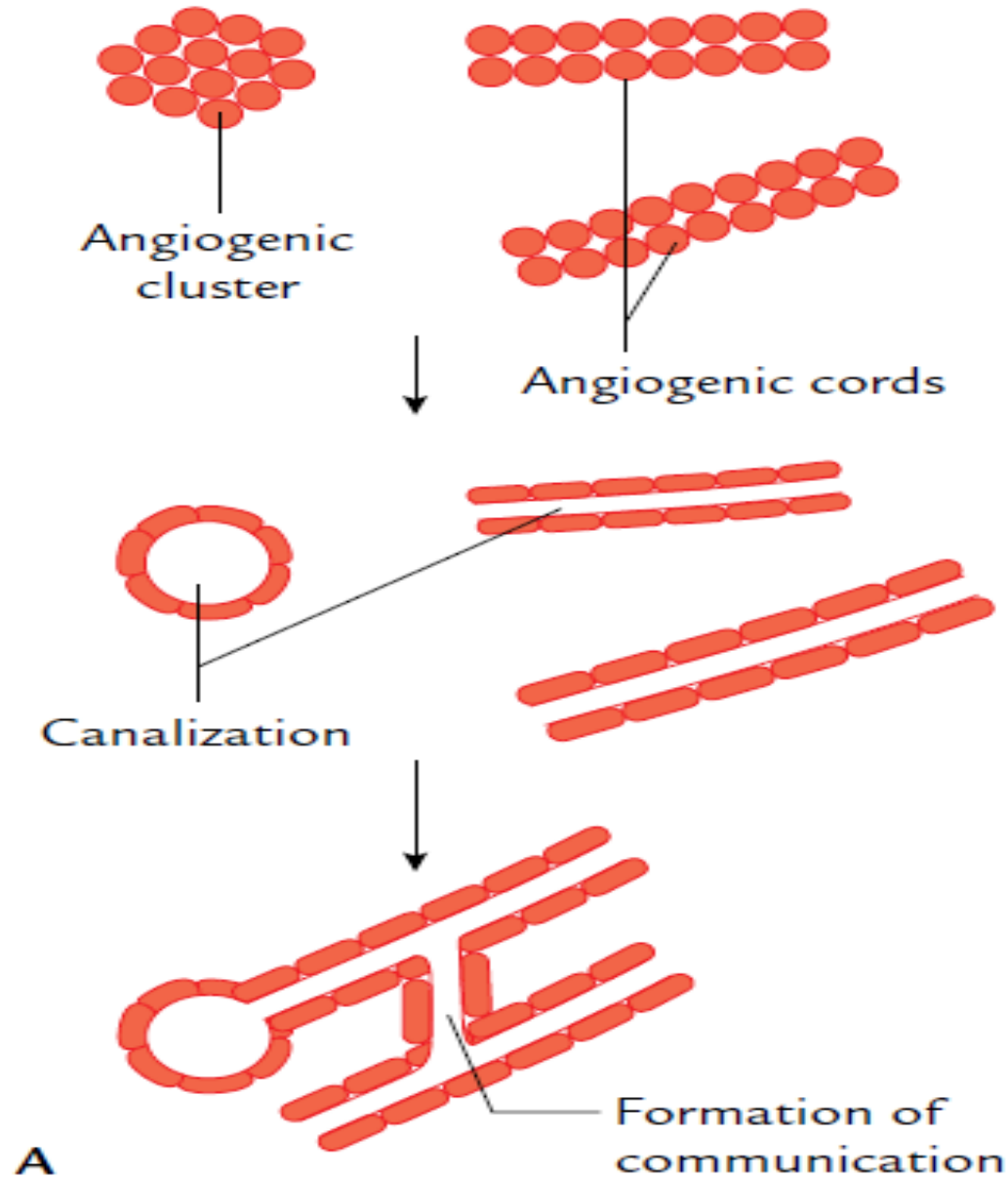
- **N.B.** Major blood vessels of the vascular system of the body, including, dorsal aortae and cardinal vessels are formed by **vasculogenesis**, and **remaining** vessels are formed by **angiogenesis**.
- Both the processes of vasculogenesis and angiogenesis are patterned, and guided by vascular endothelial growth factor and other growth factors.

Development of Blood Vessels,,,

❖ *Vasculogenesis*;

- ✓ The mesenchymal cells differentiate into angioblasts (vessel-forming cells) that aggregate to form **isolated angiogenic cell clusters or cords**.
- ✓ Lumen appears in these clusters and cords, and primitive blood vessels are formed.
- ✓ The angioblasts flatten to form endothelial cells that arrange themselves around cavities in clusters and cords to form the endothelial lining of the blood vessels.
- ✓ The primitive blood vessels get interconnected with each other to form a network of endothelial vascular channels—the blood vessels.
- ✓ **Because of hemodynamic and genetic factors certain channels enlarge to form blood vessels and channels connected to them form branches or tributaries, while other channels disappear.**

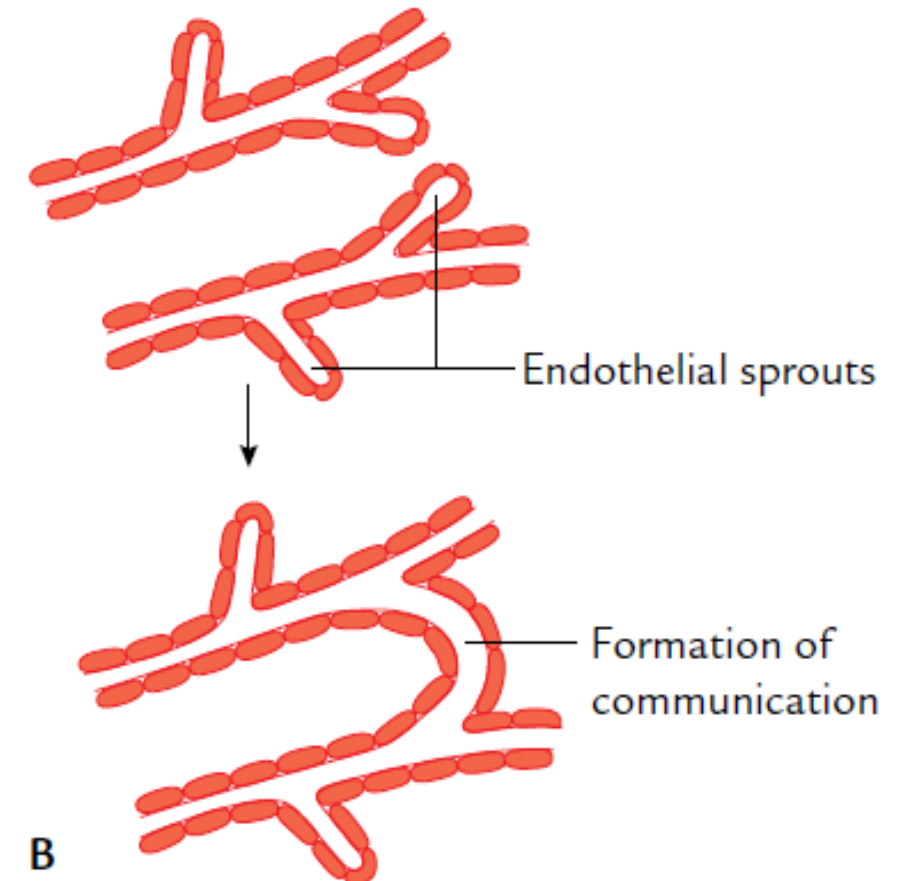
Development of Blood Vessels,,,



Development of Blood Vessels,,,

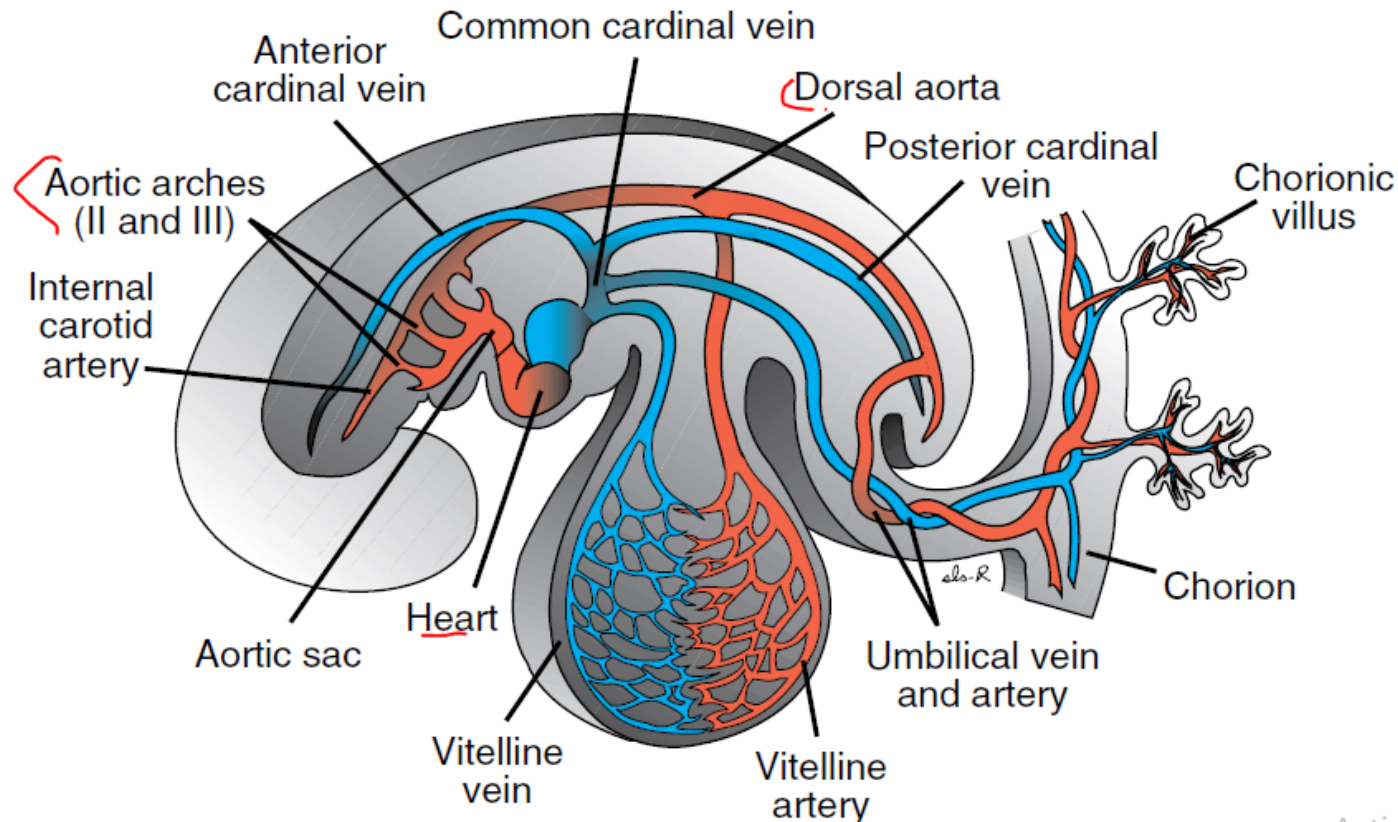
❖ *Angiogenesis;*

- ✓ Vessels sprout into the adjacent areas by endothelial budding & fuse with other vessels.
- ✓ Finally, in this way, permanent vascular system is formed.
- ✓ **N.B.** The primitive blood vessels cannot be distinguished structurally as arteries or veins. However, they are named according to their future fate and relationship to the heart.

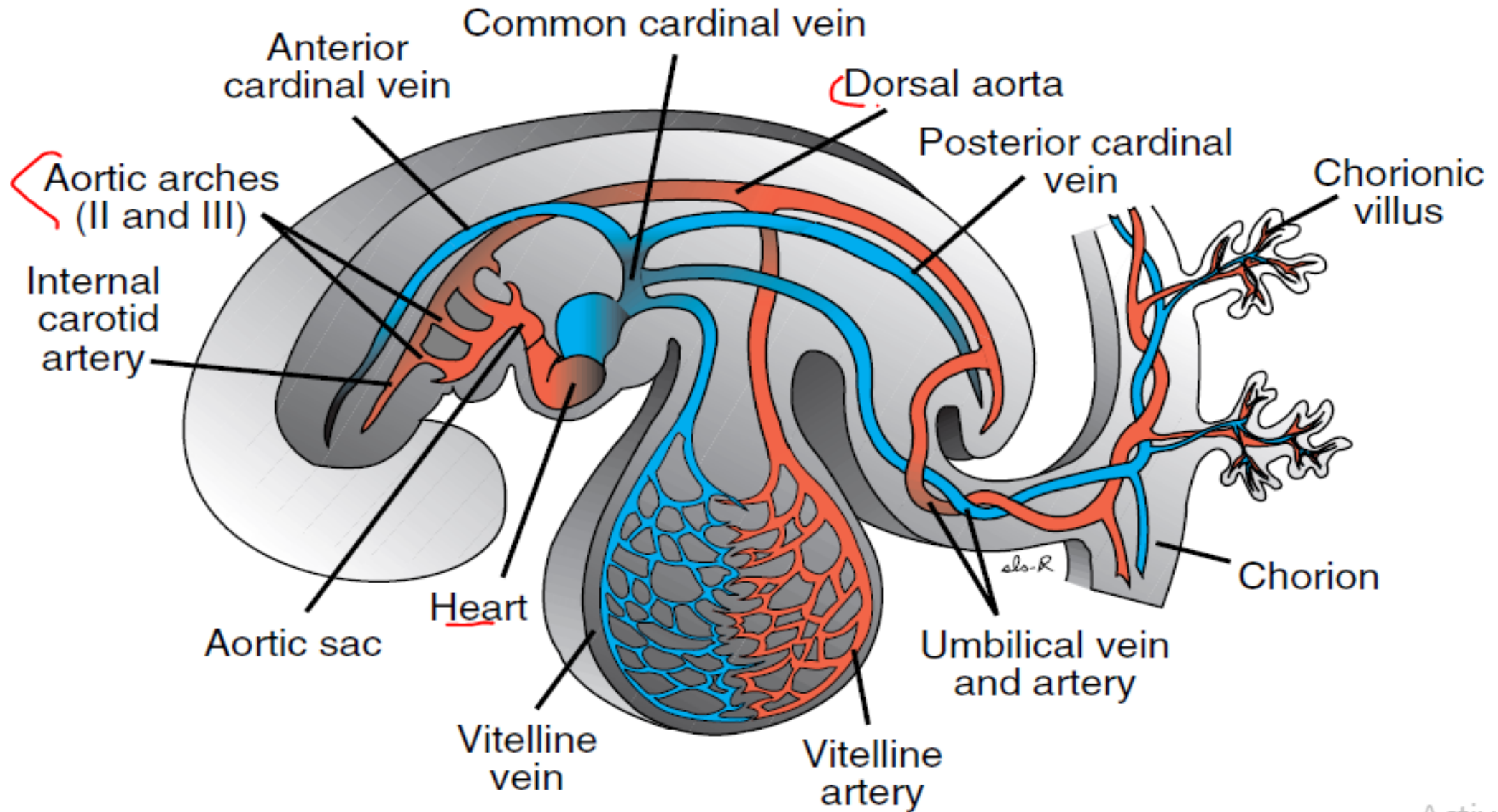


Development of Arterial System

- The arteries of body develop from two main sources:
- **pharyngeal (aortic) arch arteries** and **dorsal aorta**.
- ❖ **The aortic arches** give rise to the arteries of head and neck region.
- ❖ **The dorsal aorta** gives rise to the arteries of rest of the body.



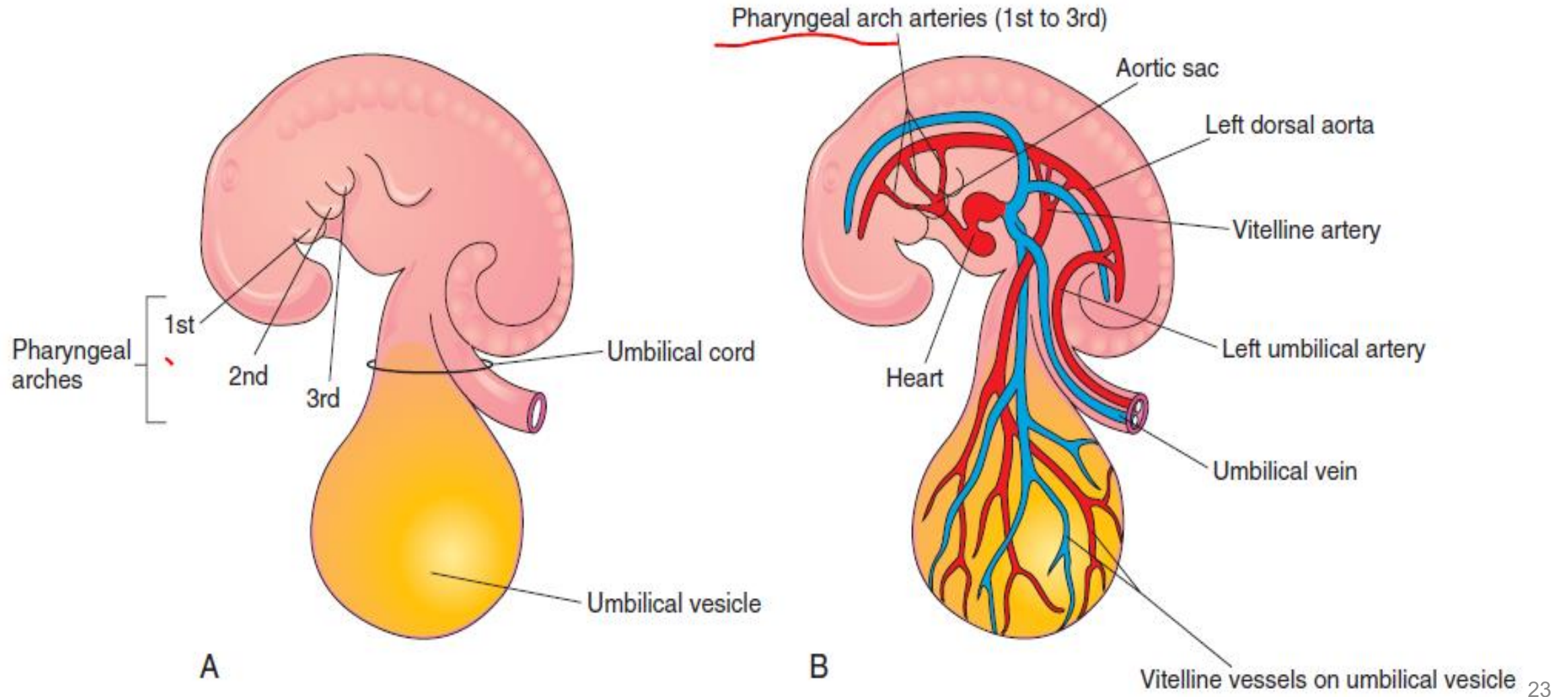
Development of Arterial System,,,



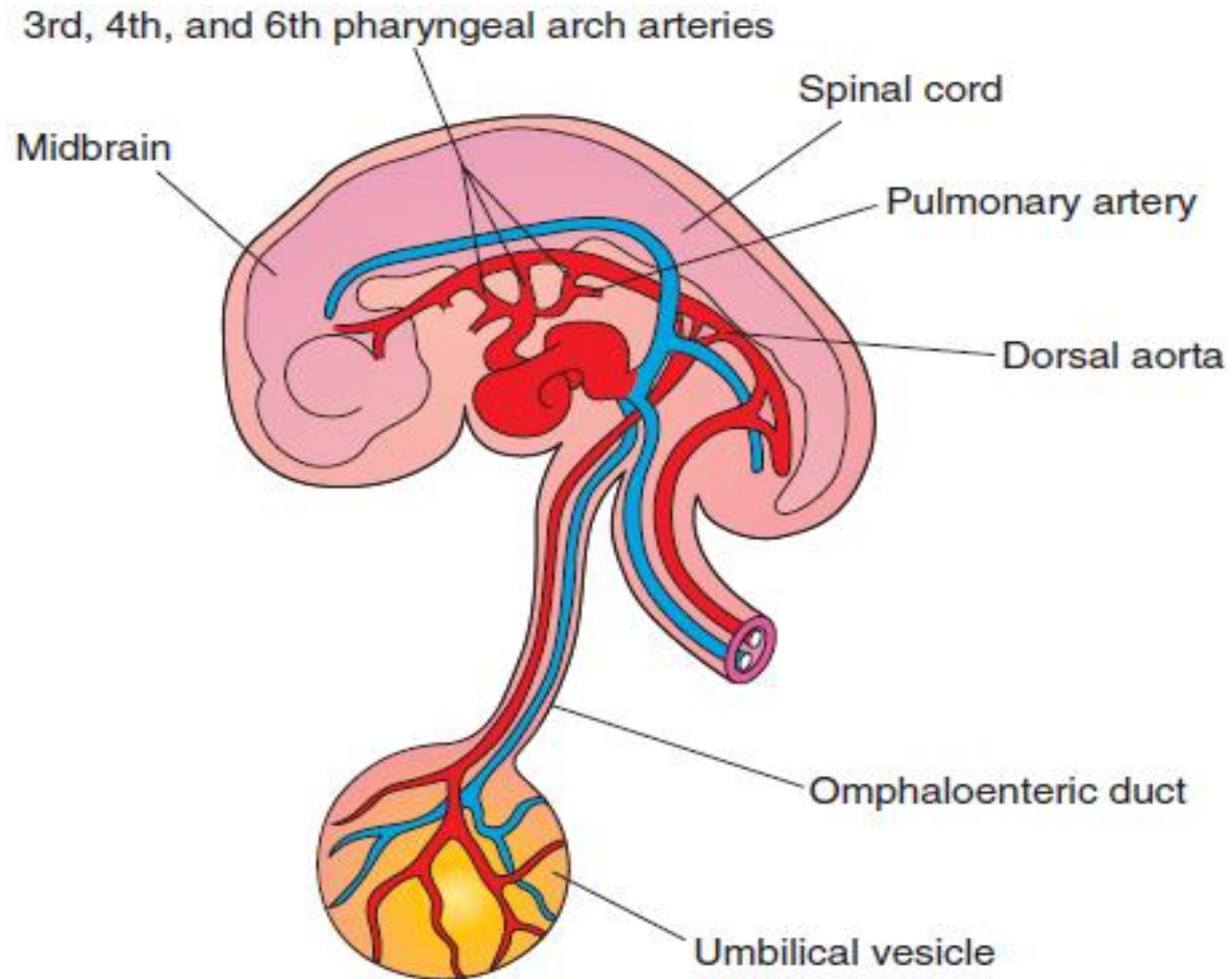
Pharyngeal Arch Arteries

- As the pharyngeal arches develop **during fourth and fifth weeks**, each arch is supplied by an artery—the **pharyngeal arch artery** derived from the aortic sac (i.e., aortic sac contributes a branch to each new arch as it forms).
- **Six pairs of pharyngeal arch arteries** arise from aortic sac (**4th wk**) to supply pharyngeal arches.
- These arteries terminate in the dorsal aorta of the ipsilateral side
- By the time the 6th pair of pharyngeal arch arteries has formed, the **first two pairs** have disappeared.
- During the 8th week, primordial pharyngeal arch arterial pattern is transformed into the final fetal arterial arrangement.
- ❖ **NOTE:** pharyngeal arches and their vessels appear in a **cranial-to-caudal sequence**, so that they are not all present simultaneously.

Pharyngeal Arch Arteries,,,



Pharyngeal Arch Arteries,,



Derivatives of pharyngeal arch arteries

- ❑ **1st pharyngeal arteries:** Most of these pharyngeal arch arteries disappear, but remnants of them form part of the **maxillary arteries**, which supply the **ears, teeth, and muscles of the eye and face.**
- ✓ These arteries may also contribute to the formation of the **external carotid arteries.**
- ❑ **2nd pharyngeal arteries;** Dorsal parts of these arteries persist and form the stems of **stapedial arteries** in the middle ear.
- ❑ **3rd pharyngeal arteries, Proximal parts** form the **common carotid arteries**
- ✓ Distal parts join with dorsal aortae to form **internal carotid arteries** which supply the middle ears, orbits, brain and its meninges, and the pituitary gland.

Derivatives of pharyngeal arch arteries,,,

❑ **Lt 4th pharyngeal arteries:** forms **part of the arch of the aorta**.

- The proximal part of the artery develops from the **aortic sac** and the distal part is derived from the **left dorsal aorta**.

❑ **Rt 4th pharyngeal arteries;** Form **proximal part** of Rt subclavian artery

❖ **Note;**

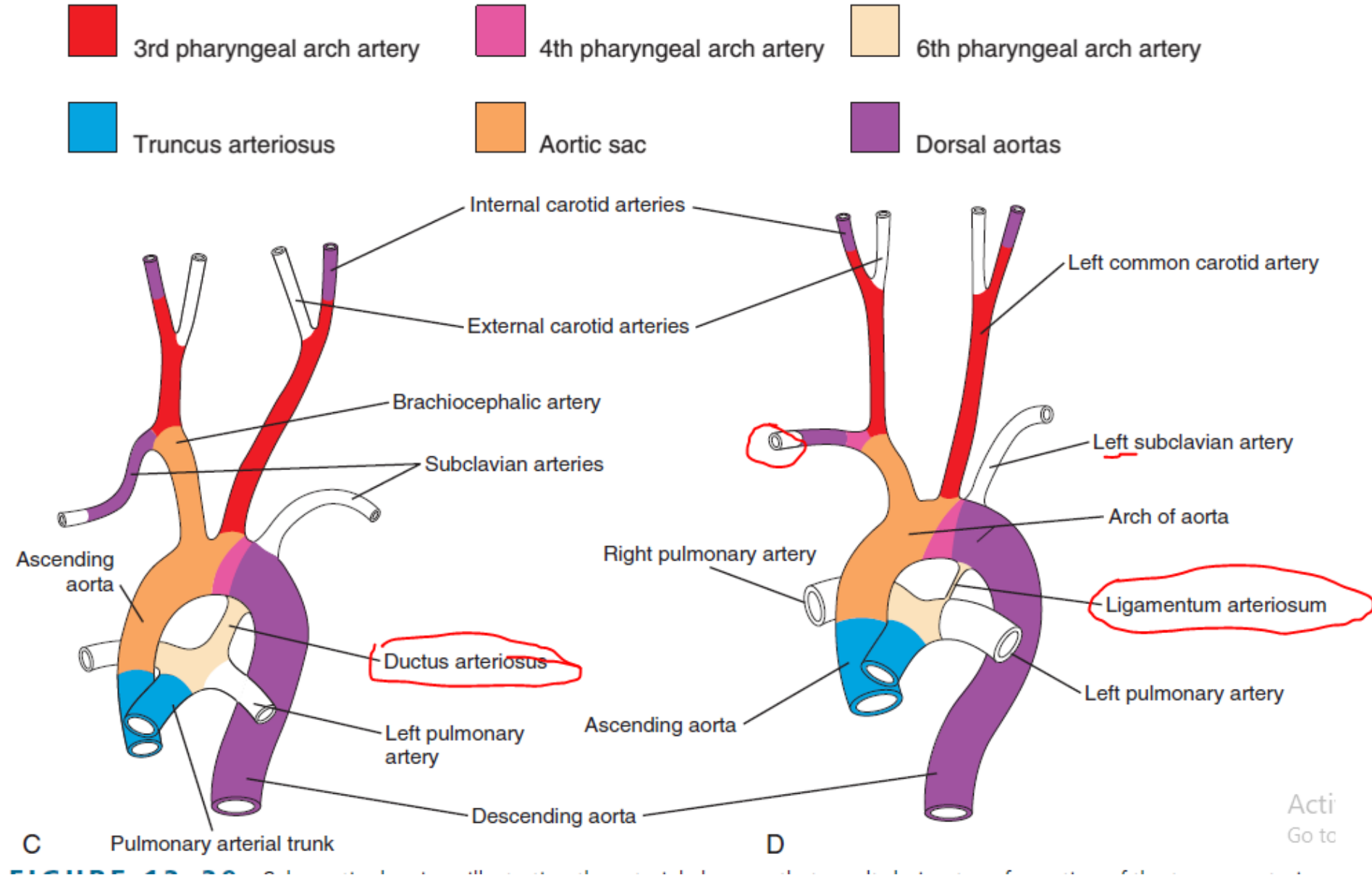
❖ **Distal part** of Rt subclavian artery is formed from Rt dorsal aorta & right 7th intersegmental artery.

❖ **Lt subclavian artery** is derived from the left 7th intersegmental artery

- As development proceeds, differential growth shifts the origin of left subclavian artery cranially.

➤ Consequently, it lies close to the origin of the left common carotid artery

Derivatives of pharyngeal arch arteries,,,



Derivatives of pharyngeal arch arteries,,,

□ 5th pharyngeal arteries

- Approximately 50% of the time, the 5th pair of pharyngeal arch arteries consists of rudimentary vessels that soon degenerate without vascular derivatives.
- In the other 50%, these arteries do not develop at all

□ Left 6th pharyngeal arteries

- The **proximal** part persists as the **proximal part** of the **left pulmonary artery**.
- The **distal part** passes from the left pulmonary artery to the dorsal aorta and forms a prenatal shunt called the **ductus arteriosus(DA)**

Derivatives of pharyngeal arch arteries,,,

❖ Right 6th pharyngeal arteries

- The **proximal part** persists as the proximal part of the right pulmonary artery.
- The **distal part** degenerates

❖ Note

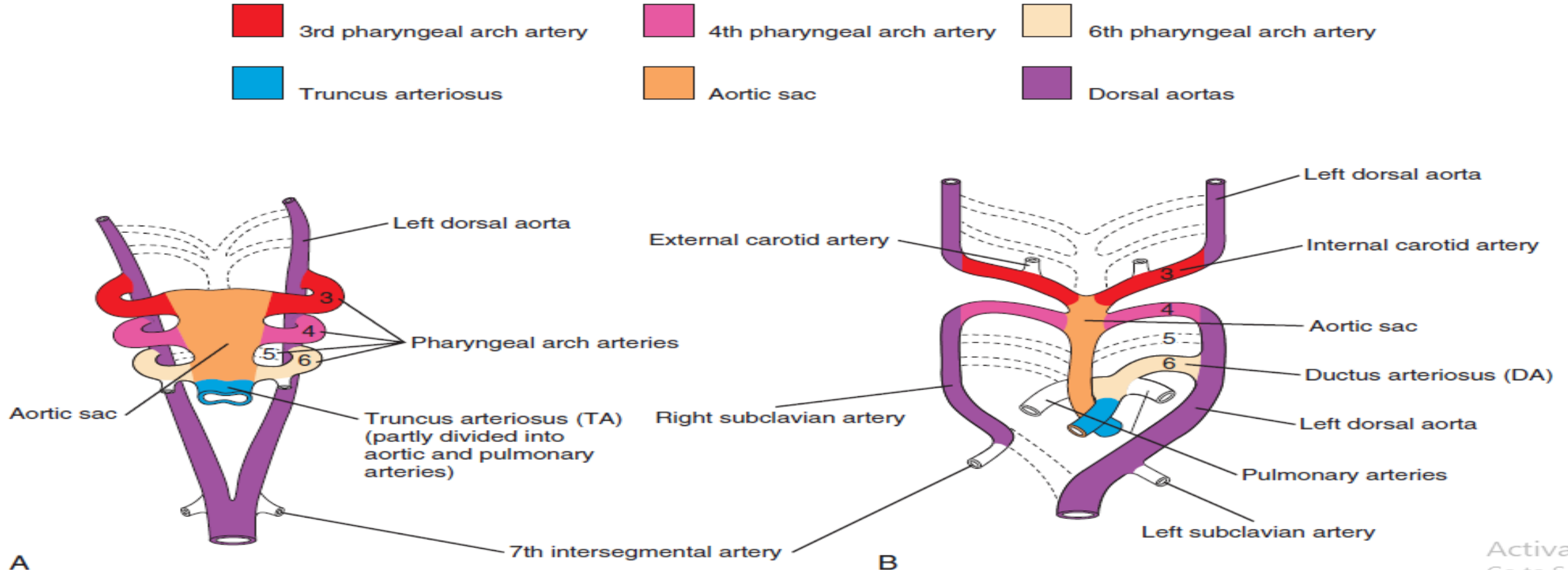
- Transformation of the 6th pair of pharyngeal arch arteries explains why the course of **recurrent laryngeal nerves** differs on the two sides.
- These nerves supply the 6th pair of pharyngeal arches & hook around the 6th pair of pharyngeal arch arteries on their way to the **developing larynx**.

Derivatives of pharyngeal arch arteries,,,

❑ Schematic drawings illustrating the arterial changes that result during transformation of the truncus arteriosus, aortic sac, pharyngeal arch arteries, and dorsal aortae into the adult arterial pattern. The vessels that are not colored are not derived from these structures.

A, Pharyngeal arch arteries at 6 weeks; by this stage, the first two pairs of arteries have largely disappeared.

B, Pharyngeal arch arteries at 7 weeks; the parts of the dorsal aortae and pharyngeal arch arteries that normally disappear are indicated with *broken lines*.



Derivatives of pharyngeal arch arteries,,,



3rd pharyngeal arch artery



4th pharyngeal arch artery



6th pharyngeal arch artery



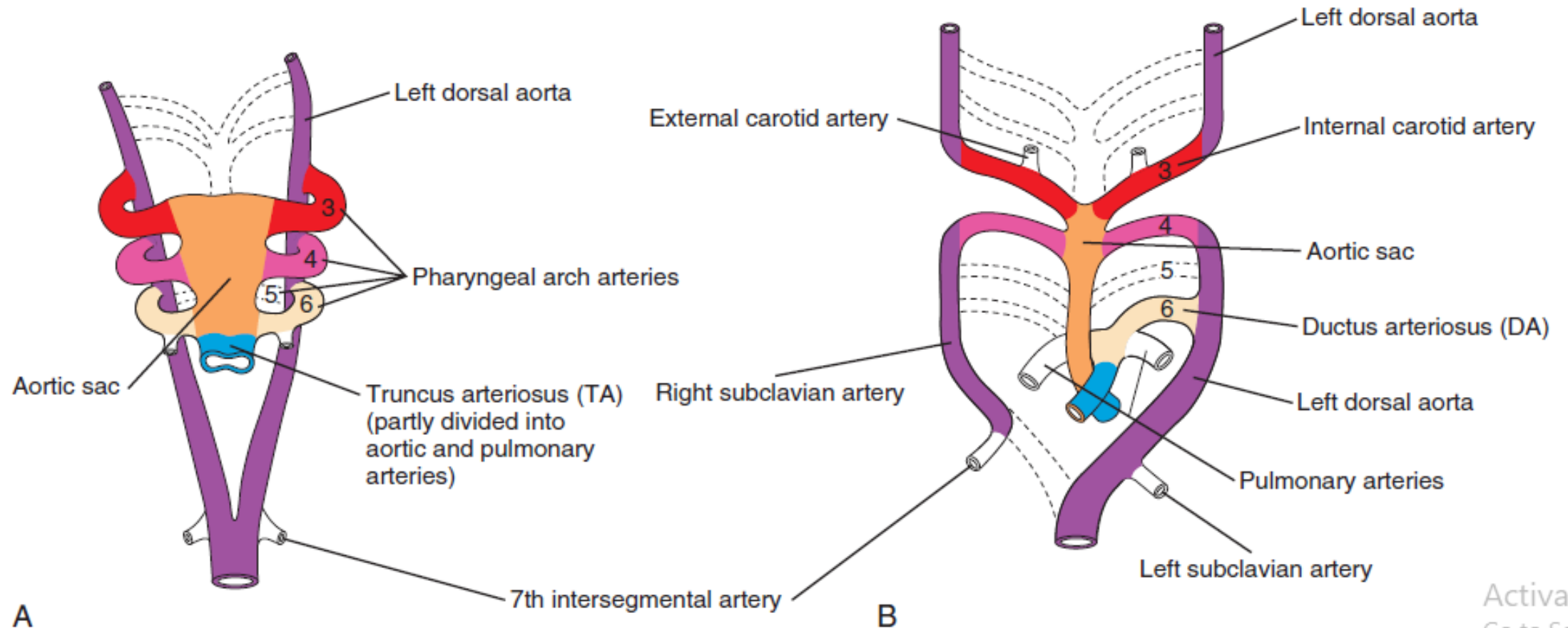
Truncus arteriosus



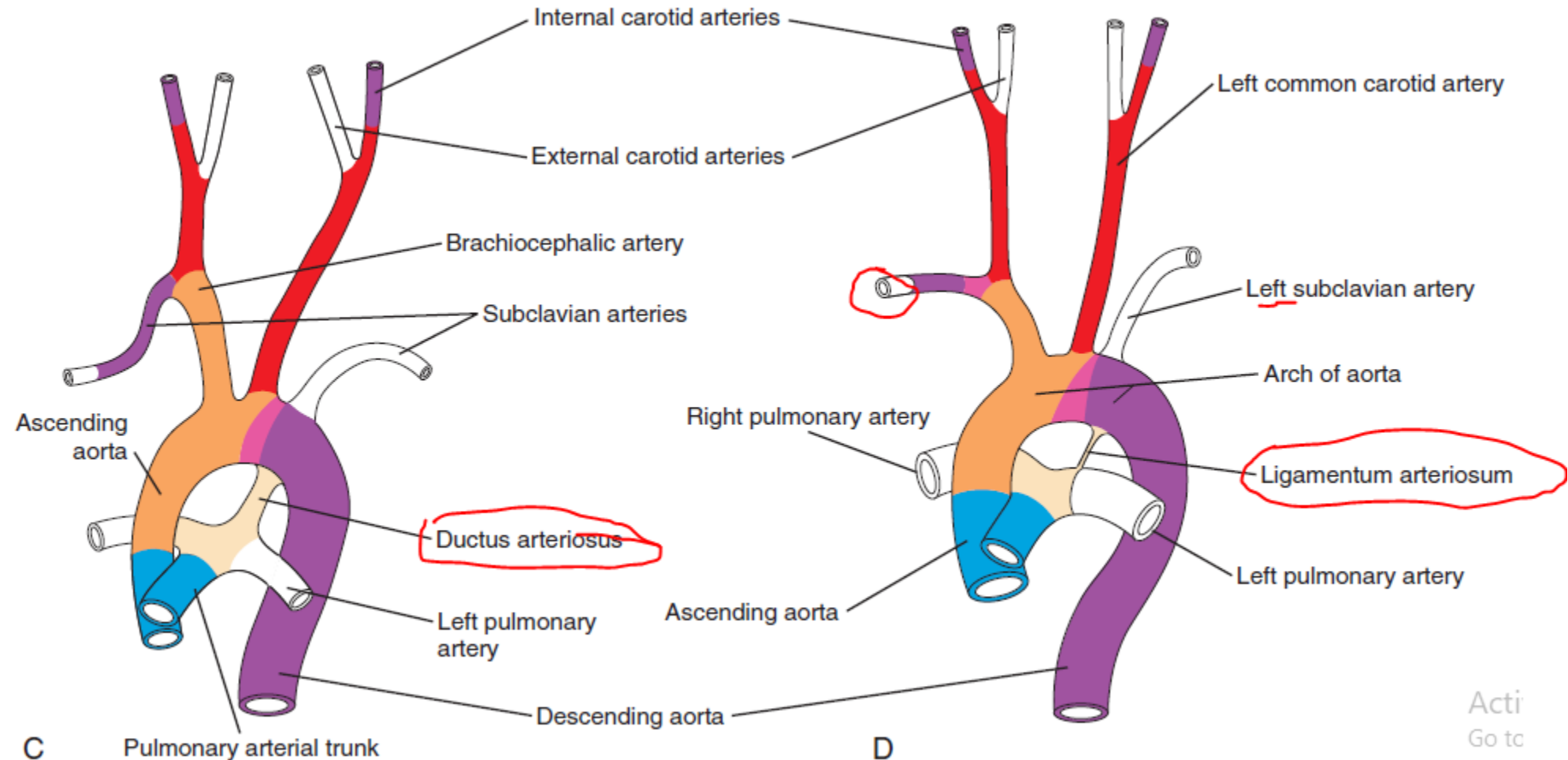
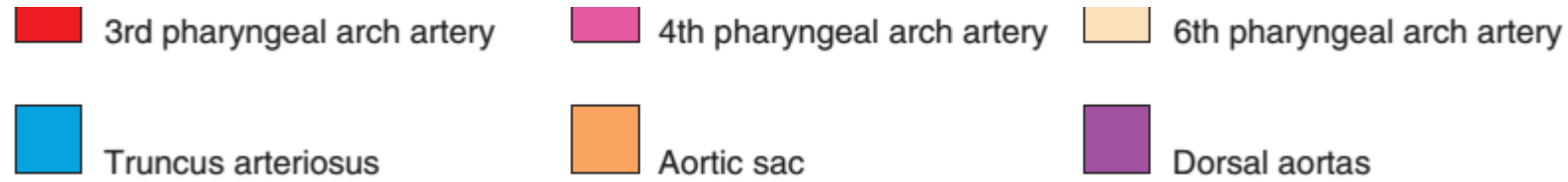
Aortic sac



Dorsal aorta



Derivatives of pharyngeal arch arteries,,,



Development of Other Arteries of the Body

- They develop from primitive dorsal aortae.

❑ **The primitive dorsal aorta gives off three sets of branches:** dorsolateral (somatic intersegmental), lateral splanchnic, and ventral splanchnic.

- I. The **dorsolateral (somatic intersegmental) branches** form arteries of upper and lower limbs, intercostal lumbar, and lateral sacral arteries.
- II. The **lateral splanchnic branches** form phrenic, suprarenal, renal, & gonadal vessels.

III. 3. The **ventral splanchnic branches** form *vitelline (celiac, superior, and inferior mesenteric)* and *umbilical* arteries. ***The arteries for bronchi and esophagus also arise from ventral splanchnic branches.***

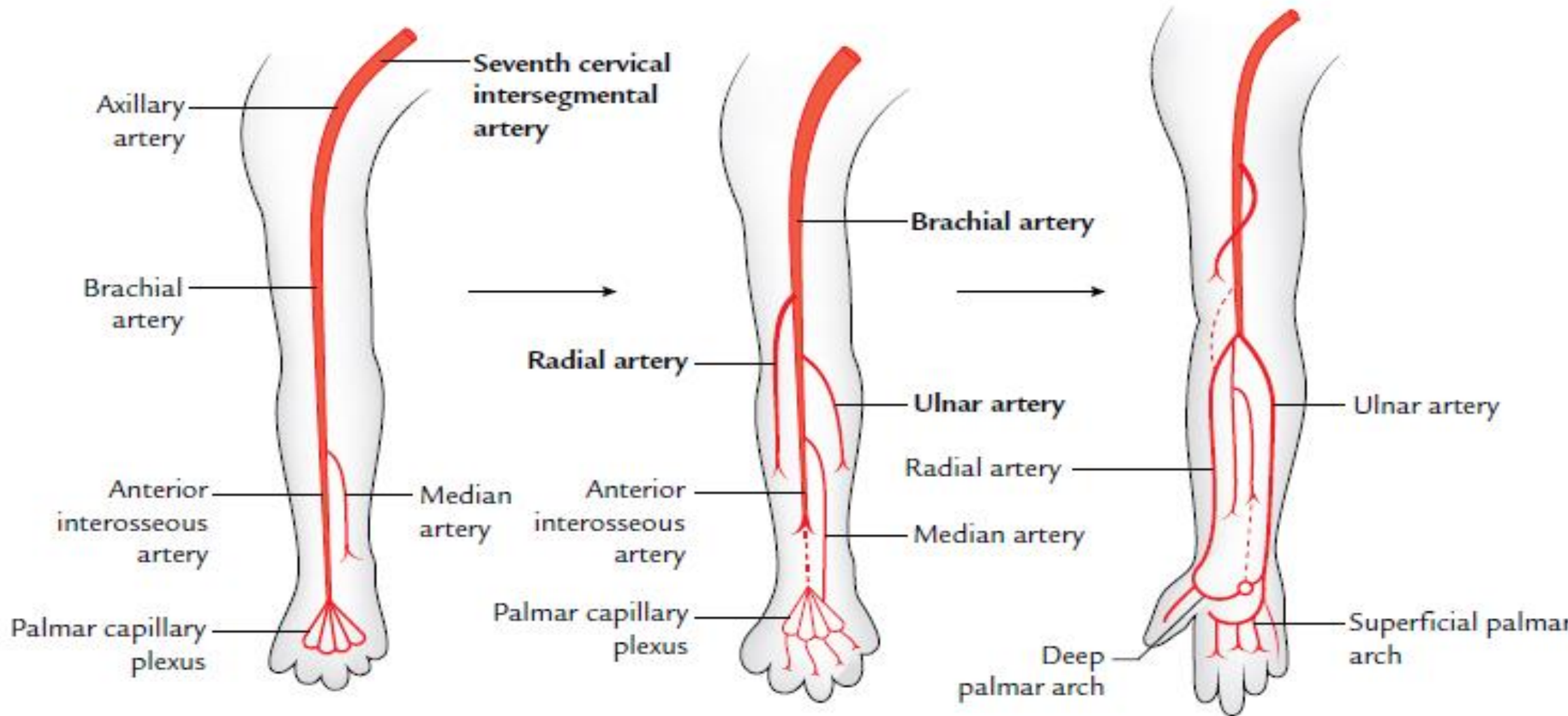
Development of the Arteries of the Limbs

- Each limb is supplied by an **axis artery** that is derived from intersegmental arteries. The axis artery runs along the central axis of the limb.

□ Axis Artery of the Upper Limb:

- The axis artery of the upper limb is derived from **seventh cervical intersegmental (subclavian) artery**.
- This artery grows distally along the ventral axial line and terminates in a palmar capillary plexus in hand.
- Main trunk of axis artery forms *axillary artery*, *brachial artery*, *anterior interosseous artery*, and *deep palmar arch*.

Development of the Arteries of the Limbs,,



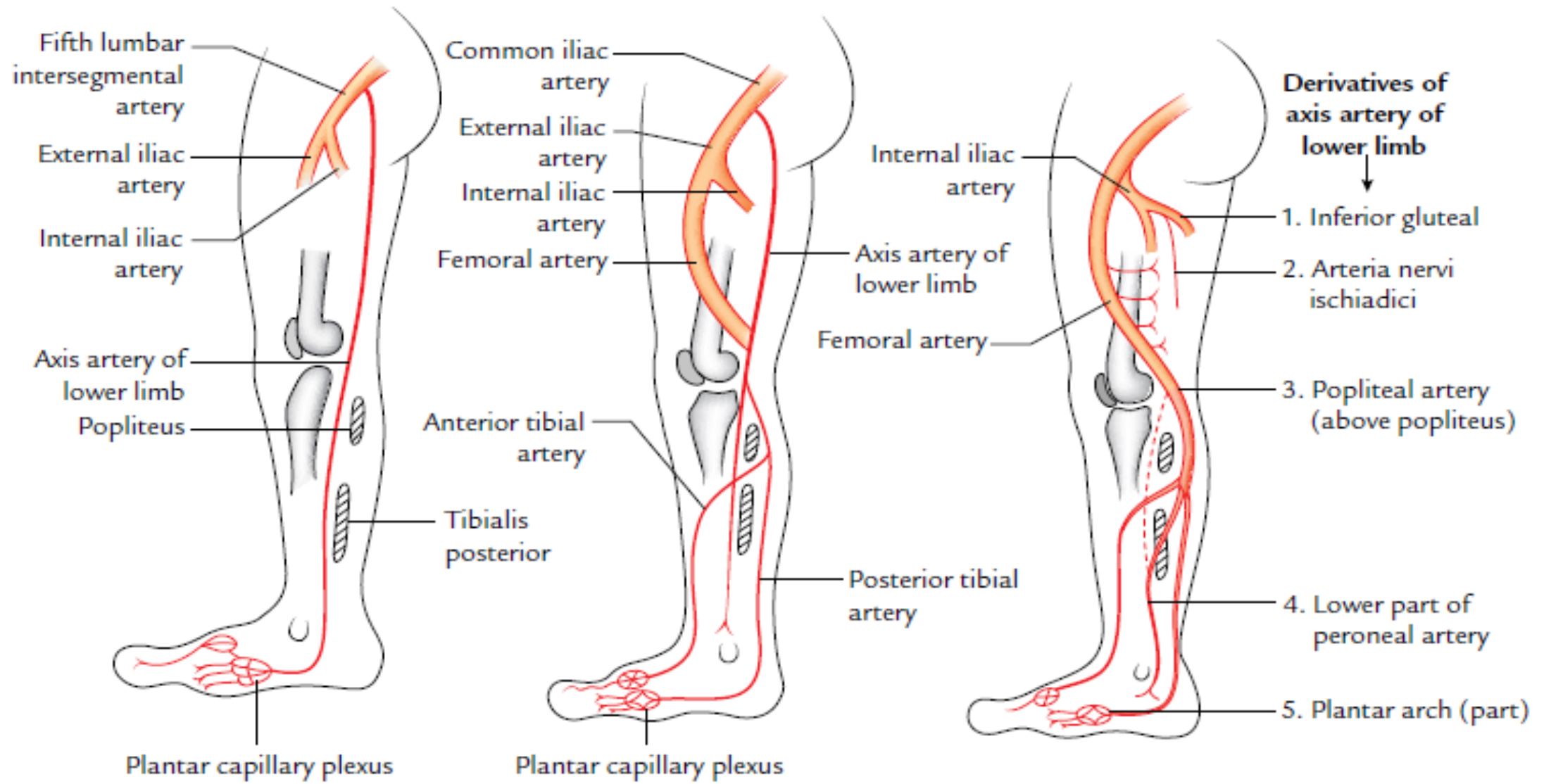
➤ *Stages in the development of the arteries of the upper limb*

Development of the Arteries of the Limbs,,

❑ Axis Artery of the Lower Limb;

- The axis artery of the lower limb is derived from **fifth lumbar intersegmental artery**.
- The axis artery runs on the posterior aspect of the lower limb. It forms **inferior gluteal artery**—a small artery accompanying the sciatic nerve (called *ischiodic artery/arteria comitans nervi ischiadici*)—part of **popliteal artery** above the popliteus muscle, **distal part of peroneal artery**, and **part of plantar arch**.
- The **femoral artery** appears as an entirely new vessel formed on the ventral aspect of thigh. It develops a connection with the external iliac artery above and popliteal artery below.

Development of the Arteries of the Limbs,,



➤ *Stages in the development of arteries of the lower limb.*

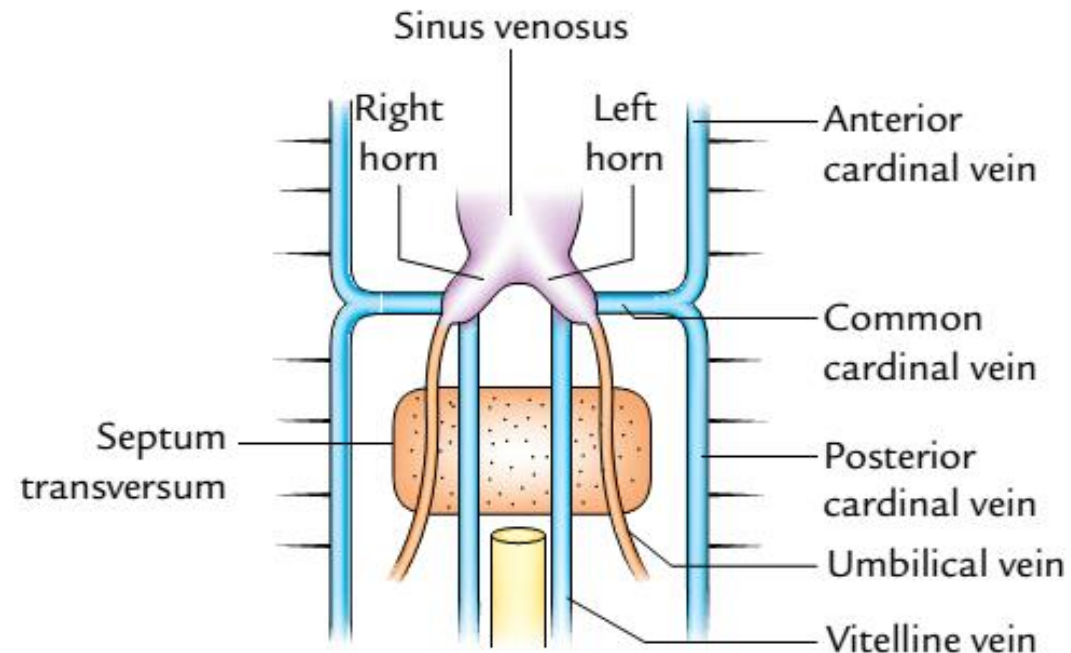
Development of the Veins

Development of the Veins

- The veins of the body develop from three sets of primitive veins:

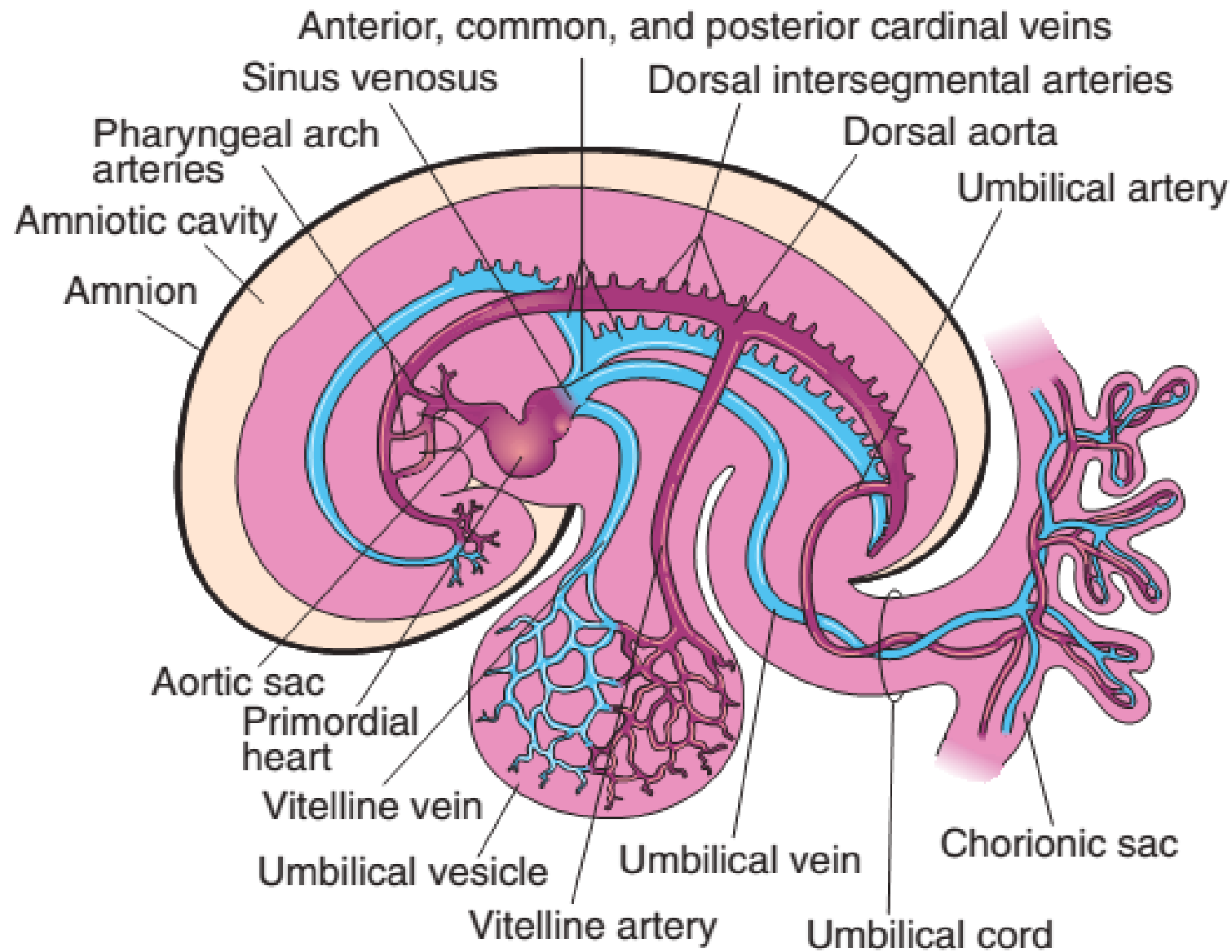
- Vitelline
 - Umbilical
 - Cardinal
- Drain into horn of sinus venosus**

- These veins undergo **remodeling** and form adult veins



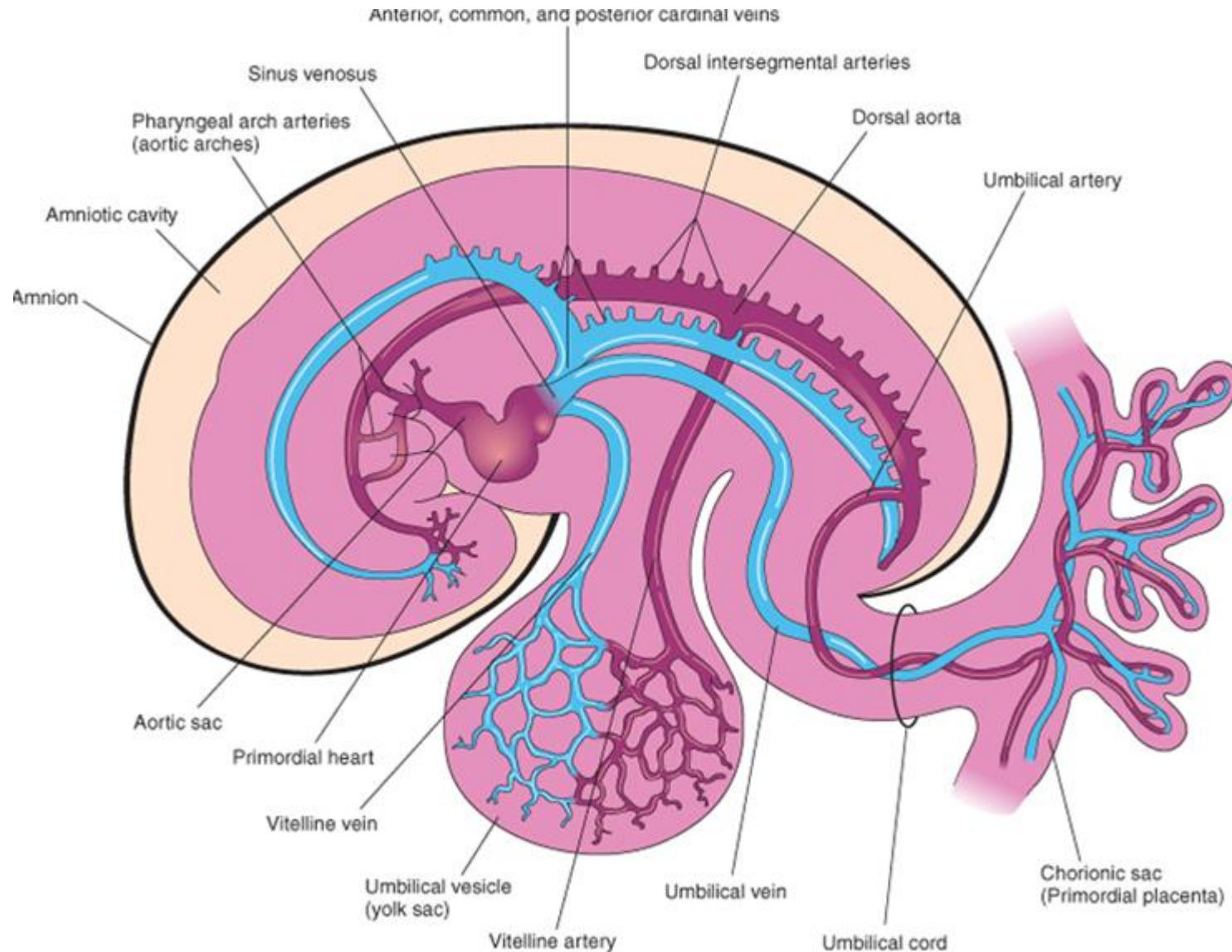
Development of the Veins

- The veins develop from the three major venous circuits, **vitelline, umbilical** and **cardinal**
- Like the arteries they develop in a cephalocaudal direction
- The precursors to the veins are **never all present** at the same time.
- In addition, as new structures develop the course of veins changes
- The formation of the **liver** and the **mesonephric kidney** has profound affects in redirecting blood flow.
- Most of the venous blood is channeled from the **left side to the right side** of the body through the **anastomosing vessels**.



✓ Drawing of the embryonic cardiovascular system (at approximately 26 days) showing vessels on the left side only

Development of the Veins,,,



Left side embryonic cardiovascular system (26 days) 4 –week embryo :

❖ Changes in Sinus venosus:

➤ **It** consists of body and 2 hornes, right & left, each horn receives 3 veins

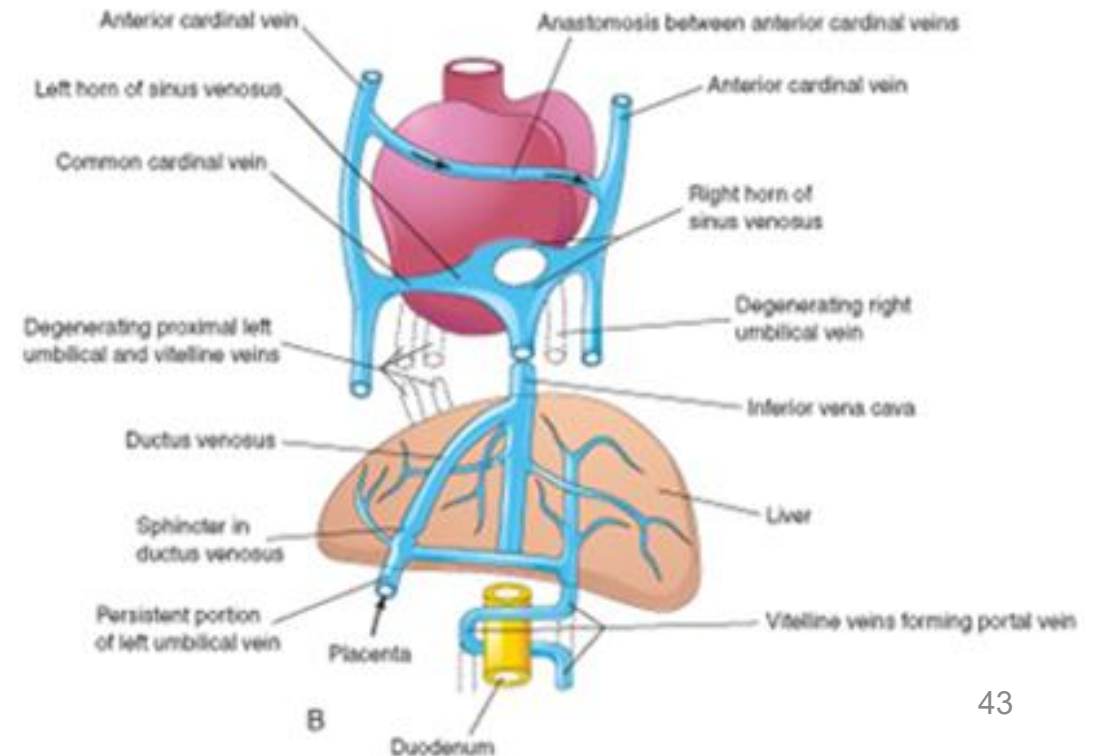
1- Vitelline vein from yolk sac.

2- Umbilical vein from placenta,

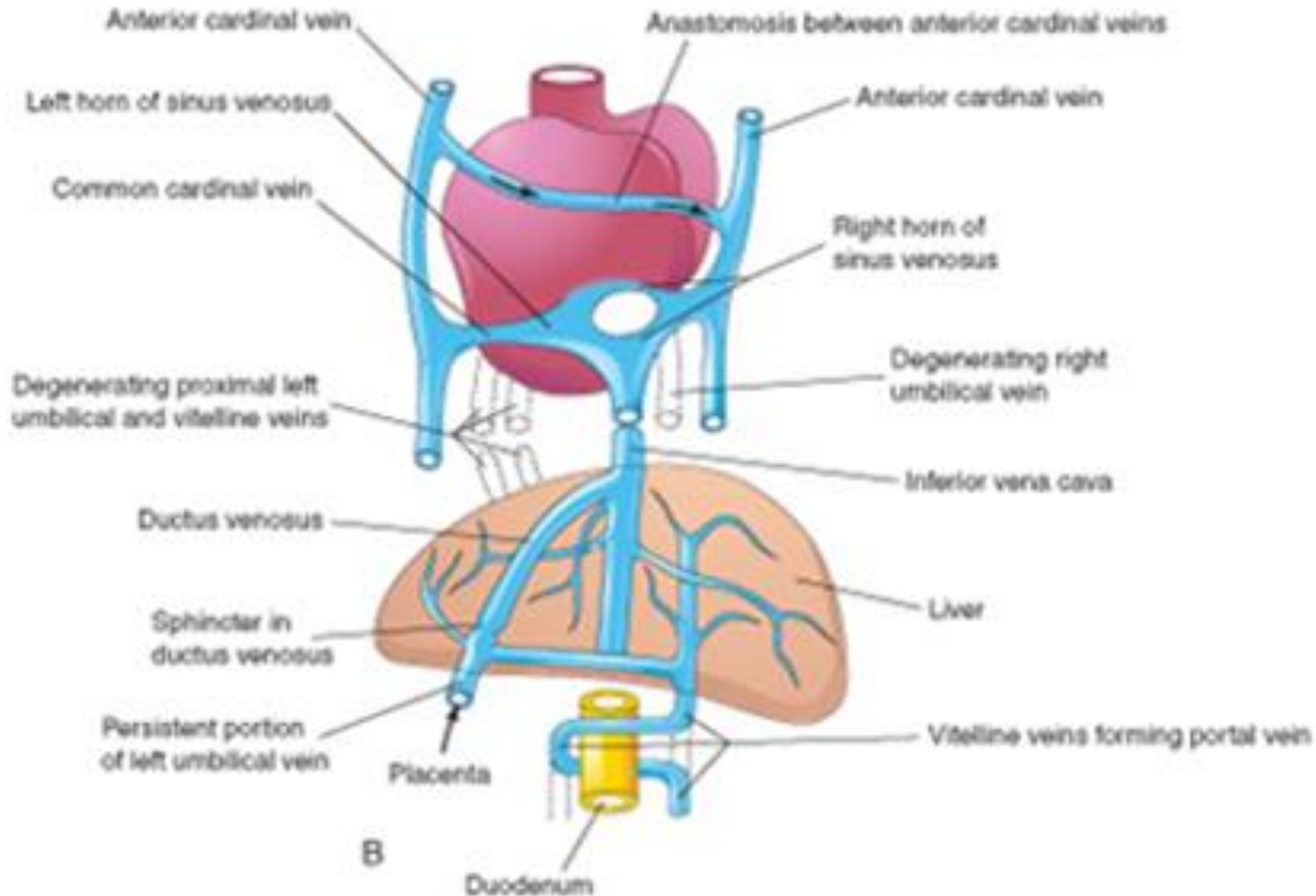
3-Common cardinal vein from body of embryo.

➤ **(B) Later** , due to shunting of blood from **left side to right side** in the connection by anastomosis between the 2 anterior cardinal veins. this shunt.

becomes **left brachiocephalic vein...** (C)



Changes in Sinus venosus,,,

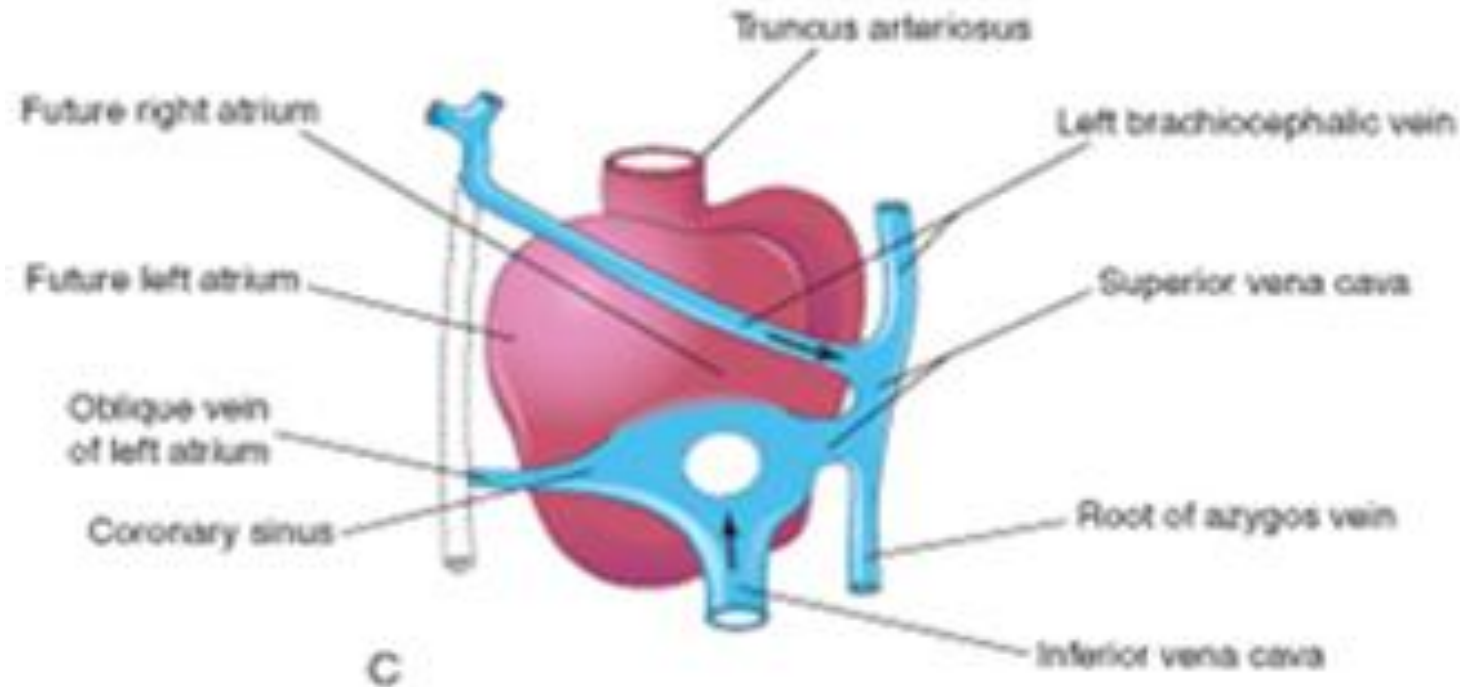


Changes on left side:

1- left horn & body of sinus venosus form the **coronary sinus**.

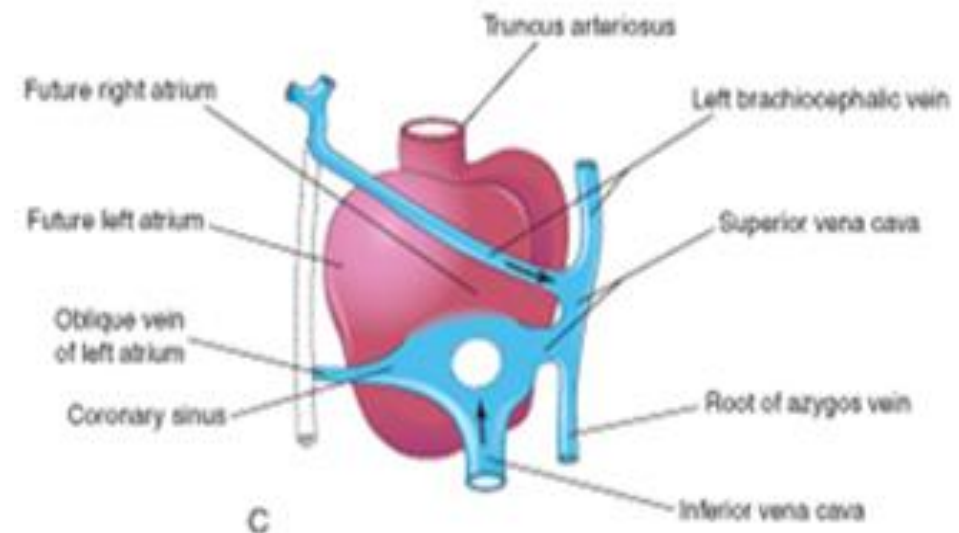
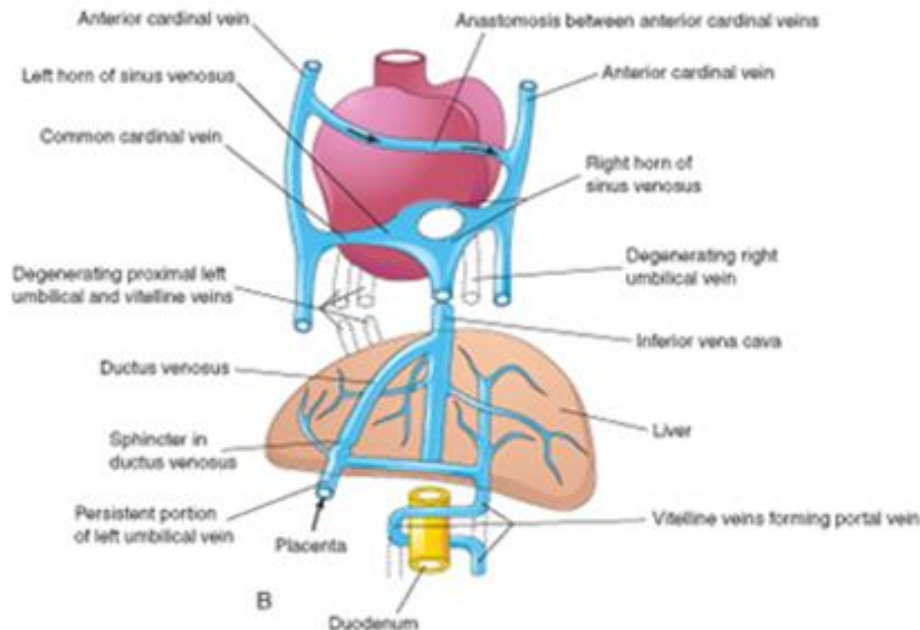
2- left common cardinal vein becomes small to form **oblique vein of left atrium**.

3- left vitelline & right umbilical veins, **degenerate**



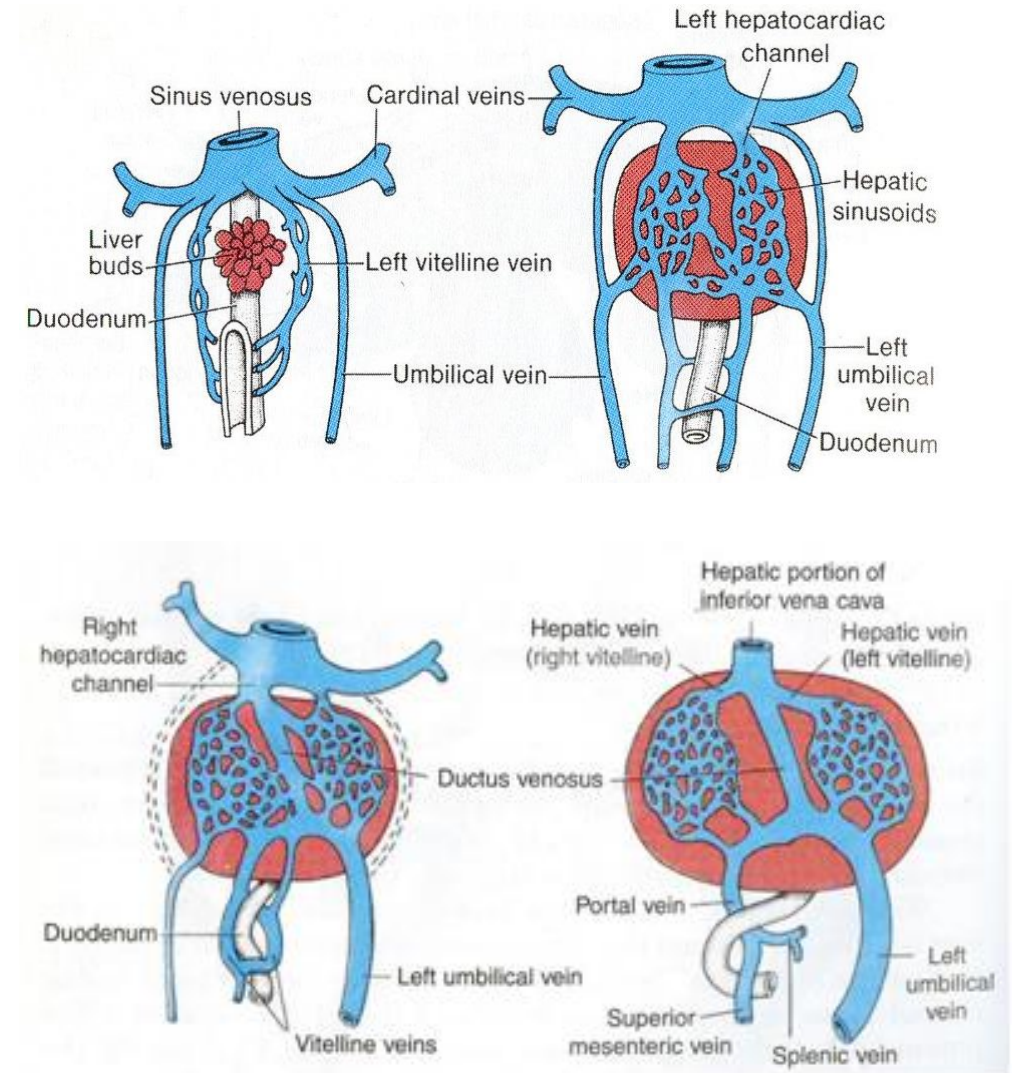
Changes on right side:,,,

- 1- The right horn becomes absorbed into right atrium to form its smooth part, sinus venarum.
2. Right common cardinal vein enlarges to form **SVC**.
- 3- Right vitelline vein becomes **IVC**.
- 4- Right umbilical vein **disappears**



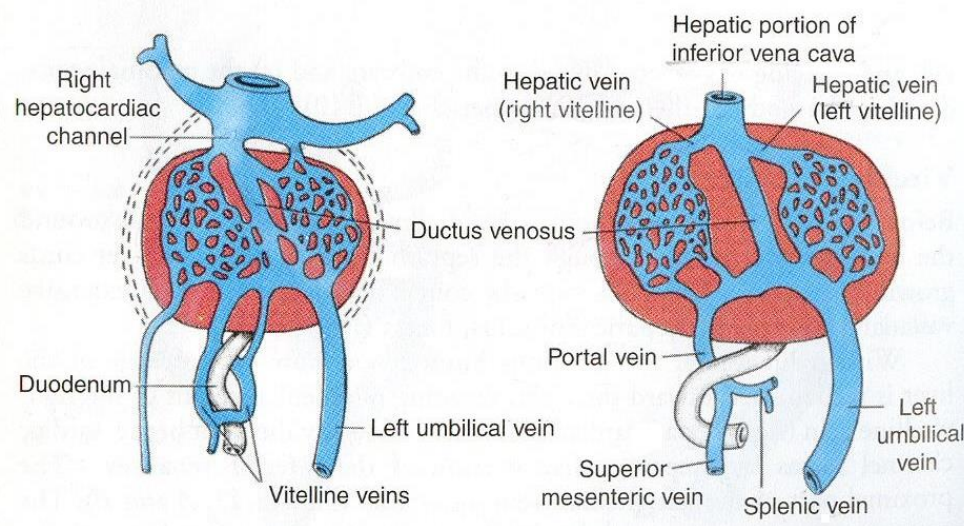
Vitelline Veins

- Pass through the **septum transversum** and drain into the **sinus venosus**
- In relation to the liver developing within the **septum transversum**, the vitelline veins are divided into:
 - **Pre-hepatic part:** forms anastomosis around the **duodenum** which later on gives rise to the **portal vein**



Vitelline Veins,,,

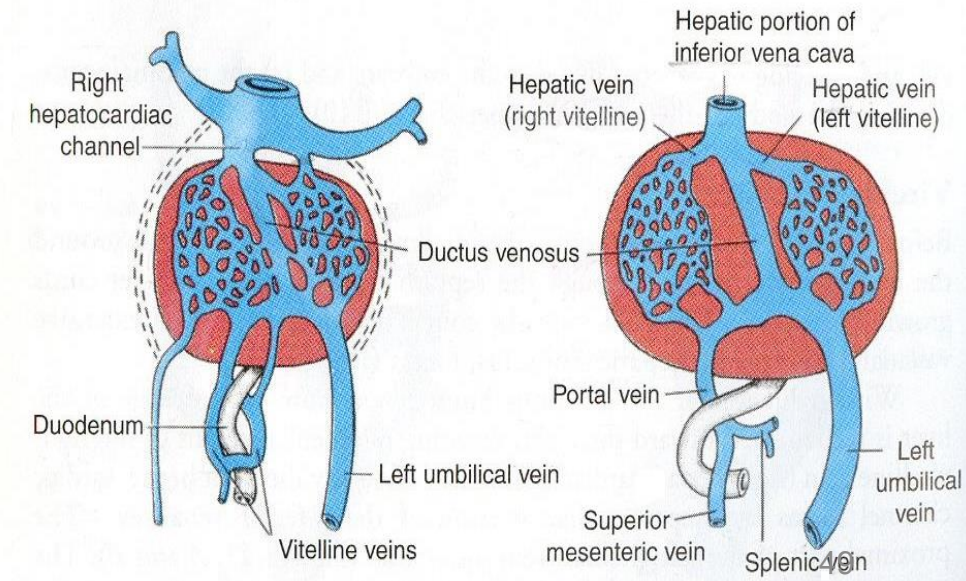
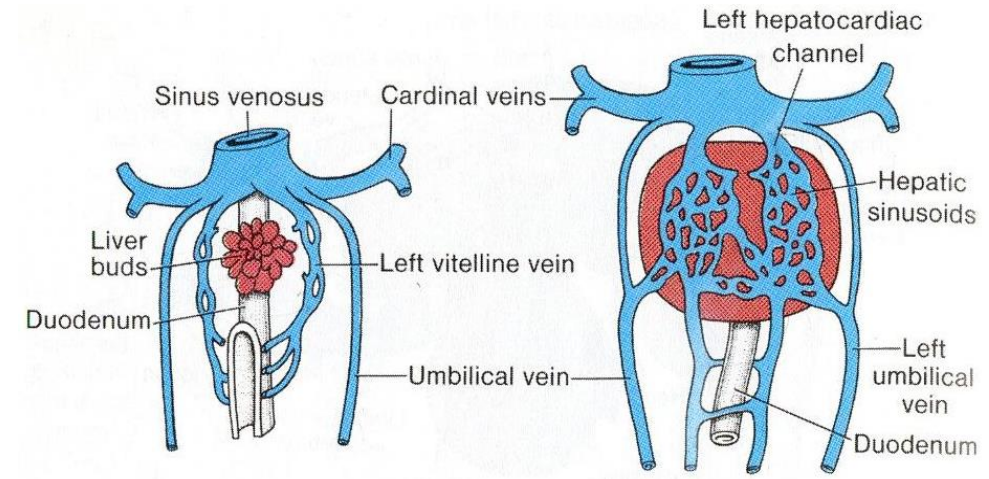
- **Hepatic part:** interrupted by the liver cords, forms an extensive vascular network called the **hepatic sinusoides**
- **Post-hepatic part:**
 - **Left vein disappears**
 - **Right vein forms the:**
 - **Hepatic veins &**
 - **Hepatic segment of inferior vena cava**



Umbilical Veins

❑ Bring oxygenated blood from the placenta

- Initially run on each side of the developing **liver** and drain into the **sinus venosus**
- As the liver grows, the umbilical veins lose their connection with **heart** and open into the **liver**.
- The **right umbilical vein** disappears by the end of the embryonic period.
- The left **umbilical vein** persists.

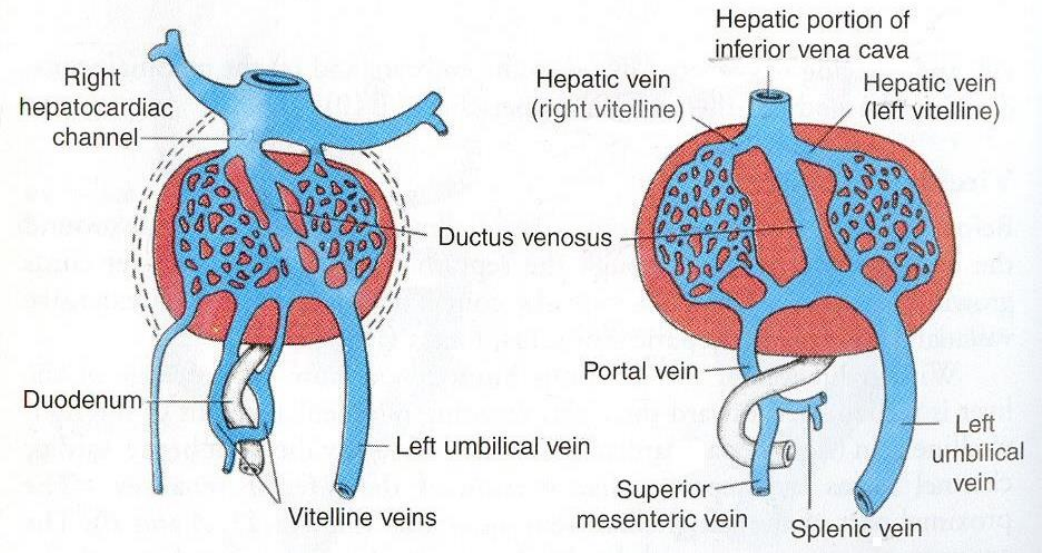


Umbilical Veins,,

- ❖ A wide channel, the **ductus venosus**, appears through the substance of liver to connect the **left umbilical vein** with the **inferior vena cava**

➤ After birth:

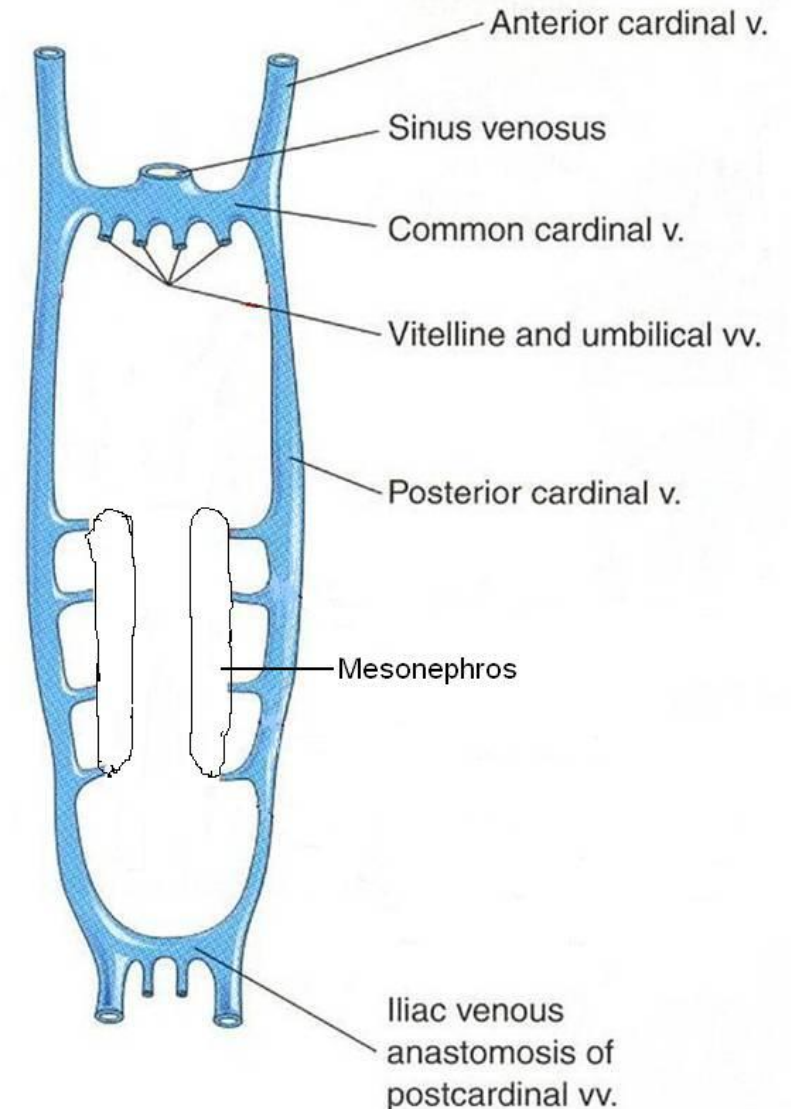
- ❖ The left umbilical vein obliterate to form the **ligamentum teres** of the liver
- ❖ The ductus venosus obliterate to form the **ligamentum venosum**



Cardinal Veins

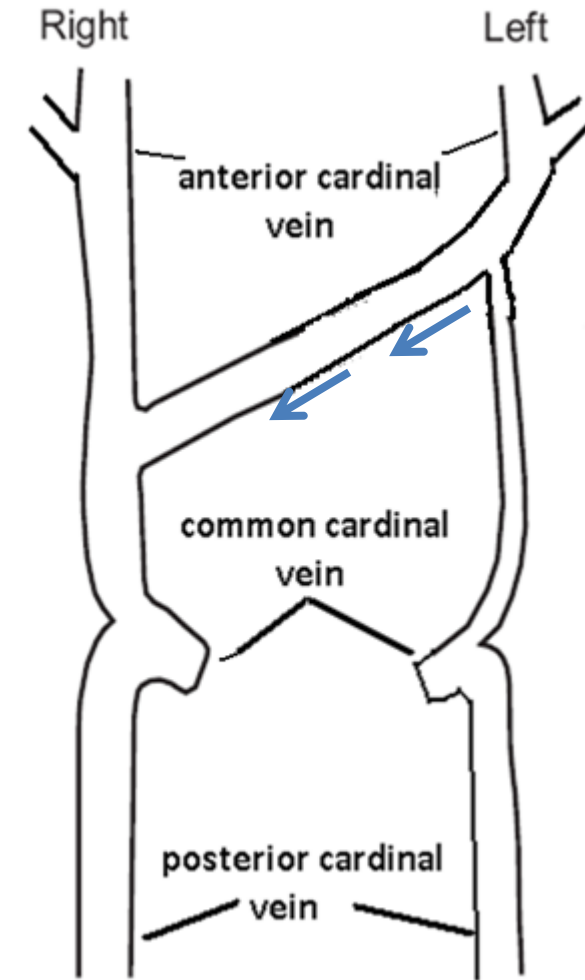
❖ **Are responsible to drain the body of the embryo**

- The **cranial part** of the embryo is drained by **paired anterior cardinal veins**
- The **caudal part** of the embryo is drained by **paired posterior cardinal veins**
- The anterior & posterior cardinal veins join to form **common cardinal veins**, which drain into the **sinus venosus**.



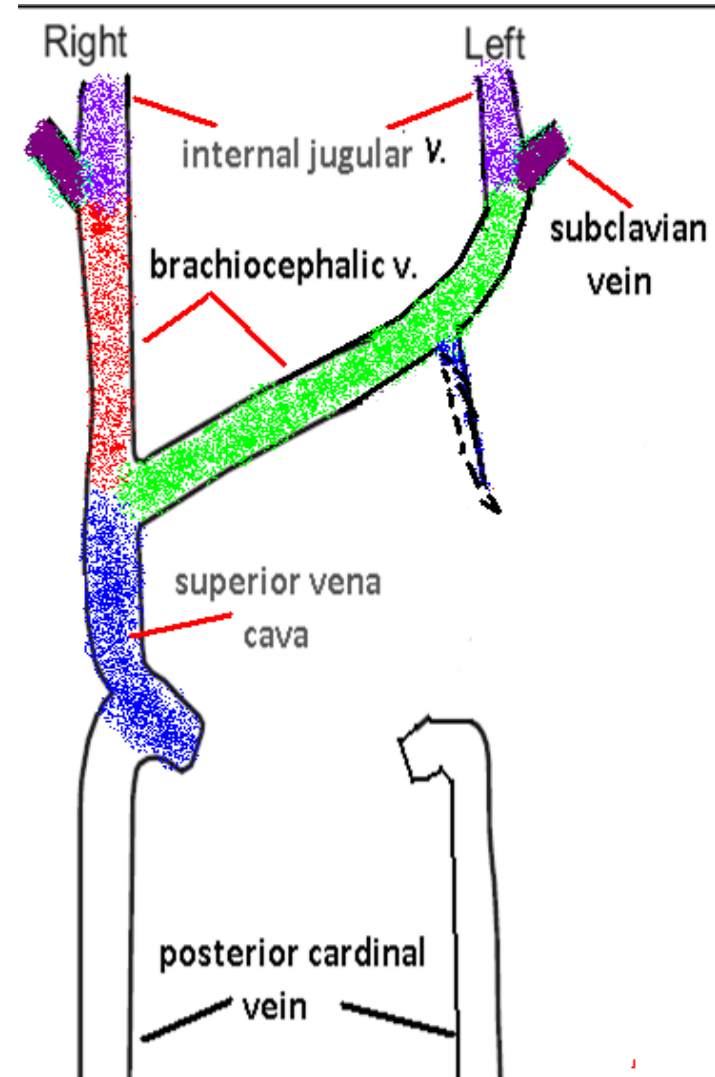
Anterior Cardinal Veins

- Become connected by an oblique anastomosis which shunts blood from left to right
- This anastomosing channel becomes the **left brachiocephalic vein**
- Left anterior cardinal vein
 - **Cranial part:** becomes the **left internal jugular vein**
 - **Caudal part:** degenerates



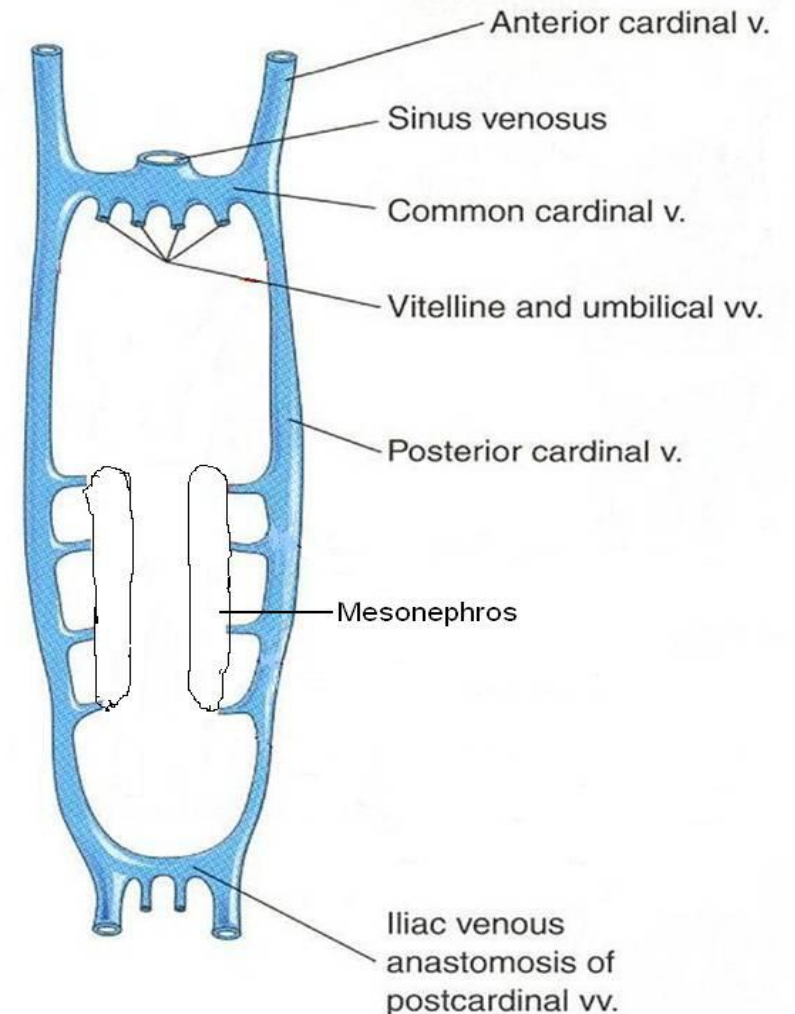
Anterior Cardinal Veins,,,

- Right anterior cardinal vein
 - **Cranial part:** becomes the right internal jugular vein
 - **Middle part:** gives rise to the right brachiocephalic vein
 - **Caudal part** of right anterior cardinal vein and the right common cardinal vein form the **superior vena cava**



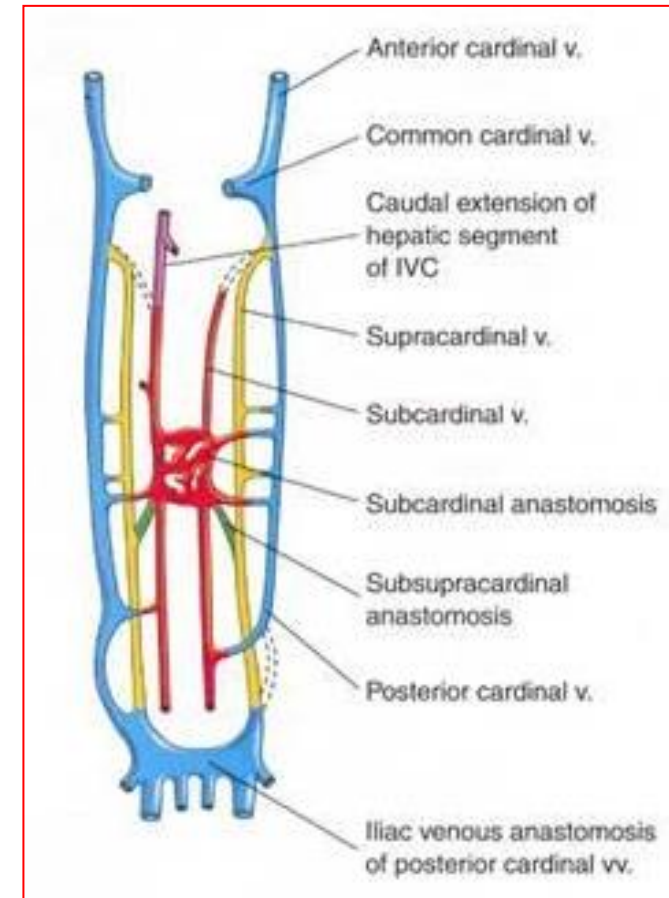
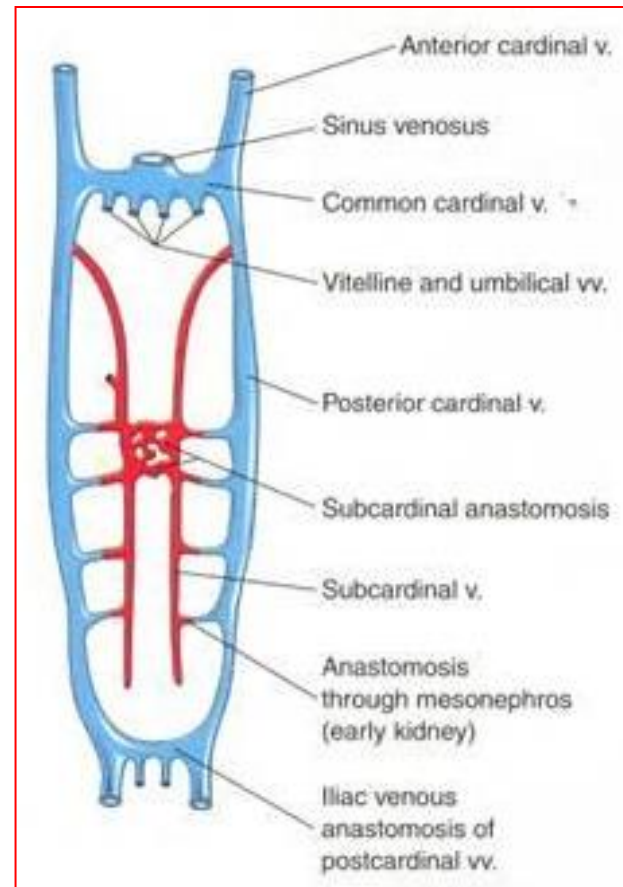
Posterior Cardinal Veins

- Drain the **caudal part** of the body of embryo including the developing **mesonephroi** and largely disappear with this transitory kidneys.
- Caudally the two veins get connected by an **anastomosing channel** that directs the blood from the **left to the right** vein
- The only adult derivatives of the posterior cardinal veins are the root of the azygos vein and the common iliac veins



Posterior Cardinal Veins,,,

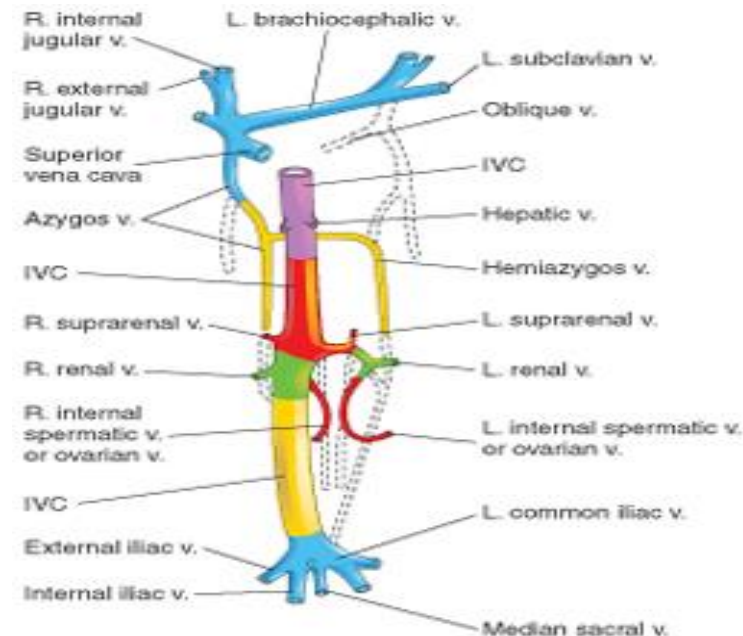
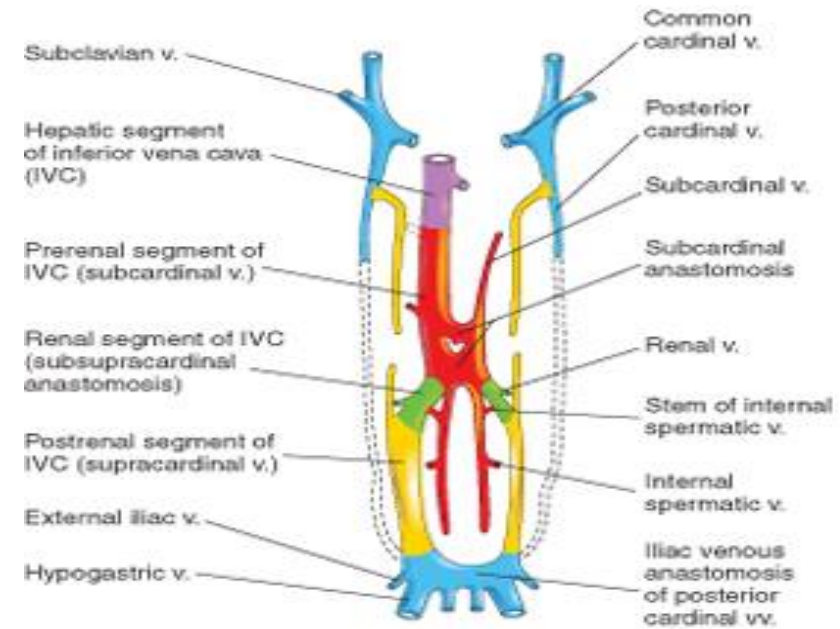
- Gradually the posterior cardinal veins are replaced by two new veins: **subcardinal** & **supracardinal**



Posterior Cardinal Veins,,

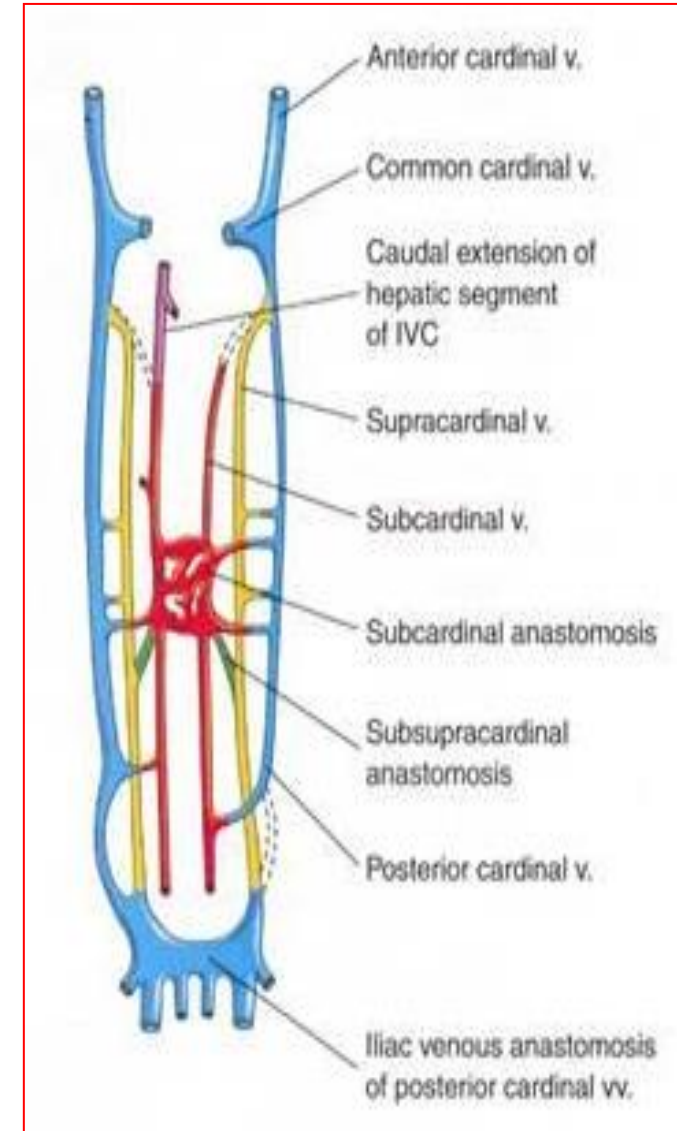
- The adult derivatives of the **posterior cardinal veins** are the:

- Root of the azygos vein &
- Common iliac veins



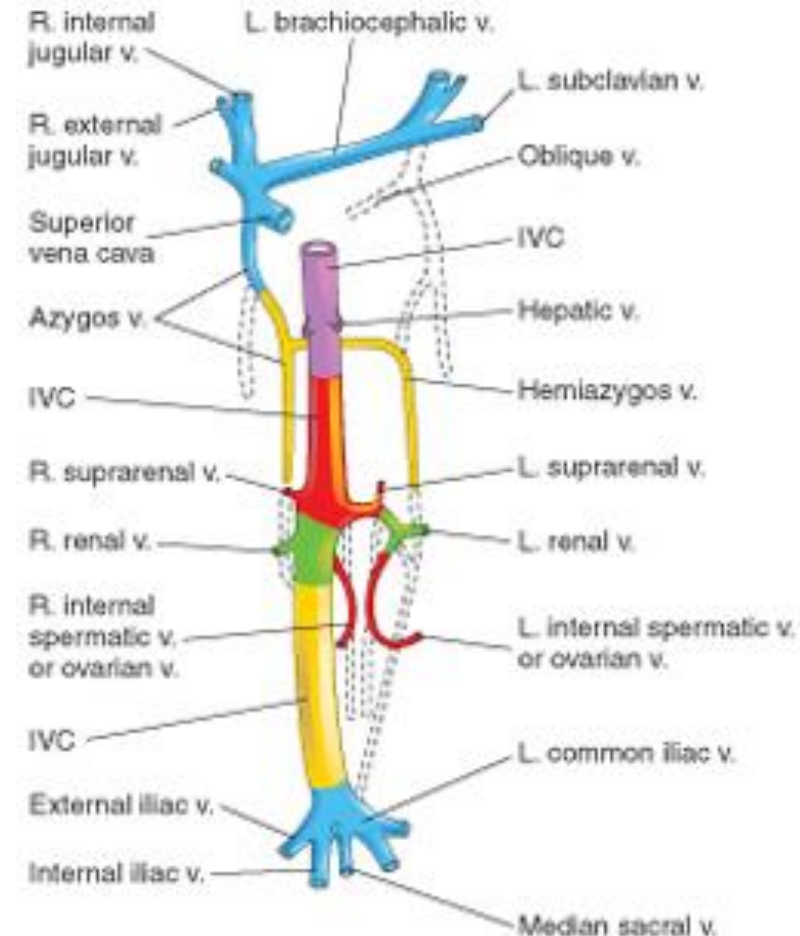
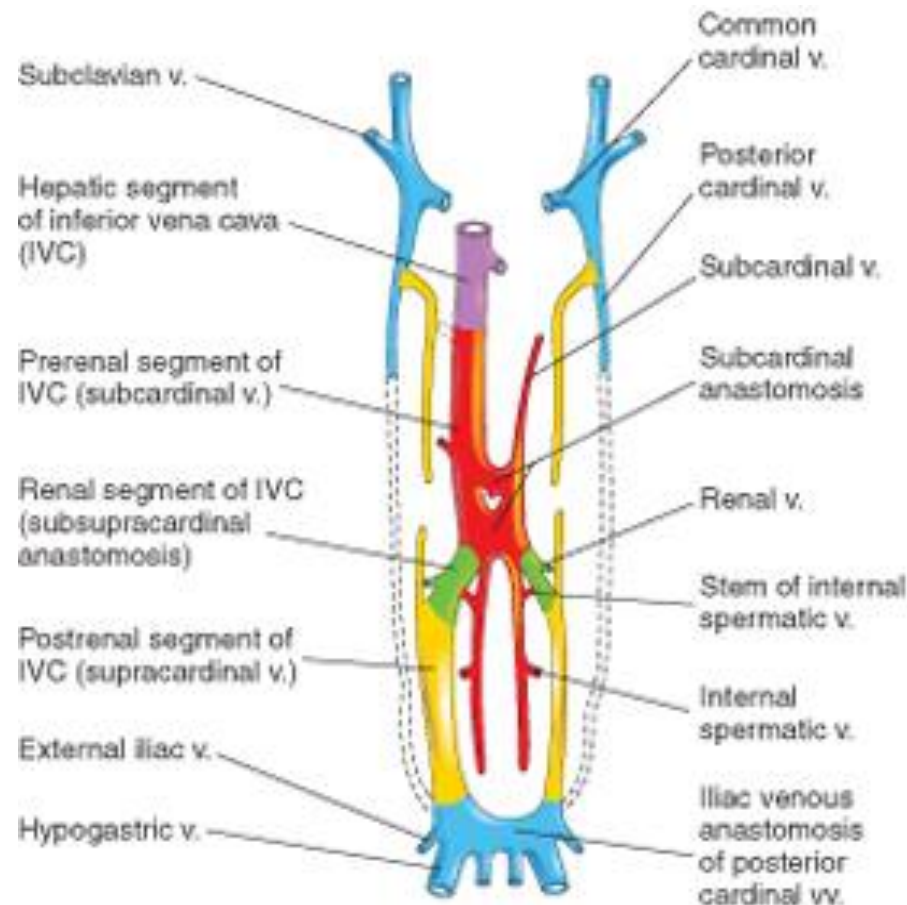
Subcardinal Veins

- Appear before the supracardinal veins
- Become connected:
 - To each other by the **subcardinal anastomosis**
 - With the posterior cardinal veins through the mesonephros
 - With supracardinal veins through subsupracardinal anastomosis



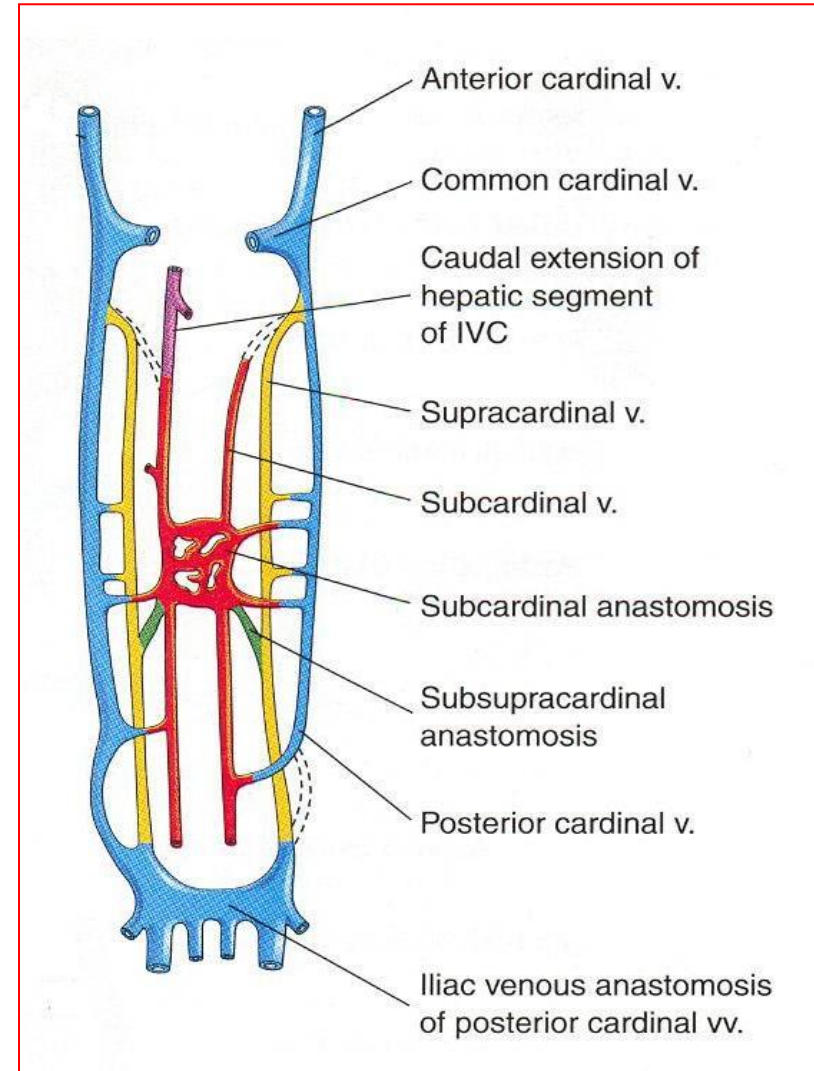
Subcardinal Veins,,,

- The adult derivatives of the subcardinal veins are the:
 - Stem of the left renal vein, Suprarenal veins, Gonadal veins and Prerenal segment of IVC



Supracardinal Veins

- Last pair of veins to develop
- Appear **lateral** to the **subcardinal veins**
- Become connected at both ends, to the posterior cardinal veins
- Get disrupted in the region of the kidney



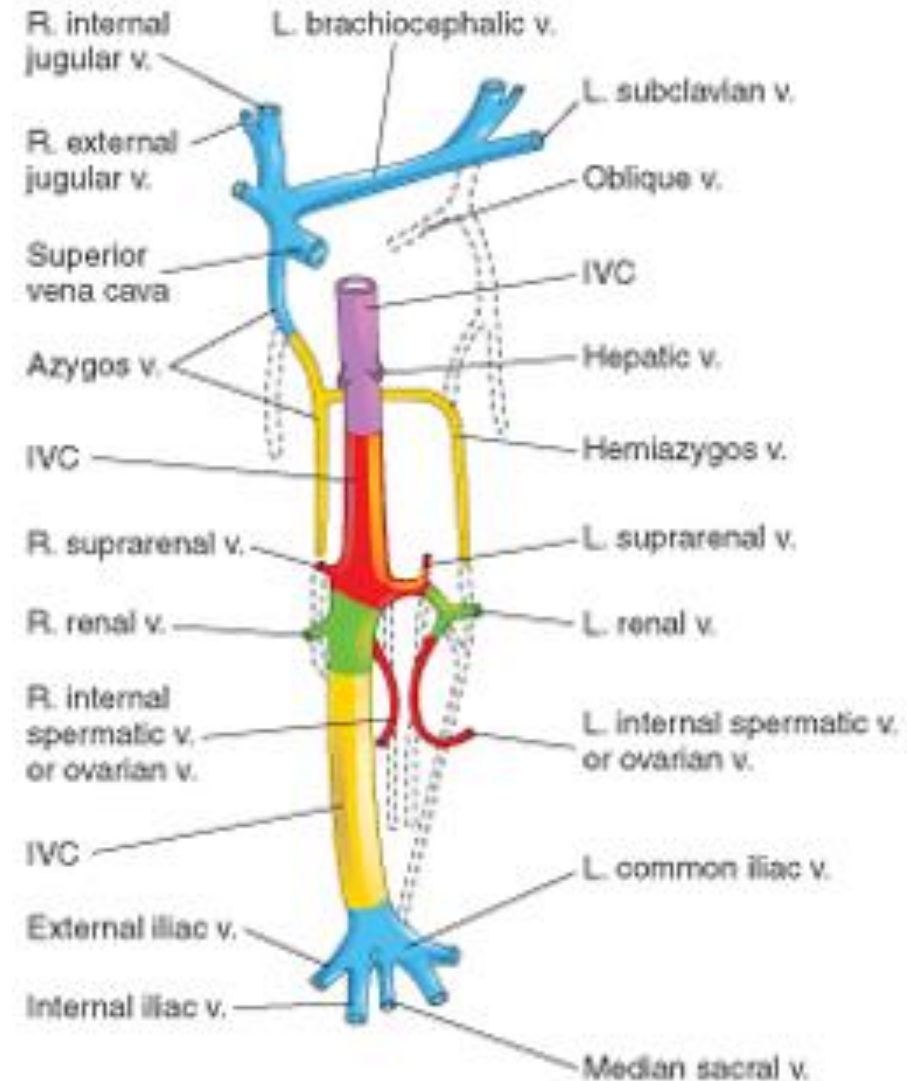
Supracardinal Veins,,,

➤ Cranial to the kidney:

- Gives rise to **azygos** and **hemiazygos** veins

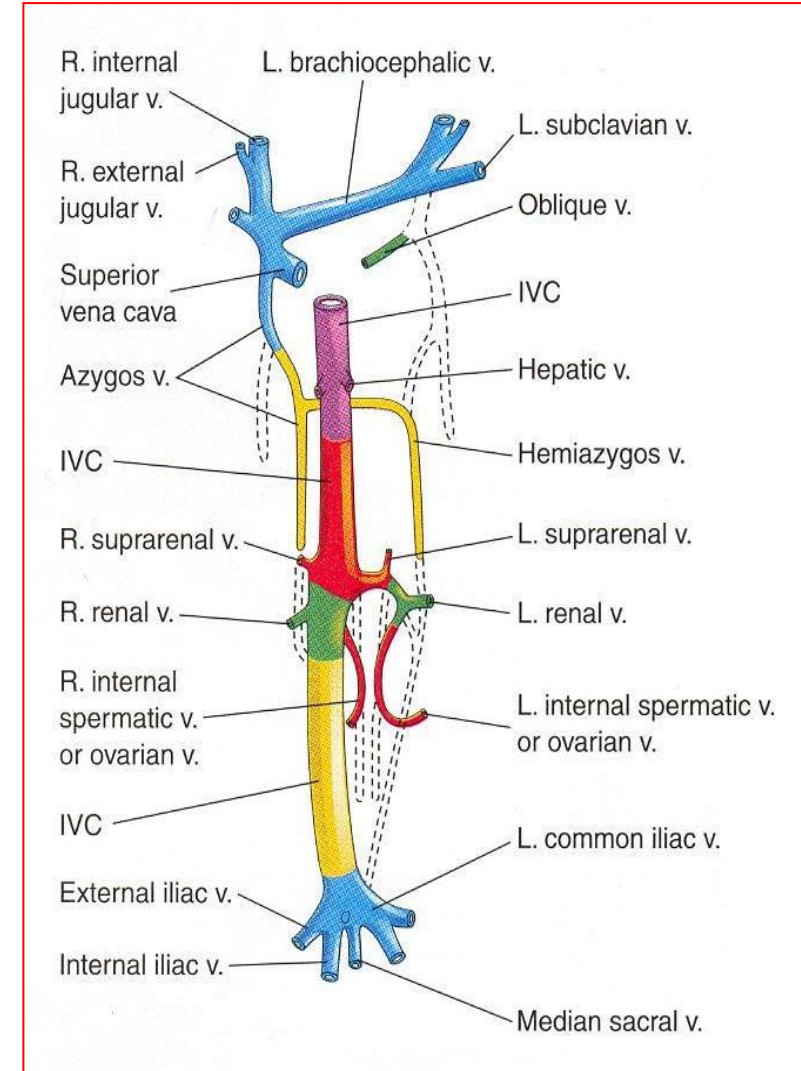
➤ Caudal to the kidney:

- Left vein degenerates
- Right vein forms **postrenal** segment of IVC



Development of Azygos Veins

- **Azygos vein** is derived from the:
 - **Cranial part of the right supracardinal vein** &
 - **Terminal part of the right posterior cardinal vein**
- **Hemiazygos vein** is derived from the **cranial part of the left supra-cardinal vein**

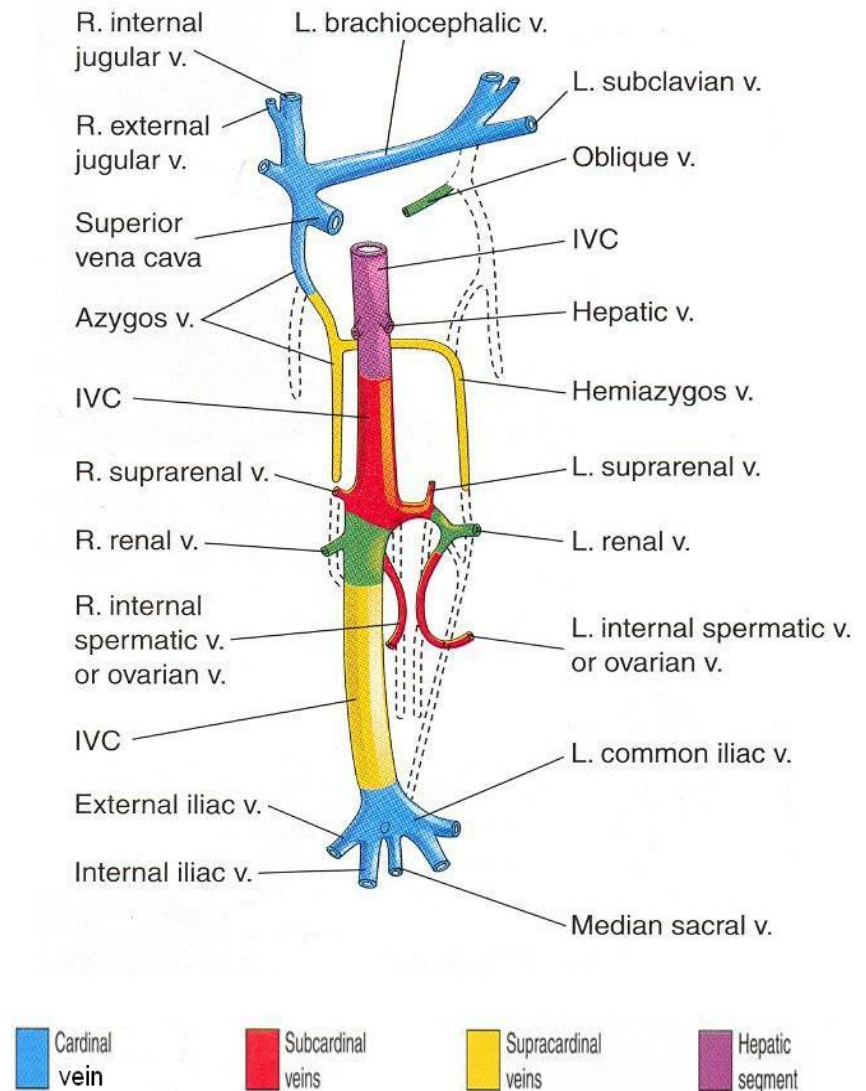


Development of Inferior Vena Cava

- The **IVC** develops during a series of changes in the primordial veins

➤ Composed of:

- Hepatic segment derived from the right vitelline vein
- Prerenal segment derived from the right subcardinal vein
- Renal segment derived from the subcardinal-supracardinal anastomosis
- Postrenal segment derived from the right supr-acardinal vein



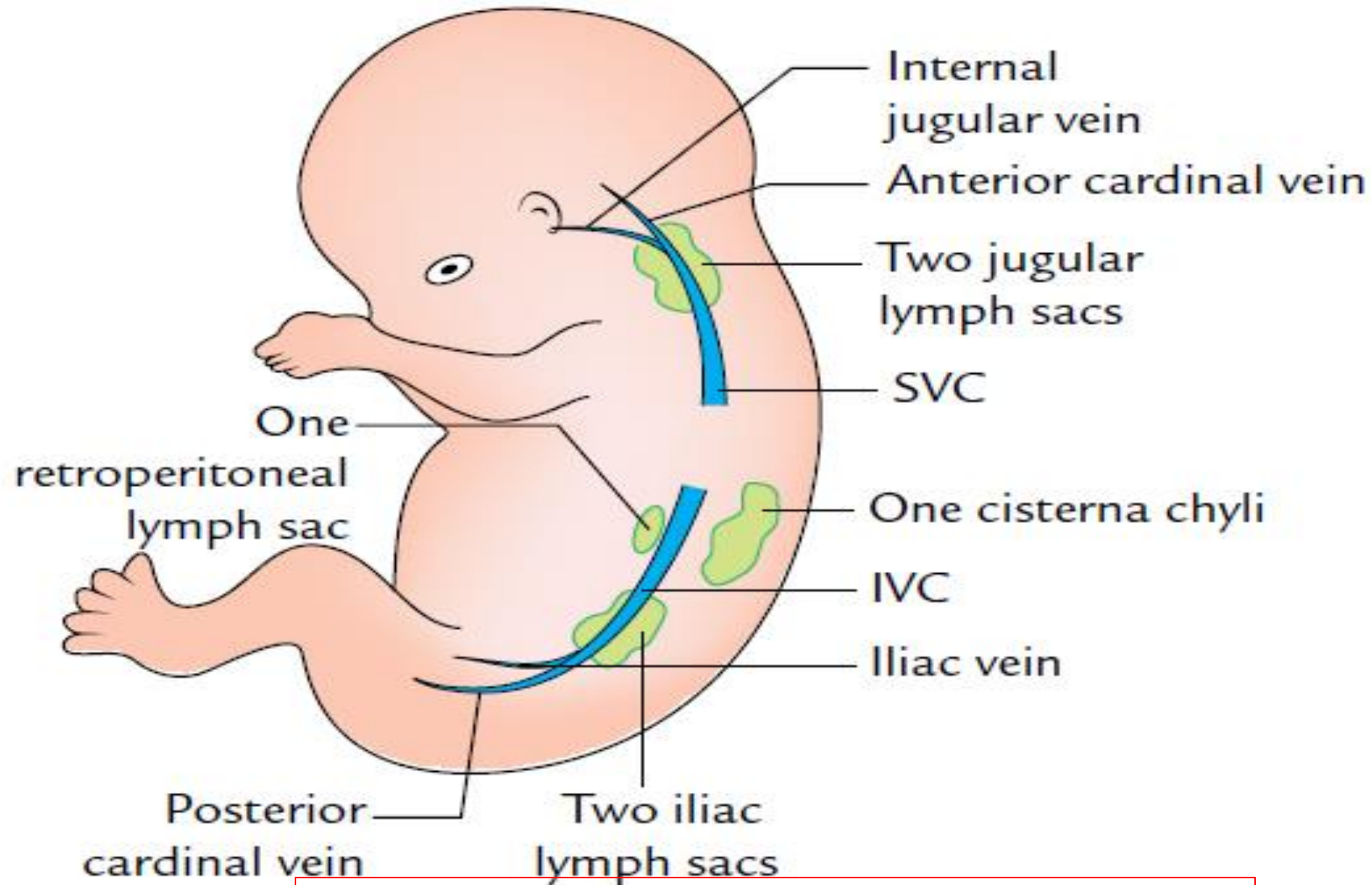
Development of Lymphatic System

- The lymphatic system begins its development later than the cardiovascular system, not appearing until the fifth week of gestation.
- The lymphatic system begins to develop at the end of the sixth week, approximately 2 weeks after the primordia of the cardiovascular system are recognizable.
- The exact development of lymph vessels is not clear but they may develop in the following two ways:
 - **They may form from mesenchyme in situ or**
 - **they may arise as sac-like outgrowths from the endothelium of veins.**
- In whatever way they develop they connect themselves with the venous system.

Development of Lymph Sacs and Lymph Vessels

- ❑ The following 6 primary lymph sacs are formed during the embryonic period:
1. **Two jugular lymph sacs** at the junction between the subclavian veins and anterior cardinal veins (future internal jugular veins).
 2. **Two iliac lymph sacs** at the junction of iliac and the posterior cardinal veins.
 3. **One retroperitoneal lymph sac** in the root of the mesentery on the posterior abdominal wall.
 4. **One cisterna chyli** lying posterior to the retroperitoneal lymph sac.
 5. Numerous small lymph channels/vessels connect the lymph sacs with each other and pass along the main veins to drain lymph from limbs, body wall, head, and neck.
 6. Two large lymph channels (right and left lymph channels) connect the jugular lymph sacs with the cisterna chyli. **Soon large anastomosis forms between these ducts.**

Development of Lymph Sacs and Lymph Vessels,,,



➤ **Development of lymph sac.**

Development of Thoracic Duct

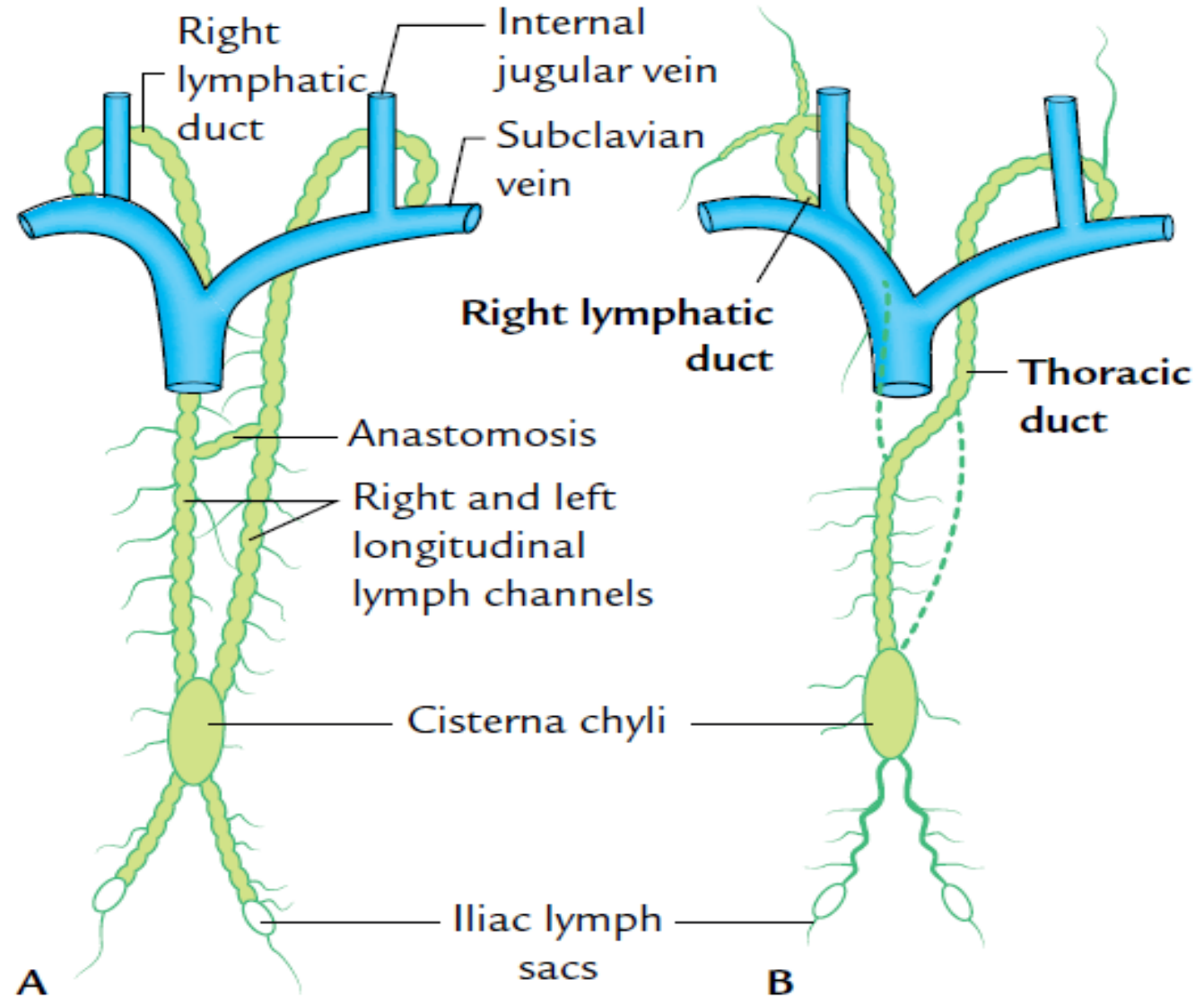
❖ **There are three steps in the development of the thoracic duct:**

- **1. Stage I:** A network of lymph channels is formed in front of the thoracic part of the vertebral column.
 - **2. Stage II:** In this network of lymph channels, two large longitudinal channels appear—one left and another right with a number of cross communications.
 - **3. Stage III:** Formation of cross communication between two longitudinal channels opposite to the body of fifth thoracic vertebra.
- Now the right longitudinal channel below cross communication and the left longitudinal channel above communication persists and gives rise to the thoracic duct.
- All other channels disappear.**

Development of Thoracic Duct,,

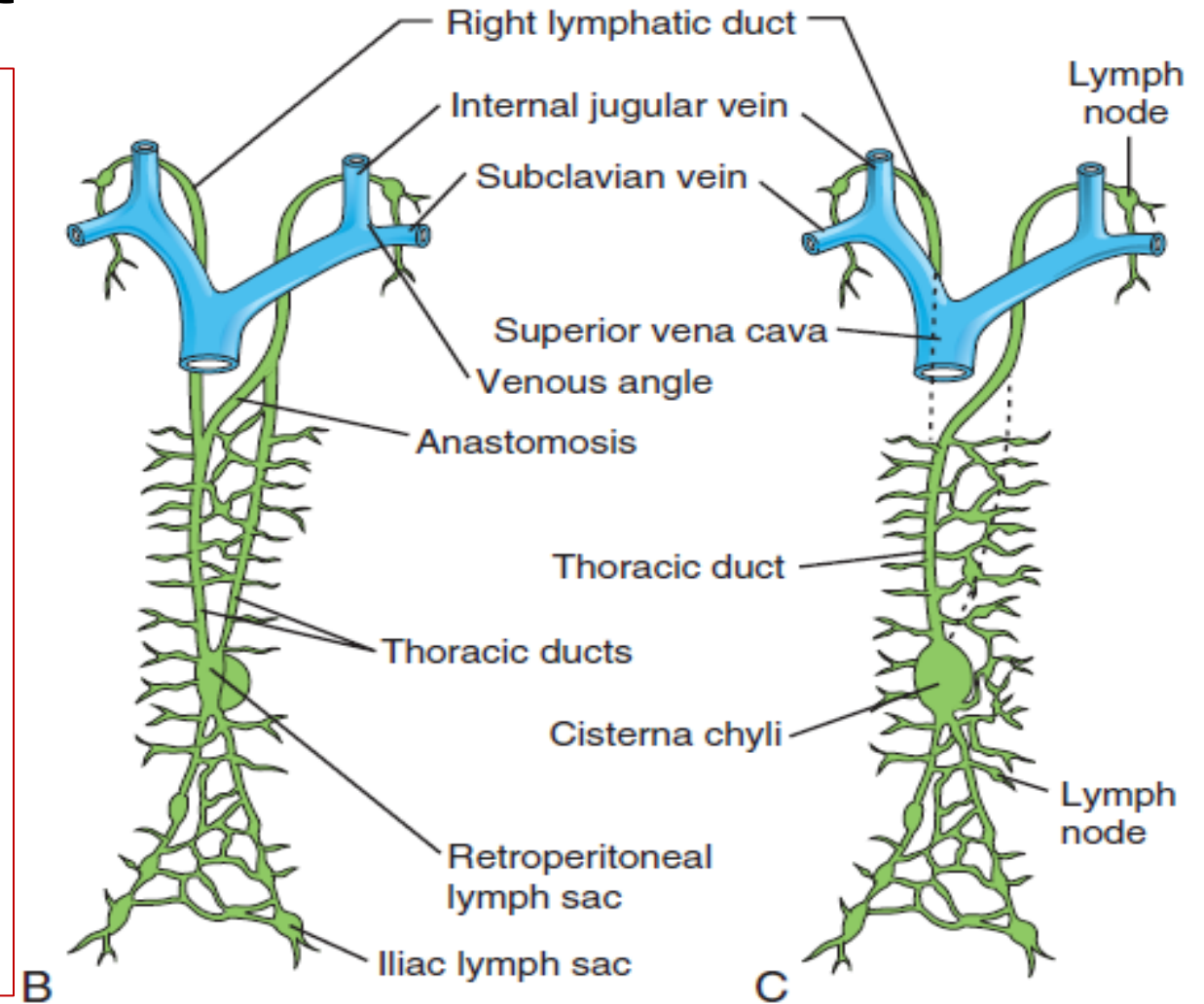
❑ Thus thoracic duct is derived from the following three sources:

- I. Caudal part of the right longitudinal lymph channel.
- II. Anastomosis between the right and left longitudinal lymph channel.
- III. Cranial portion of the left longitudinal lymph channel.



Development of Right Lymphatic Duct

- It develops from the cranial portion of the right longitudinal lymph channel.
- Both thoracic and right lymphatic ducts open into the junction of the internal jugular and subclavian vein of the left and right side, respectively



- B, Ventral view of the lymphatic system at 9 weeks showing the paired thoracic ducts.
- C, Later in the fetal period, illustrating formation of the thoracic duct and right lymphatic duct

Development of Lymph Nodes

- The lymph nodes develop from the mesenchymal cells in relation to the primary lymph sacs.
- Except for the upper part of cisterna chyli all the lymph sacs are transformed into groups of lymph nodes during early fetal life as follows:
 - ❖ Mesenchymal cells invade each lymph sac and break up its cavity into a network of lymph channels to form *lymph sinuses*.
 - ❖ Other mesenchymal cells give rise to the capsule and connective tissue network of the lymph nodes.

Development of Lymphocytes

- Early on the lymphocytes first develop from stem cells in the mesenchyme of the yolk sac and later on from the liver and spleen.
 - The early lymphocytes enter into bone marrow, from where they divide to form the **lymphoblasts**.
 - The lymphocytes that appear in the lymph nodes before birth are derived from thymus.
 - The small lymphocytes leave the thymus to reach other lymphoid organs through circulation. Later some mesenchymal cells of lymph nodes also differentiate into the lymphocytes.
- ❖ **N.B.** The lymph nodules do not appear in the lymph nodes until just before and/or just after birth, as there is no exposure to foreign antigens.

Development of Spleen and Tonsils

1. **Spleen** develops from an aggregation of mesenchymal cells between the two layers of dorsal mesogastrium.
 2. **Palatine tonsil** develops from the endodermal lining of second pharyngeal pouch and underlying mesenchyme.
 3. **Tubal tonsil** develops from an aggregation of the lymph nodules around the opening of pharyngotympanic tube in the nasopharynx.
 4. **Pharyngeal tonsil** develops from an aggregation of the lymph nodules in the posterosuperior wall of the nasopharynx.
 5. **Lingual tonsil** develops from an aggregation of the lymph nodules on dorsum of the posterior onethird of the tongue.
- **N.B. The lymph nodules also develop into the mucosa of gastrointestinal and respiratory tracts.**

Common anomalies of the heart and great vessels

Common anomalies of the heart and great vessels

- Like other systems most **abnormalities in the development of CVS** happen during the **third to eighth weeks of gestation**.
- Congenital cardiovascular malformations account for about 20% of all congenital defects observed in live-born infants.
- They occur in about 5 to 8 of every 1000 live births, and the percentage in stillborn infants is probably even higher.
- Heart and vascular abnormalities make up the largest category of human birth defects.

Ectopia cordis

- ***Ectopia cordis*** is a rare condition in which the heart lies exposed on the surface of the thorax.
- It occurs due to nonunion of two sternal plates of developing sternum.
- Death occurs in most of the cases during first few days of the birth.

❑ If there are not severe cardiac defects, surgical therapy usually consists of covering the heart with skin

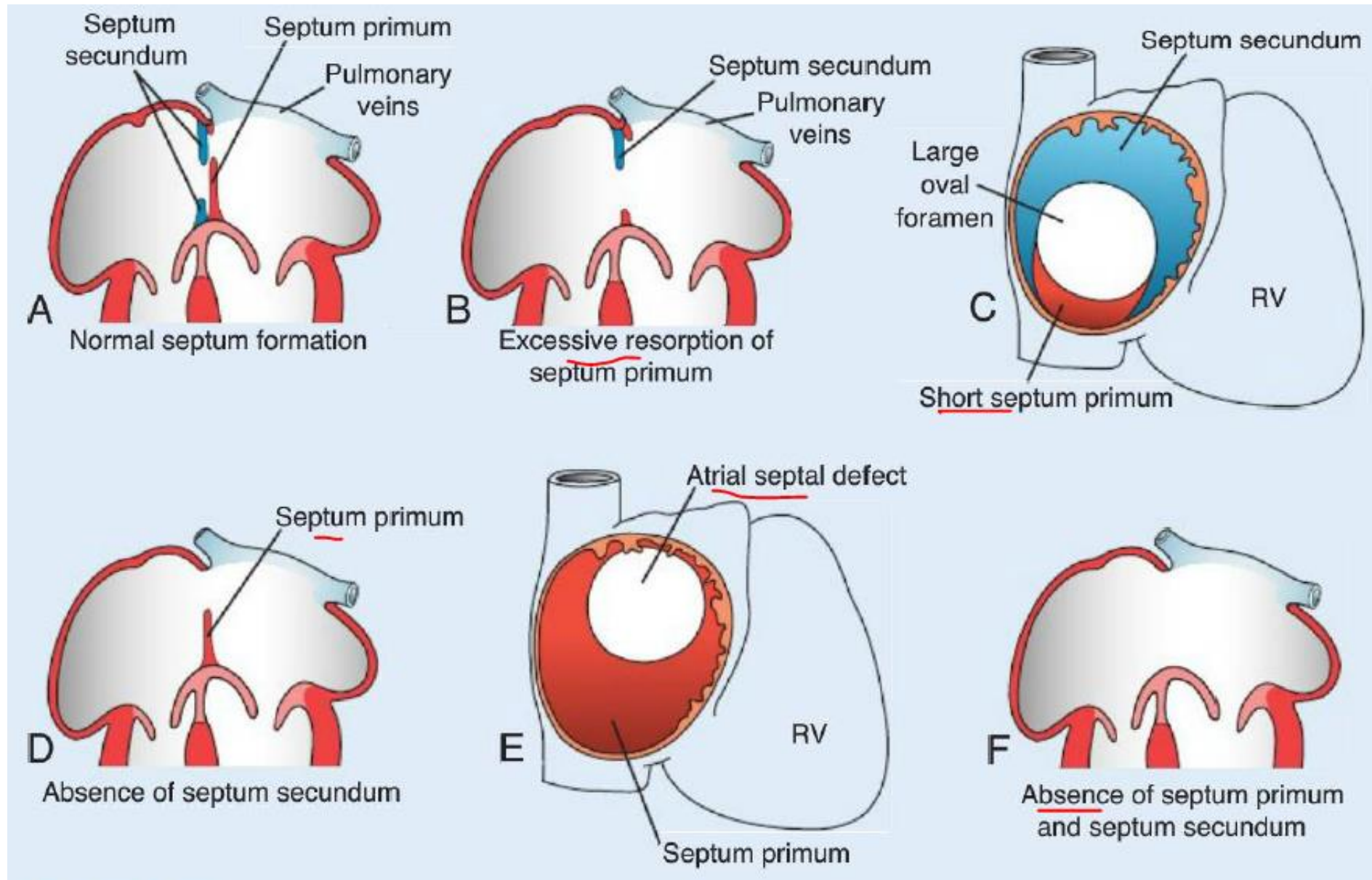
❑ The most common thoracic form of ectopia cordis results from faulty development of the sternum and pericardium because of failure of complete fusion of the lateral folds in the formation of the thoracic wall during the fourth week.



Atrial septal defect (ASD)

- ASD is a common congenital heart defect and occurs more frequently in females than in males.
- One of the most significant defects is the **ostium secundum** defect, characterized by a large opening between the left and right atria.
- It is caused either by excessive cell death and resorption of the septum primum or by inadequate development of the septum secundum.
- Depending on the size of the opening, considerable intracardiac shunting may occur from left to right.
- ***N.B. The ASD is most common congenital anomaly of the atria. If the ASD is small, the clinical symptoms may be delayed up to the 30 years of age.***

Atrial septal defect (ASD),,,



Atrial septal defect (ASD),,,

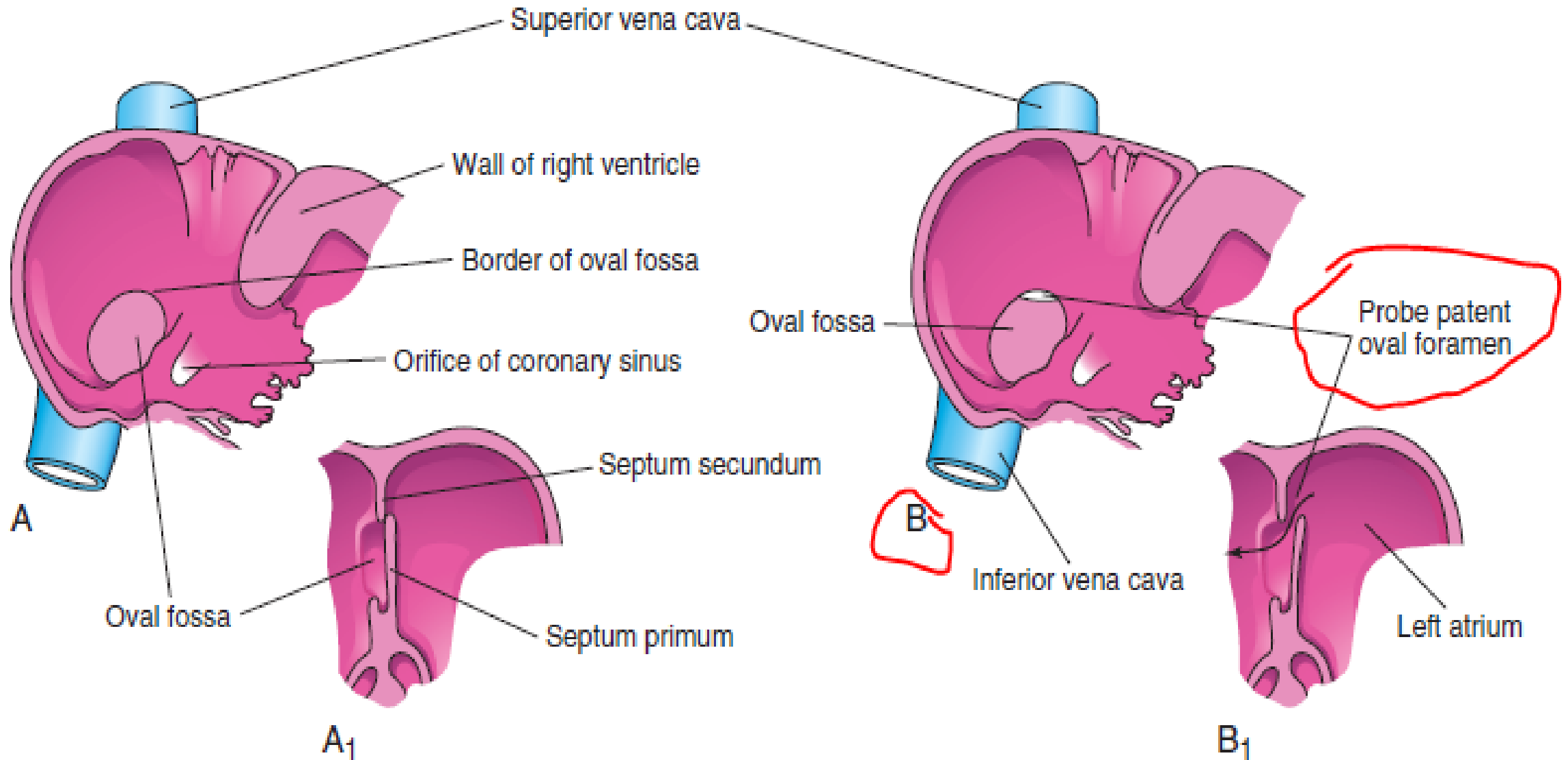
- In about 6 of 10,000 live-born infants, the septum secundum is too short to cover the foramen secundum completely.
- The most serious abnormality in this group is complete absence of the atrial septum.
- This condition, known as common atrium or **cor triloculare biventriculare**, is always associated with serious defects elsewhere in the heart.
- Occasionally, the oval foramen closes during prenatal life.
- This abnormality, **premature closure of the oval foramen**, leads to massive hypertrophy of the right atrium and ventricle and underdevelopment of the left side of the heart. Death usually occurs shortly after birth.

Atrial septal defect (ASD),,,

❑ **The most common form of ASD is patent oval foramen.**

- A probe patent oval foramen is present in up to 25% of people.
- In this circumstance, a probe can be passed from one atrium to the other through the superior part of the floor of the oval fossa.
- This defect is not clinically significant, but a probe patent oval foramen may be forced open because of other cardiac defects and contribute to the functional pathology of the heart.
- Probe patent oval foramen results from incomplete adhesion between the flap-like valve of the oval foramen and the septum secundum after birth.

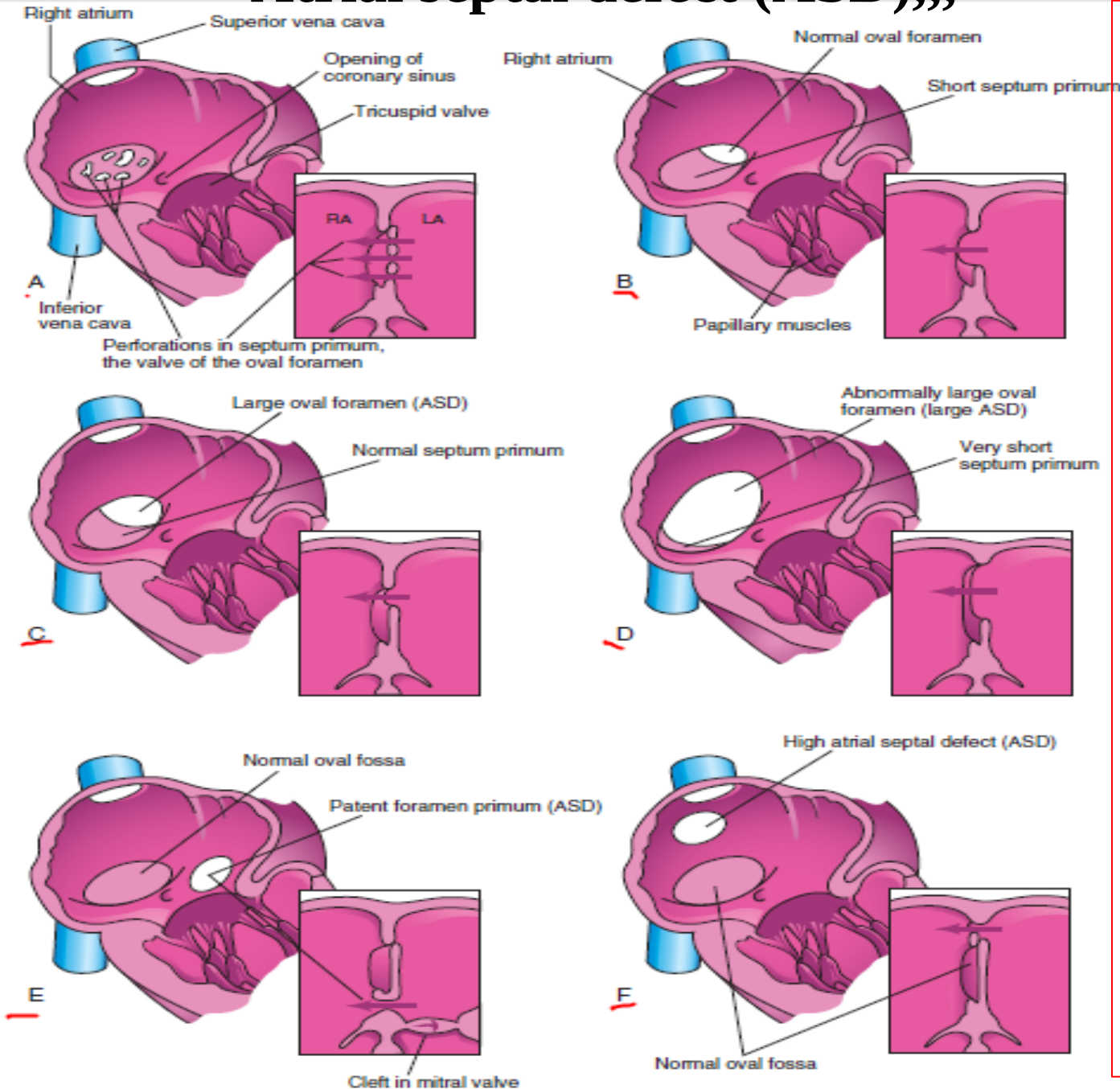
Atrial septal defect (ASD),,,



Atrial septal defect (ASD),,,

- **There are four clinically significant types of ASD:** ostium secundum defect, endocardial cushion defect with ostium primum defect, sinus venosus defect, and common atrium. The first two types of ASD are relatively common.
- **Ostium secundum ASDs** are in the area of the oval fossa and include both defects of the septum primum and septum secundum.
- Ostium secundum ASDs are well tolerated during childhood; symptoms such as pulmonary hypertension usually appear in the 30s or later.
- The defects may be multiple and, in symptomatic older children, defects of 2 cm or more in diameter are not unusual. **Females with ASD outnumber males 3:1.**

Atrial septal defect (ASD),,,

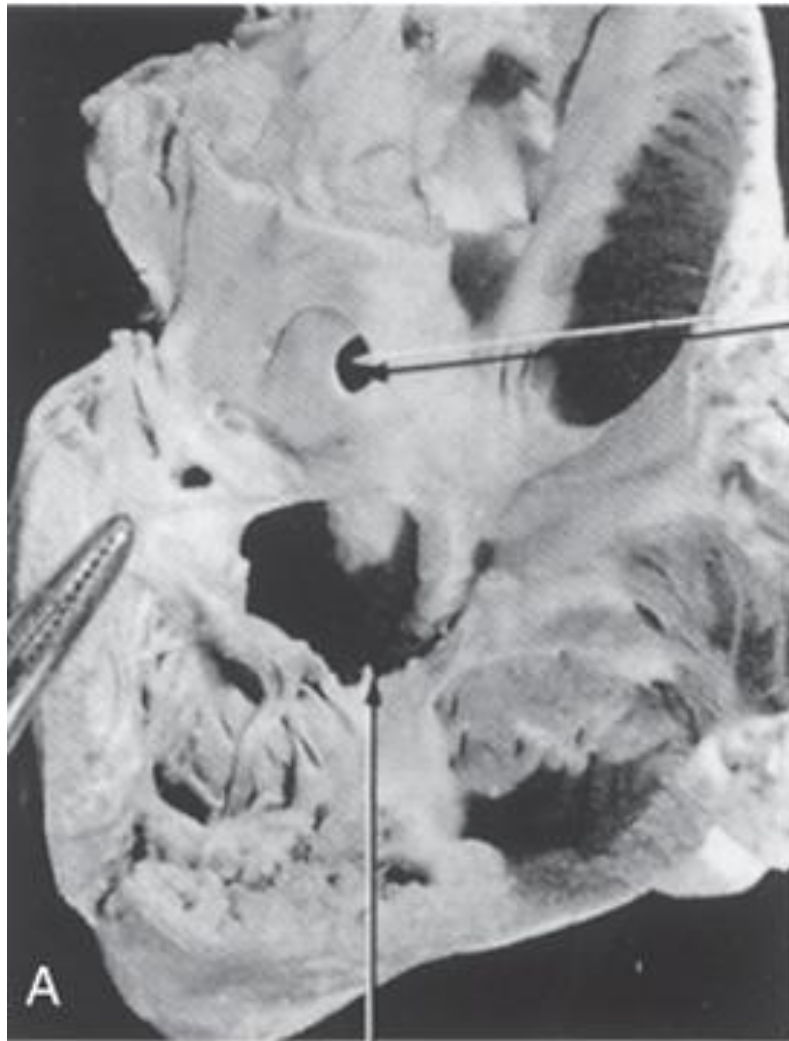


- ✓ **A, Patent** oval foramen resulting from resorption of the septum primum in abnormal locations.
- ✓ **B, Patent** oval foramen caused by excessive resorption of the septum primum (short flap defect).
- ✓ **C, Patent oval** foramen resulting from an abnormally large oval foramen.
- ✓ **D, Patent** oval foramen resulting from an abnormally large oval foramen and excessive resorption of the septum primum.
- ✓ **E, Endocardial** cushion defect with primum-type ASD. The adjacent section shows the cleft in the anterior cusp of the mitral valve.
- ✓ **F, Sinus** venosus ASD

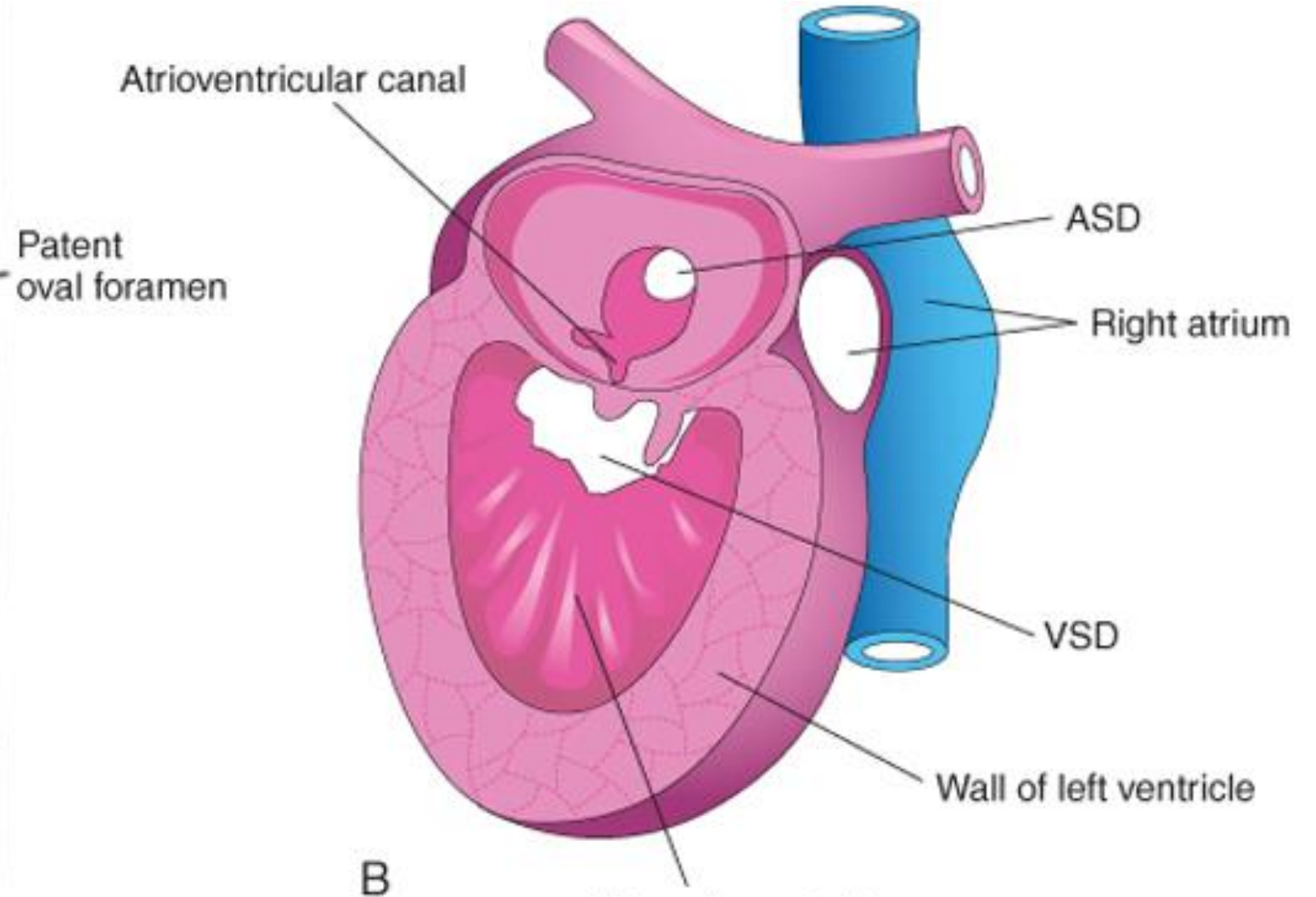
Atrial septal defect (ASD),,,

- ❑ **Endocardial cushion defects with ostium primum ASDs:** are less common forms of ASD.
- ❑ Several cardiac abnormalities are grouped together & they result from the same developmental defect, a deficiency of the endocardial cushions and the AV septum.
 - The septum primum does not fuse with the endocardial cushions; as a result, there is a **patent foramen primum-ostium primum defect**.
 - Usually there is also a cleft in the anterior cusp of the mitral valve.
 - In the less common complete type of endocardial cushion and AV septal defects, fusion of the endocardial cushions fails to occur.
 - As a result, there is a large defect in the center of the heart known as an **AV septal**

Atrial septal defect (ASD),,,



Atrioventricular
septal defect



B

Atrial septal defect (ASD),,,

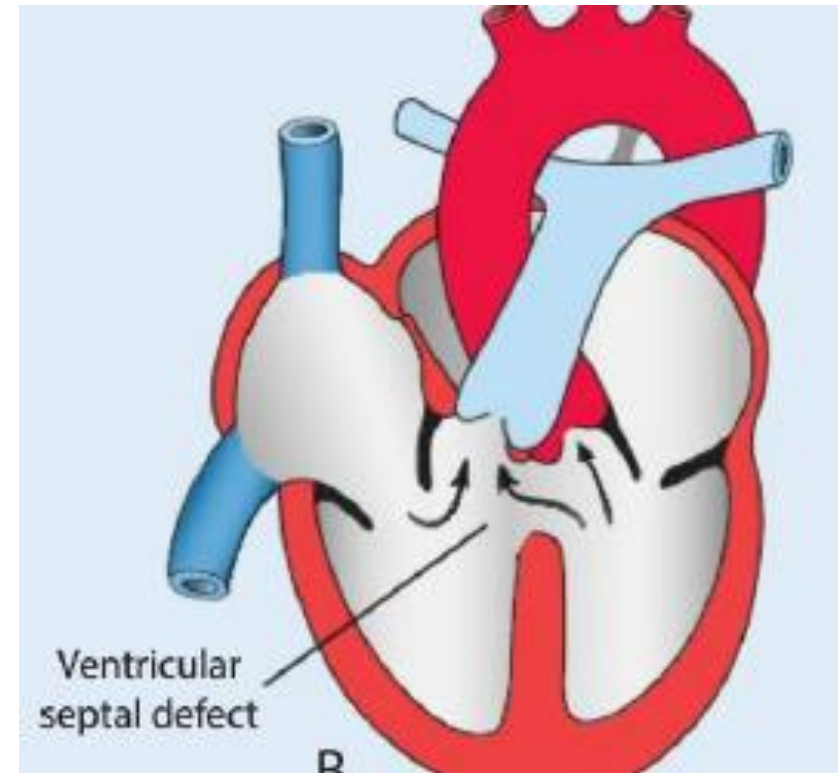
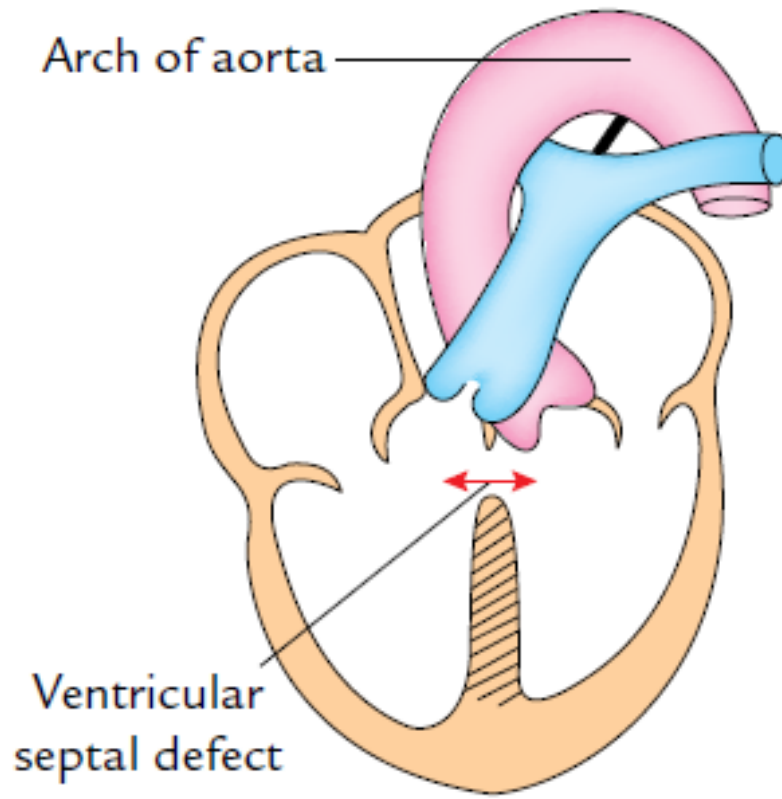
- ❑ **All sinus venosus ASDs** (high ASDs) are located in the superior part of the interatrial septum close to the entry of the SVC.
- ❑ A sinus venosus defect is one of the rarest types of ASD.
- ❑ It results from incomplete absorption of the sinus venosus into the right atrium and/or abnormal development of the septum secundum.
- ❑ This type of ASD is commonly associated with partial anomalous pulmonary venous connections.
- **Common atrium** is a rare cardiac defect in which the interatrial septum is absent.
- This situation is the result of failure of the septum primum and septum secundum to develop (combination of ostium secundum, ostium primum, and sinus venosus defects).

Ventricular septal defect (VSD)

- ***Ventricular septal defect (VSD)*** is most common congenital anomaly of the heart. This defect is most common in the membranous part of the IV septum.
- It is caused by failure of fusion of right and left bulbar ridges with the AV cushions, which form the membranous part of the IV septum.
- This defect results a communication between right and left ventricles.
- Blood flows from the left to the right ventricle. As a result of left-to-right shunting of blood, the output from the left ventricle is reduced.
- Consequently, the patient complaints of excessive fatigue on exertion. Rarely, the defect may be present in muscular part of the IV septum.

Ventricular septal defect (VSD),,,

- ❖ Absence of the IV septum-**single ventricle** or common ventricle-resulting from failure of the IV septum to form, is extremely rare and results in a **three-chambered heart** (L. *cor triloculare biatriatum*).



☐ ***Ventricular septal defect (VSD)***

Ventricular septal defect (VSD),,,

❑ **A ventricular septal defect can arise from several causes:**

- ✓ Deficient development of the proximal conotruncal swellings,
- ✓ Failure of the muscular and membranous ventricular septa to fuse,
- ✓ Failure of the dorsal and ventral endocardial cushions to fuse (atrioventricular septal defect), and
- ✓ Insufficient development of the interventricular muscular septum.
- Whatever the origin of a ventricular septal defect, its most serious consequence is the left-to-right shunting of blood and the consequent increased blood flow to the pulmonary circulation. ***It can be repaired surgically.***

Fallot's tetralogy (tetralogy of Fallot)

- ***In Fallot's tetralogy [tetralogy of Fallot]*** many cardiac defects occur together more often than in isolation.
- In some cases, such associated defects are actually components of the same malformation as, for example, a ventricular septal defect & persistent truncus arteriosus.
- Fallot used the term tetralogy to refer to the four classic malformations in this syndrome:

I. pulmonary stenosis,

II. overriding aorta (i.e., displacement of aortic orifice to Rt to override the IV septum),

III. IV septal defect, and

IV. hypertrophy of the right ventricle.

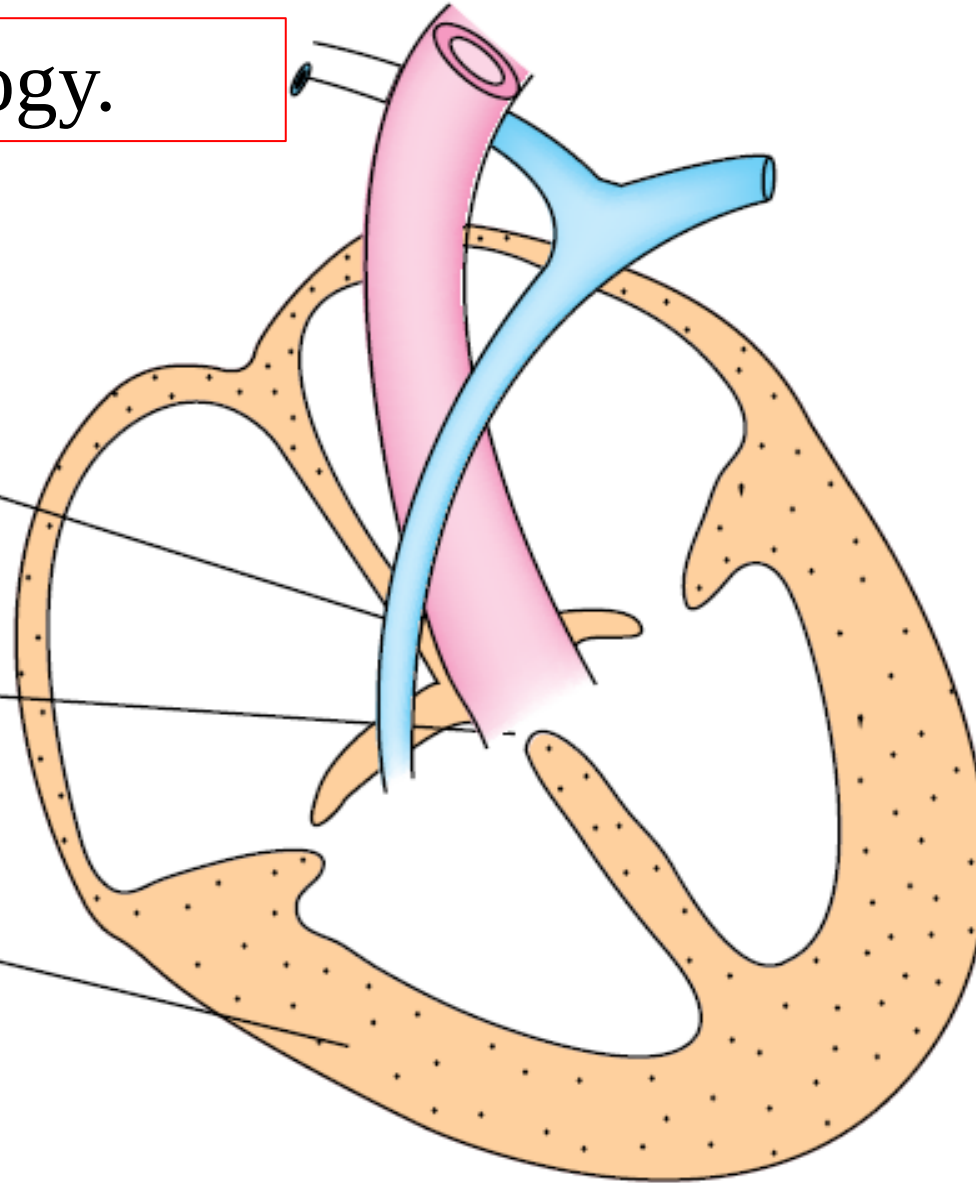
V. The resultant right-to-left shunting of blood leads to cyanosis.

❑ ***The Fallot's tetralogy is the commonest congenital cyanotic heart disease.***

Fallot's tetralogy (tetralogy of Fallot),,,

☐ Fallot's tetralogy.

1. Stenosis of pulmonary trunk
2. Ventricular septal defect
3. Overriding of aorta
4. Right ventricular hypertrophy



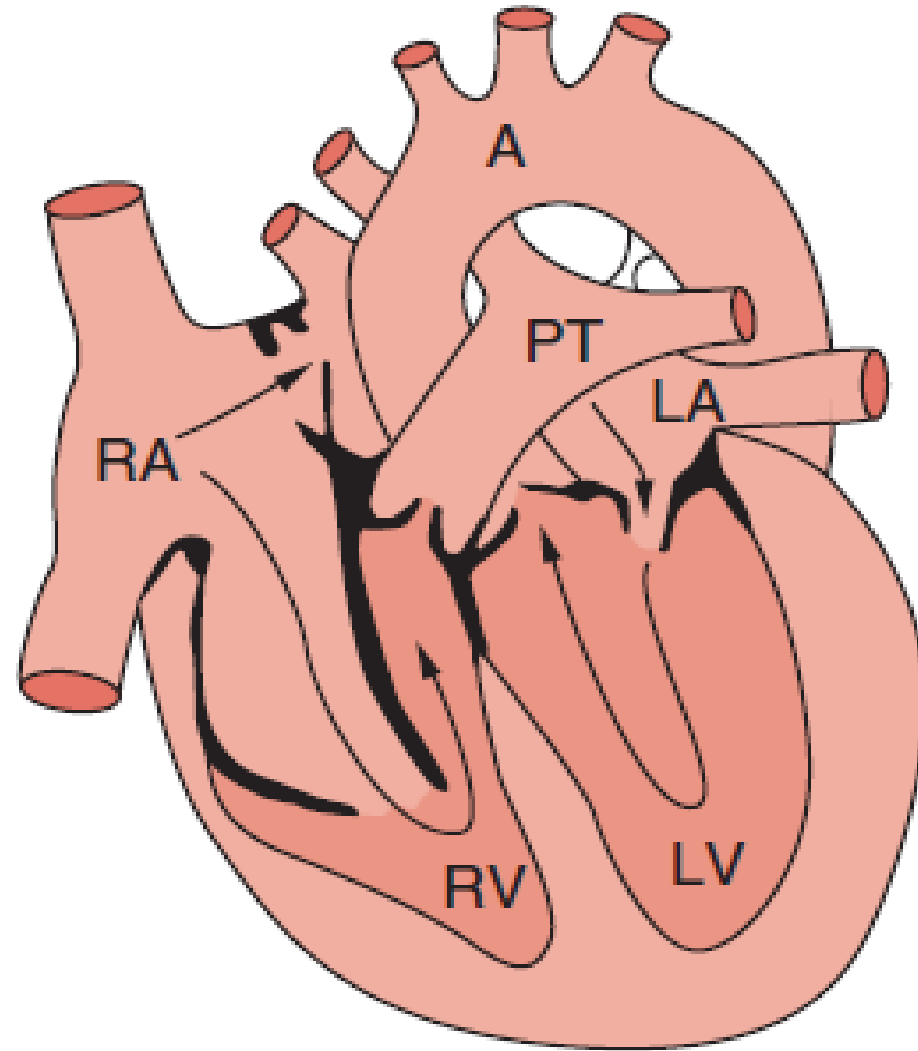
Atrioventricular Valve Defects

- Atrioventricular valve defects arise from errors in the remodeling necessary for forming the valve leaflets, chordae tendineae, and papillary muscles.
- If the atrioventricular septum does not form, sometimes the wedging and remodeling required for aligning the atrioventricular canals with the appropriate ventricle fails.
- As a consequence, a ventricle can end up having a double inlet (having the inflow from both atria). Likewise, malalignment of the outflow tract may lead to a ventricle having a double outlet (having both the aorta and pulmonary artery).
- In double-outlet right ventricle malformation, both the aortic and pulmonary outflow tracts connect to the right ventricle, and this malformation is almost always accompanied by a ventricular septal defect. All arterial blood flow leaves from the right ventricle and there is mixing of oxygenated blood with unoxygenated blood within the right ventricle.

Atrioventricular Valve Defects,,,

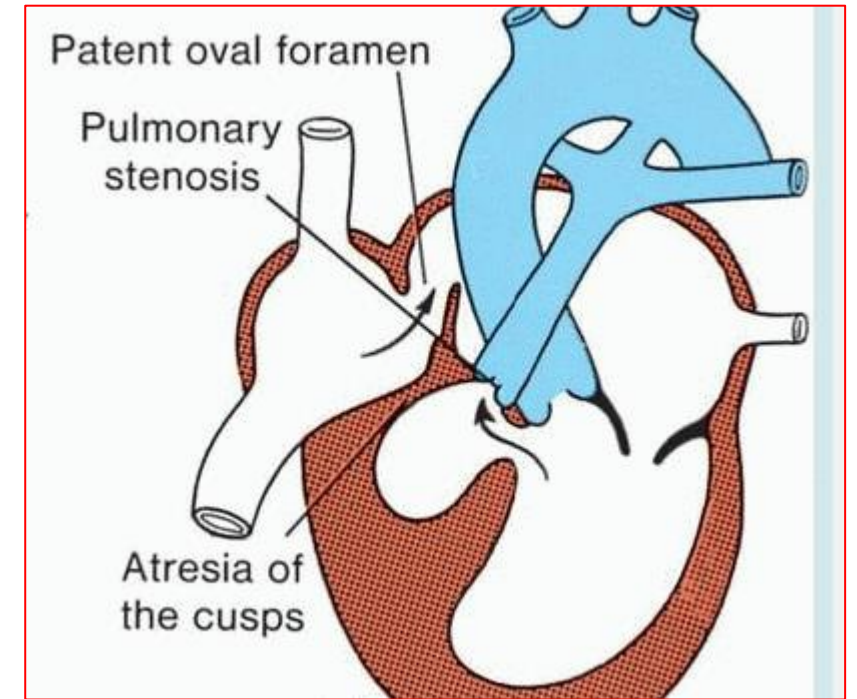
- **Ebstein anomaly** is caused by the failure of the posterior and septal leaflets of the tricuspid valve to attach normally to the annulus fibrosus; instead they are displaced inferiorly into the right ventricle.
- It results in a division of the right ventricle into a large, upper, “atrialized” portion and a small, lower, functional portion.
- There is a reduced amount of blood available to the pulmonary trunk due to the small, functional portion of the right ventricle.
- It is usually associated with an ASD and maternal lithium exposure.

Atrioventricular Valve Defects,,,



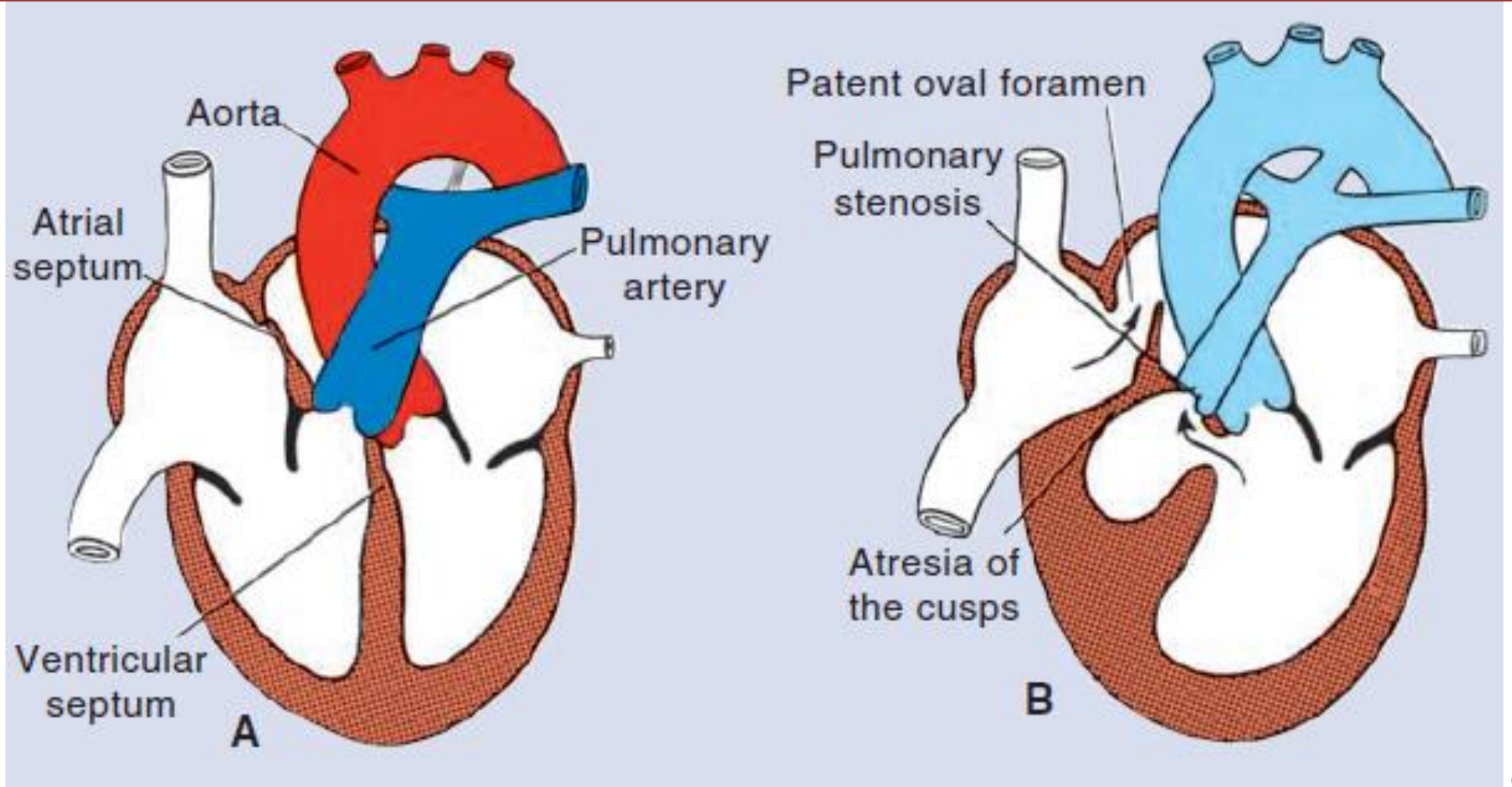
Atrioventricular Valve Defects,,,

- ***Tricuspid atresia:*** In this condition, there is complete agenesis of tricuspid valves. It is caused by an insufficient amount of tissue in AV cushion for the formation of tricuspid valve.
- **The tricuspid atresias are always associated with**
 - (a) patent foramen ovale,
 - (b) VSD,
 - (c) underdeveloped right ventricle, and
 - (d) hypertrophy of the right ventricle.
- **N.B.** The blood reaches the right ventricle through patent foramen ovale and IV foramen.



Atrioventricular Valve Defects,,,

- **Tricuspid atresia. A, normal Tricuspid B, Tricuspid atresia**
- **Note the small right ventricle and the large left ventricle**



Persistent Atrioventricular Canal

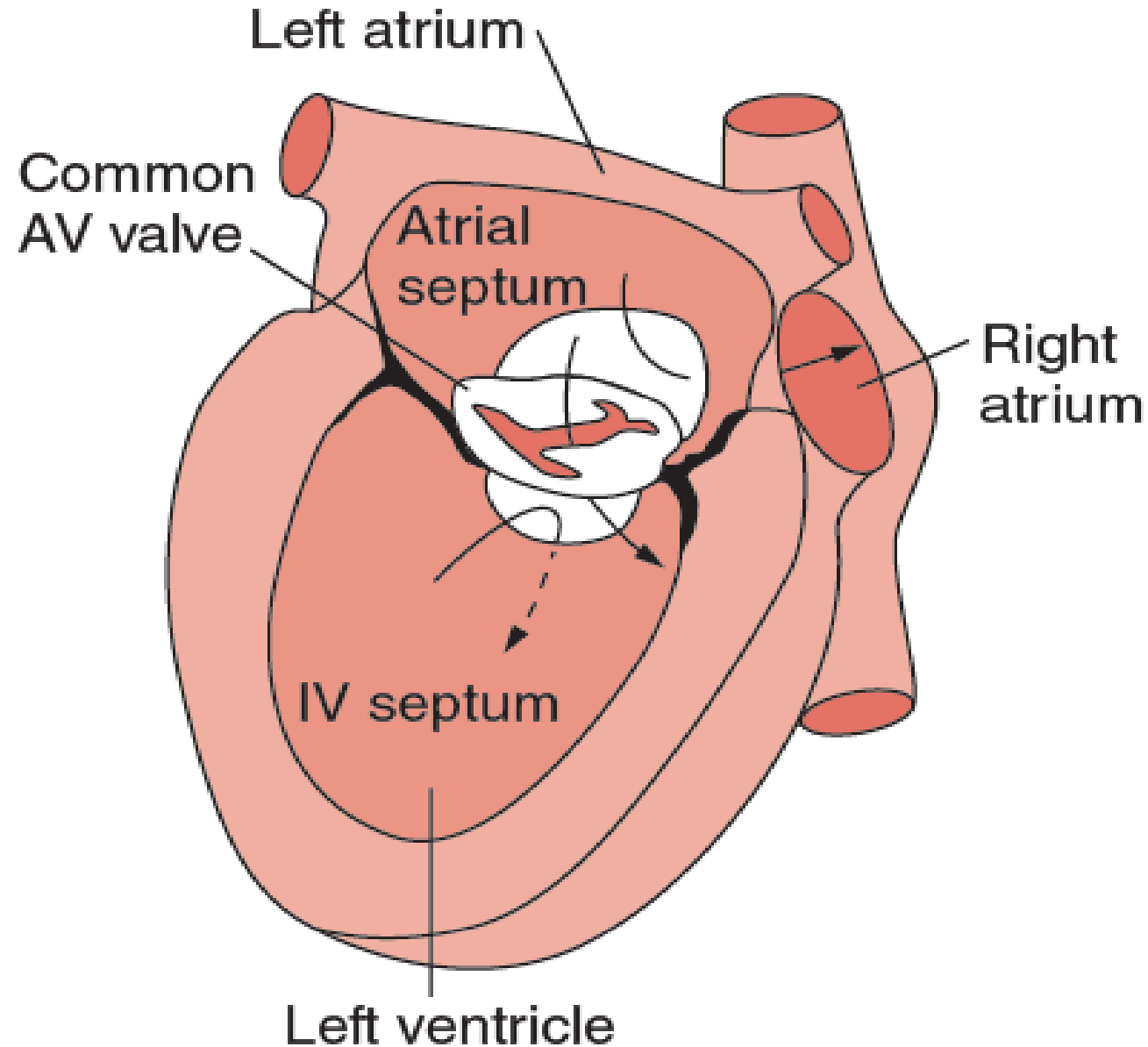
- **Persistent common AV canal** is caused by failure of fusion of the dorsal and ventral AV cushions.
- It results in a condition in which the common AV canal fails to partition into the right and left AV canals, so that a large opening exists in the center of the heart.

Consequently, the tricuspid and bicuspid valves are represented by one valve (common AV valve) common to both sides of the heart.

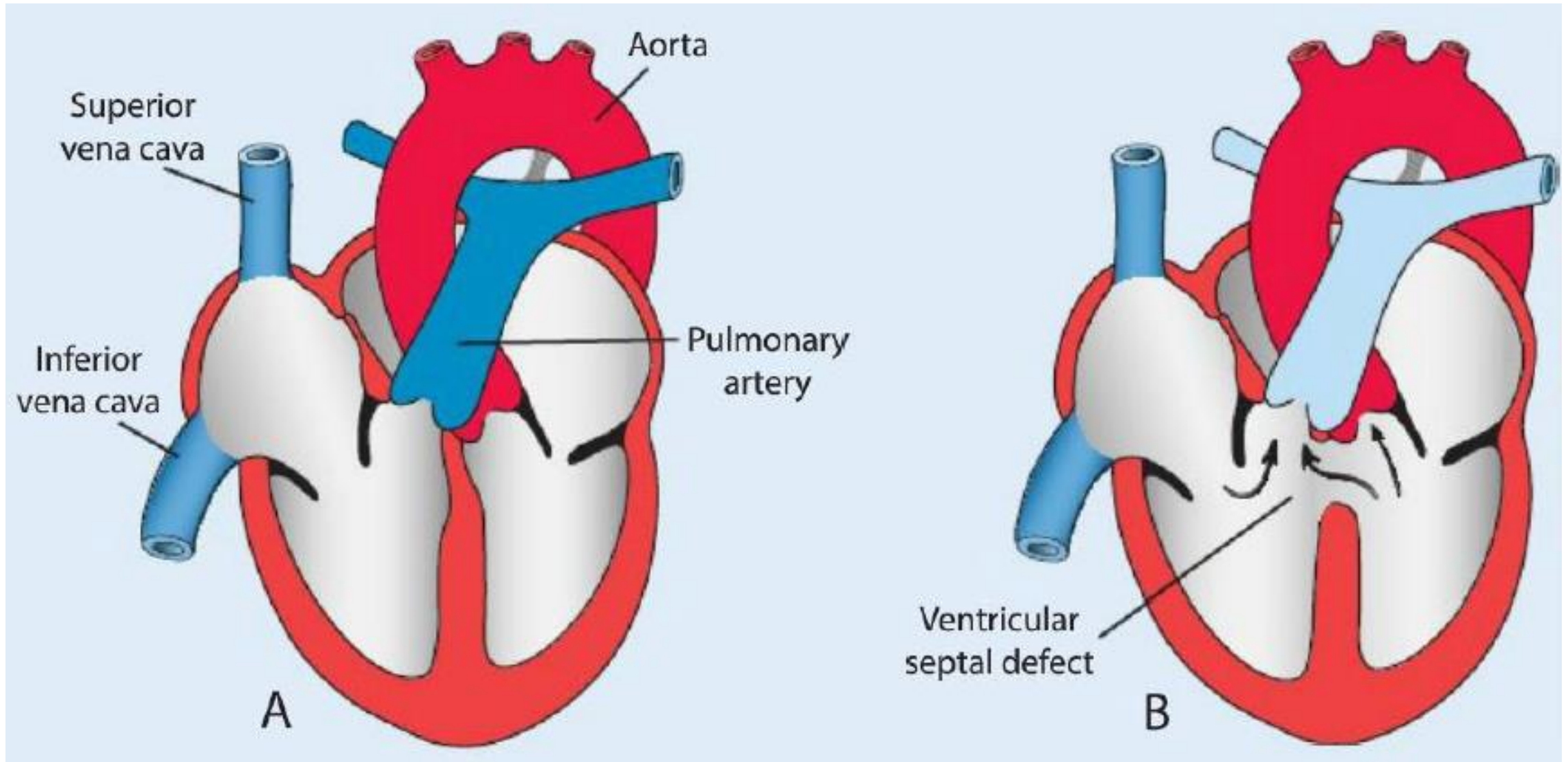
❑ **Two common hemodynamic abnormalities are found:**

- I. **Left–right shunting** of blood from the left atrium to the right atrium causes an enlarged right atrium and right ventricle.
- II. **Mitral valve regurgitation** causes an enlarged left atrium and left ventricle.

Persistent Atrioventricular Canal,,,



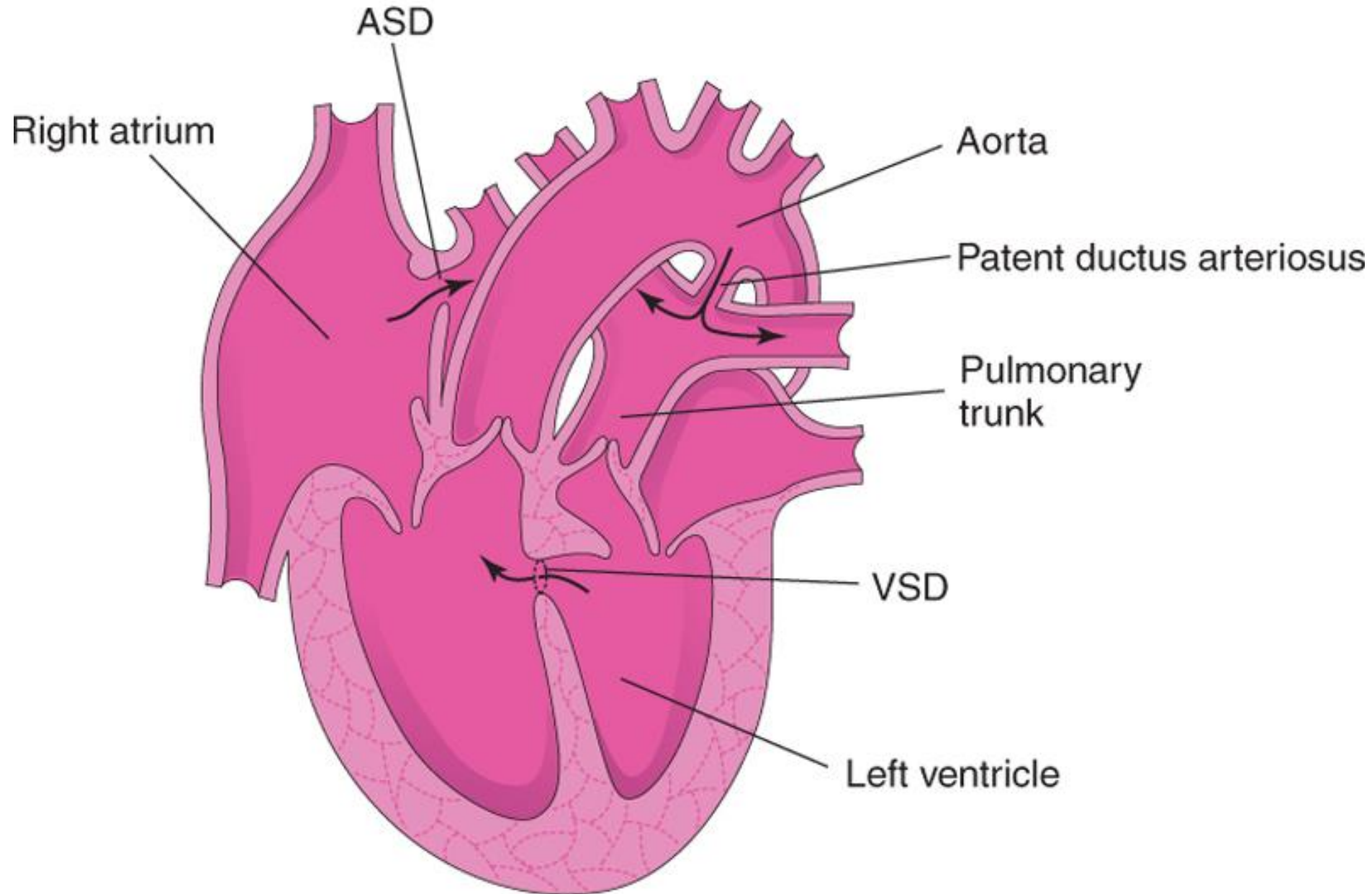
Persistent Atrioventricular Canal,,,



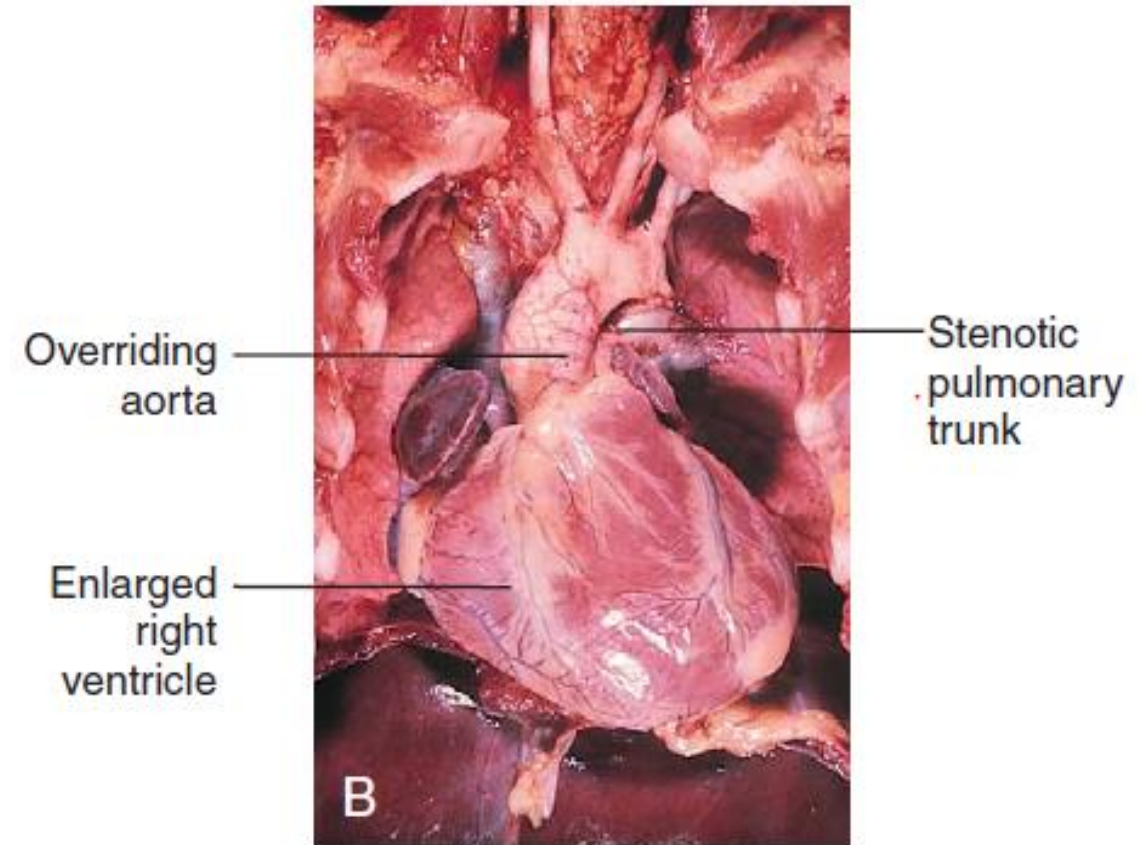
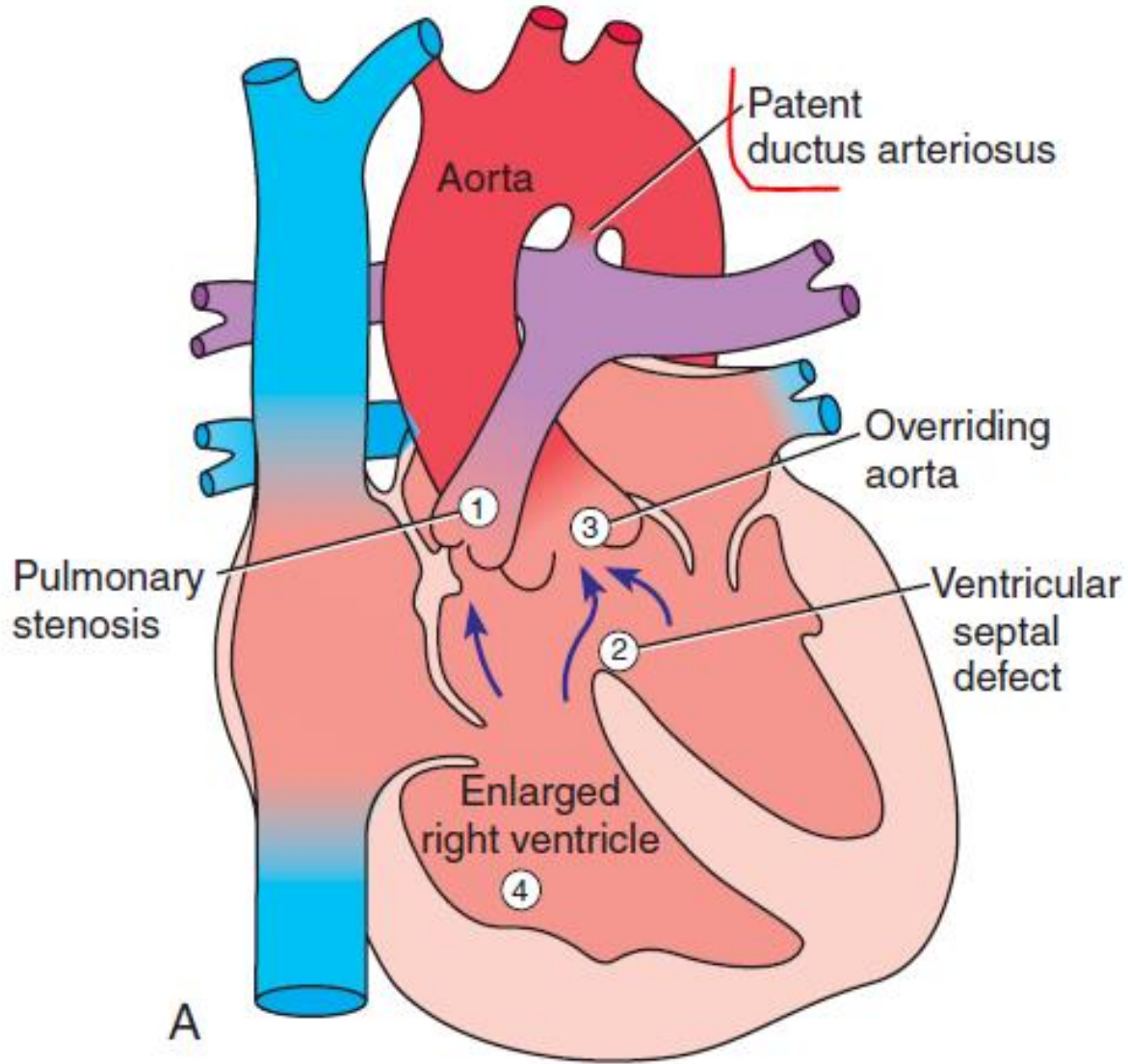
Persistent truncus arteriosus

- Persistent truncus arteriosus results from failure of the truncal ridges and aortopulmonary septum to develop normally and divide the TA into the aorta and pulmonary trunk.
- A single arterial trunk, the TA, arises from the heart and supplies the systemic, pulmonary, and coronary circulations. A VSD is always present with a TA defect; the TA straddles the VSD
- Recent studies indicate that developmental arrest of the outflow tract, semilunar valves, and aortic sac in the early embryo is involved in the pathogenesis of TA defects.
- The common type of TA is a single arterial vessel that branches to form the pulmonary trunk and ascending aorta.
- In the next most common type, the right and left pulmonary arteries arise close together from the dorsal wall of the TA

Persistent truncus arteriosus,,,



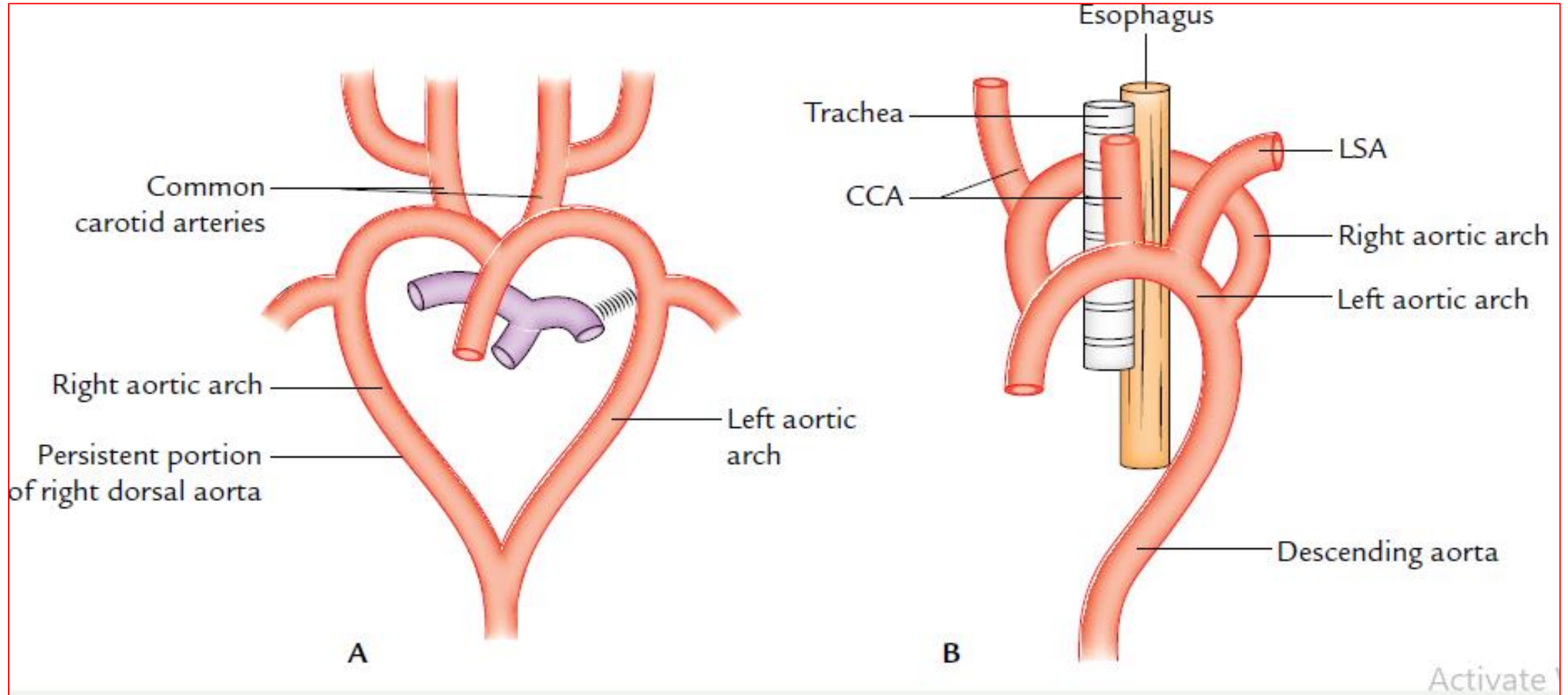
Persistent truncus arteriosus,,



Double aortic arch

- ***Double aortic arch*** In this anomaly, the right dorsal aorta between the origin of seventh cervical intersegmental artery and its junction with the left dorsal aorta does not disappear.
- As a result, there is formation of arch of the aorta on both sides.
- In this anomaly, the right arch of aorta passes behind the trachea and esophagus, while the left arch passes in front of them.
- The vascular ring thus formed between right and left aortic arches commonly causes compression of trachea and esophagus, leading to difficulty in breathing and swallowing.

Double aortic arch,,,



- Double aortic arch. A. Persistence of the distal portion of right dorsal aorta.
- B. Double aortic arch forming vascular ring around trachea and esophagus. LSA = left subclavian artery, CCA = common carotid arteries

The end!